

UNITEC INSTITUTE OF TECHNOLOGY

MASTER THESIS

Analysis and Reconstruction of Distorted Speech Using Deep Learning

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A thesis submitted in partial fulfillment of the requirements

for the degree of Master of Applied Technologies

in the

School of Computing, Electrical & Applied Technology

August 14, 2024

Abstract

Communication through whispering is essential for individuals who have undergone laryngectomy, a surgical procedure involving removing part or all of the vocal box. Whispering is unique due to its absence of fundamental frequency (pitch) and can be considered as an alternative mode of communication for laryngectomy patients. While it is generally a quiet form of communication in healthy people, it often results in a hushed (and sometimes unintelligible) speech in laryngectomised individuals, necessitating the use of prosthetics or specialised treatments. Current prosthetic solutions have inherent limitations. The medical treatments pose a risk of getting infections after surgery, and the generated speech by prosthetics sounds mechanical, following these computational methods developed. Some state-of-the-art deep learning algorithms have been developed in computational methods to generate natural-sounding speech. However, they were focused on reconstructing whispered to normal speech, not on laryngectomised or distorted speech.

This thesis focuses on analysing these Deep Learning algorithms using objective evaluation metrics and aims to apply these existing algorithms to a laryngectomised dataset for the first time in literature. We discuss the results of these evaluations and perform a comparative analysis between the models.

We are starting our analysis with GAN-based models. Following this, we are moving to WESPER, a prediction-based model. Lastly, we are going to analyse the voice conversion-based models developed to convert speech from one speaker style to another and translate one language into another.

Our initial analysis comprises 198 tests on 11 models and 6 objective evaluation metrics. This evaluation is going to be done on the testing dataset, which

has three patient categories, namely Partial Laryngectomy (PL), Total Laryngectomy (TL), and Total Laryngectomy with Trachea Esophageal Puncture (TEP).

Based on the results of such evaluation, I propose modifications to the architecture of five GAN-based models. I particularly adjust the models and loss functions to improve the outcome for laryngectomy patients. In addition, they are cross-compatible with each other and propose a total of 25 models. For a better understanding of the features of laryngectomised speech, we are including the laryngectomised dataset combined with the wTIMIT dataset in the training process. Using the same set of objective evaluation metrics, these proposed models demonstrate a better denoising effect in reconstructed speech, spectral features, and intelligibility than the existing models.

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List of Abbreviations

AE	Auto-Encoder
BCE	Binary Cross Entropy
BERT	Bidirectional Encoder Representations from Transformers
CD	Cepstral Distortion
CELP	Code-Excited Linear Prediction
CNN	Convolutional Neural Network
DBLSTM	Two Deep Bidirectional LSTM
DCNN	Deep Convolutional Neural Networks
DL	Deep Learning
DTW	Dynamic Time Warping
EL	Electrolarynx
ES	Esophageal speech
F0	Fundamental Frequency
Fv	Feature Vector
GAN	Generative Adversarial Network
GMM	Gaussian Mixture Model
HuBERT	Hidden Unit BERT
IS	Itakura-Saito Distance
KL	Kullback-Leibler
Leaky-ReLU	Leaky Rectified Linear Unit
LLR	Log Likelihood Ratio
LPC	Linear Predictive Coding
MCC	Mel Cepstral Coefficients

MCD	Mel Cepstral Distortion
MELP	Mixed Excitation Linear Prediction
MFCC	Mel-Frequency Cepstral Coefficients
ML	Machine Learning
MLP	Multi Layer Perceptron
NAM	Non-Audible Murmur
NN	Neural Network
NZ	New Zealand
PeLU	Parametric Exponential Linear Unit
PESQ	Perceptual Evaluation of Speech Quality
PL	Partial Laryngectomy
PSD	Power Spectral Density
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Networks
RMSE	Root Mean Squared Error
RBM	Restricted Boltzmann Machines
SWS	Source-Filter Waveform Synthesis
STFT	Short-Time Fourier Transform
STOI	Short-Time Objective Intelligibility
STU	Speech-to-Unit
SSNR	Segmental Signal-to-Noise Ratio
TEP	Tracheoesophageal Puncture
TL	Total Laryngectomy
TL-TEP	Total Laryngectomy - Tracheoesophageal Puncture
TTS	Text-to-Speech
UTS	Unit-to-Speech
VUV	Voice-Unvoice
WSC	Whisper-to-Speech

Chapter 1

Introduction

1.1 Background and Inspiration

In a laryngectomy, the larynx also referred to as the voice box, is removed entirely or partially during surgery [1]. The larynx, a significant neck physical component, is responsible for making speech sounds, guarding the airway while swallowing, and permitting air movement to and from the lungs, among other vital tasks [2], [3]. Surgery is frequently used to treat diseases like laryngeal cancer as they are unresponsive to conventional therapies like radiation therapy or chemotherapy [4]–[7]. A laryngectomy involves the removal of all or a portion of the larynx, including the vocal cords, surrounding tissues, and perhaps neighboring lymph nodes [4], [8].

There are two types of laryngectomy procedures: Partial Laryngectomy (PL) and Total Laryngectomy (TL) [9], [10]. Only a section of the larynx is removed during a partial laryngectomy [11]–[13]. This surgery is considered when the cancer is contained in a particular larynx region [14], [15]. The goal is to efficiently treat the malignancy while preserving as much of the patient's natural voice and swallowing abilities as feasible. A total laryngectomy involves removing the larynx in its entirety. This operation is frequently required when cancer has spread far or all other treatment options have been tried. Natural

voice function is permanently lost after a total laryngectomy [16]. Following a laryngectomy, the airway is redirected to form a stoma or neck opening. This stoma eliminates the typical passage through the nose and mouth and allows air to travel straight from the trachea (windpipe) to the outside [5], [17], [18]. However, because the larynx is absent, patients cannot use their vocal cords to make speech sounds or to humidify and filter air before it enters their lungs [17]–[19]. Quality of life is significantly affected after laryngectomy, and patients identified speech, appearance, and activity as the most important issues following laryngectomy [20].

Certain medical techniques were devised to help with communication following laryngectomy, but they sounded mechanical and came with a risk of infection [12]. After that, electrolarynx devices were created, but the speech they produced sounded artificial [21], [22]. While computational algorithms have been developed recently, they primarily focus on whispered-to-normal speech reconstruction.

1.2 Objectives

This thesis investigates cutting-edge Machine Learning (ML) and Deep Learning (DL) algorithms for reconstructing whispers into normal speech. We concentrate on a thorough analysis that uses objective criteria to gauge the success of these algorithms accurately. Furthermore, we extend our analysis to adapting these state-of-the-art techniques to address the unique challenges of laryngectomised or distorted speech. We aim to modify these advanced architectures to meet the specific needs of laryngectomy patients.

1.3 Questions and Challenges

During this thesis, the following questions will be explored:

- In the context of whispered to normal speech conversion, what are the most effective metrics for evaluating the performance of deep learning algorithms?
- How can the best-performing methods identified from the evaluation of whispered to normal speech conversion be applied to improve the intelligibility of distorted or laryngectomised speech, based on the obtained results?
- Why do some state-of-the-art models fail at reconstructing distorted speech?
- What novel deep learning model can be developed or modified to enhance the reconstruction of natural-sounding speech for laryngectomised patients?

1.4 Thesis Structure

This thesis organised in four chapters and a conclusion: an introduction, objectives, and research questions were briefly discussed in Chapter 1. In Chapter 2, research on traditional and computational approaches to speech reconstruction is comprehensively discussed, and some state-of-the-art methods are presented. Chapter 2 ends with a research gap.

Chapter 3 starts with discussing datasets and then moves to objective evaluation measures. The last section discusses and analyses models discussed in Chapter 2. Chapter 4 proposes modified architectures for better alignment with laryngectomy patients' data and discusses proposed methodology, adjustments made on models, and comparative analysis with existing models before modifications. The conclusion summarises the details of the experiments.

Chapter 2

Literature Review

This chapter will first review the traditional approaches to relieve laryngectomy patients' communication problems before reviewing computational approaches, ML and DL based methods that will lead to a research gap.

2.1 Traditional Methods

Esophageal speech (ES), Tracheoesophageal Puncture (TEP), and Electrolarynx (EL) devices are three pioneer methods that have historically been used to alleviate speech difficulties within the context of conventional speech prostheses [12], [16], [23]–[25]. These methods have substantially contributed to helping people with speech difficulties resume productive communication.

ES [26] is an application of the resonating ability of the esophagus to generate speech-like sounds. People can learn to produce speech patterns that closely resemble genuine speech by carefully coordinating their muscle activities.

By making a new passageway between the trachea and the esophagus during a TEP [27], a surgical component is added to the situation. This surgery enables regulated airflow redirection, allowing people to manage their breath to produce recognisable speech sounds.

In the early 1980s, EL devices [28] began gaining attention in speech technology. These early devices produced sound using electronic vibrations, which users could shape into understandable speech. Although EL devices did not perfectly mimic natural speech, they provided a practical way for people to communicate clearly. This technology offered a reliable solution for those needing an effective means of speaking, even if the sound was different from normal speech.

This traditional method allows patients to regain limited speaking ability, but the speech sounds unnatural [21], [22]. Also, through literature, we have found that they are challenging to use, and patients are at risk of getting infections from surgical insertion in TEP [12]. To address this issues, Computational methods were developed. In the following section, this thesis focuses on computational methods.

2.2 Computational Methods

The main issue with EL devices was the generated speech sounded robotic. Alternate methods were developed to improve the naturalness of the generated speech. These methods are called computational methods. Computational methods for speech reconstruction majorly fall under two areas, which are training-based and non-training-based methods. Alternatively, they are called statistical and parameter-based methods [29]. The non-training-based methods, known as parameter-based methods [24], rely upon whisper enhancement and restore pitch [29], [30]. At the same time, training-based methods are statistical-based or machine learning algorithms that map the whispers to normal speech.

In non-training-based methods, we will discuss pioneer studies, which are SWS [24], [31] and LPC [32]. MELP [33] was introduced to improve LPC's

performance. MELP was using LPC as a foundation for the reconstruction of speech. A CELP-based system was also introduced using LPC as a foundational component [34].

Simultaneously, training-based methods were also introduced to generate natural-sounding speech. Toda introduced the use of GMMs to convert whispers to phonated speech [35]. Following that, different methods such as Cascading mapping algorithms [24] and NN-based methods were developed.

2.2.1 Non-Training Based Methods

SWS [31] is a technique for improving synthesised speech that combines a source waveform (such as the result of an EL) with a filter that simulates the features of the vocal tract. With this method, the resonance and articulation of the speech output can be adjusted, producing more realistic speech. SWS shown improvement in synthesising speech's overall quality. It mainly focuses on improving the pitch of the speech to make it more intelligible.

LPC was developed around the late 1960s for speech compression [36]. After that, LPC is being used as main method to feature extraction for signal and speech processing tasks. It converts speech waveform into a parametric representation at a relatively low data rate to be processed and analysed later [32], [37]. Due to the unchanging nature of glottal pulses, LPC, which describes speech signals as a collection of coefficients that attempt to predict the signal from past values in the time domain, performs poorly regarding natural voice quality. LPC effectively accounts for resonance and pitch movements [38]. LPC is the base for the MELP and CELP-based systems.

A speech synthesis and coding method called MELP[33] was created in the 1990s. It effectively uses a mixed excitation source (including both periodic and aperiodic components) in conjunction with LPC to encode and decode speech signals [39], [40]. This approach was good but not suited for real-time

speech reconstruction [34]. To improve the performance of MELP, a CELP-based approach was proposed.

The CELP [34] was primarily proposed for under consideration of laryngectomy patients because when working with these patients, they did not have any corresponding normal speech sample similar situations for telemedicine platforms and real-time communication applications [41]. CELP just takes the whispered speech as input and predicts the pitch and aperiodic components using LPC as core in architecture, and this approach shows better performance for real-time scenarios without parallel data.

2.2.2 Training Based Methods

The training-based methods are also known as statistical-based or ML-based techniques.

In this section, we will start by covering traditional ML methods. Then, we will move on to more recent DL approaches [42] majorly developed to reconstruct normal speech from whispered speech. Lastly, we will discuss the voice conversion (VC) techniques.

2.2.2.1 Traditional Machine Learning Based Methods

This subsection will discuss early ML or statistical-based models which are Gaussian Mixture and Cascading Mapping model.

- **Gaussian Mixture Models**

As a pioneer in the GMM-based Whispered Speech Conversion (WSC) area, Tomoki Toda is first and foremost to introduce this methods [35]. The goal of Toda's study was to produce speech that sounds more like murmur signals, which are normally inaudible. The approaches were expanded to turn

whisper-like speech into normal-sounding speech after being trained with parallel utterance data (spoken and whispered versions of the same speech) [43].

Modern variations of this approach use up to three GMMs: the first transforms the source spectral features into target spectral features, the second converts the same source spectral features into a pitch excitation, and the third creates additional aperiodic components present in the target speech to improve naturalness. The converted speech is synthesised using pitch, aperiodic components, and approximated spectral properties through the STRAIGHT vocoder [44], [45].

Although these techniques can yield decent normal speech, they frequently have problems with over-smoothing the sound characteristics. Additionally, the speech rhythm and intonation may be wrong since it is difficult to establish the fundamental frequency (f_0) from the whisper spectrum.

- **Cascading Mapping Models**

The cascading mapping algorithm [24] enhances the quality and speed of speech by reconstructing normal speech from whispers. This approach has a two-step mapping process. Initially, whispered speech is converted into "whispered speech" a form of speech that retains time alignment and spectral properties similar to natural speech but without pitch. In the second step, this "whispered speech" is then mapped to normal speech. This algorithm uses a single GMM for both F_0 estimation and voiced/unvoiced prediction using STRAIGHT [44], [45]. This approach improves processing efficiency and overall speech quality by addressing the limitations of over-smoothing and unclear speech in earlier techniques by Todo [35].

These two methods showed a way that can be used to reconstruct normal speech from whispers or nonaudible murmur (NAM), but they were trained on smaller datasets, and the ML methods were not complex enough to extract

the features. In the preceding sub-section, we will explore deep learning-based models that alleviate the complexity required for whisper or NAM to normal speech conversion tasks.

2.2.2.2 Deep Learning Based Methods

In this sub-section, we will discuss the DL-based methods that use neural networks (NN) as the base to reconstruct speech from whispers.

- **Restricted Boltzmann Machines**

RBM is a neural network variant that links input data to hidden units using a set of weights [46], [47]. These weights are modified during training to capture the connection between whispered and normal speech features [48]. The training process comprises many sequential stages. Initially, a model is trained to categorize speech frames as either voiced or unvoiced by utilising characteristics extracted from normal speech. Next, a model is trained to estimate the fundamental frequency (f_0) for the voiced frames of normal speech. The third model uses multiple RBMs to map the whispered spectral features to normal speech features. However, they had varying time alignment, and to solve it, the authors used dynamic time warping (DTW) [49] to time-align whisper and normal speech spectral features.

During the reconstruction step, the model initially assesses the voicing status of each frame of input whisper speech, classifying them as either voiced or unvoiced. The system calculates the pitch for frames with vocal sounds and transforms the spectral characteristics using the RBMs that have been trained. The mapped spectral features are subsequently merged to provide the reconstructed speech.

The evaluation of this approach shows enhancements compared to conventional GMM-based techniques in terms of objective evaluation described in the literature [50]. The RBM technique shown good accuracy in predicting

pitch and voiced frames. Subjective tests conducted with listeners also suggested that the speech reconstructed by RBM sounded more natural and clear compared to GMM-based methods.

- **Deep Partial Neural Network**

The deep partial neural network, as described in the study by McLoughlin et al. [51], is designed to reconstruct speech for patients with larynx-related dysphonia. This includes individuals who have had partial laryngectomy or have disorders that damage their vocal cords. This patients frequently has difficulties in articulating normal speech and need of methods to reconstruct this speech.

In order to tackle this challenge, the authors suggest using a semi-supervised deep neural network (DNN) methodology. The training approach entails utilising whisper and normal speech data, obtained not from patients but from the wTIMIT corpus [52], [53] as they did not have patients data. At first, two distinct Gaussian-Bernoulli RBMs are trained on the spectral characteristics of whispers and parallel normal speech in an unsupervised manner using the Contrastive Divergence technique [54]. Each RBM effectively maps the spectral features to a lower-dimensional binary space and makes it more accessible to the following layers to process data.

These trained RBMs are then stacked to form the initial layers of a DNN. The middle layers are trained in a supervised manner using the Back Propagation algorithm. This supervised training maps the binary features extracted from whispers to those of normal speech, using the pre-trained RBMs. This combination of unsupervised and supervised training enables the DNN to learn the mapping between whispered and normal speech features, even with limited data.

During reconstruction, spectral features are extracted from input whispered

speech and passed through the RBM first and then DNNs to generate corresponding speech features. These features are refined using the Maximum Output Probability Parameter Generation (MOPPG) algorithm to produce the final reconstructed speech. This method reduces overfitting because it uses pre-trained RBMs and improves the quality of the reconstructed speech compared to GMM and vanilla RBM based methods [55].

• DNN-MCC-F0

DNN-MCC-F0 paper [56] proposes a method to convert whispered speech into normal speech using low-dimensional Mel Cepstral Coefficients (MCC) and using DNNs for training on this features.

The training process involves several steps. First, spectral features are extracted from whisper and normal speech data [57]. DTW [49] is used to align these features frame-by-frame. Two separate DNNs are then trained: one for spectrum conversion and the other for pitch estimation. The proposed method uses a 60-dimensional MCC feature vector for the mapping, effectively reducing the computational complexity, and authors used 300 parallel whispers and normal speech utterances from the TIMIT database [58].

The conversion process is divided into two main modules: spectrum mapping and fundamental frequency (F0) estimation. The MCC features are used to characterise the spectrum envelope of the speech. The DNN is employed to model the relationship between the low-dimensional MCC features of whispered speech and those of normal speech.

In the spectral mapping model, the spectral envelope of the speech signal is estimated using the STRAIGHT model [44], [45], which is used as a vocoder to extract the features. The MCC features and their first-order differences are inputs to a five-layer DNN, which maps the whisper spectral features to the normal speech spectral features.

For estimating F0, whisper speech lacks that feature, so only normal speech data is used to train that model. After training both models, first, the testing whisper data passes through the MCC mapping model, then goes through the F0 model, combines the result from both models, and uses STRAIGHT to get synthesised speech.

- **SEQ2SEQ**

In this study, authors proposed a sequence-to-sequence (SEQ2SEQ) framework [59]–[61] in combination attention mechanism for whisper-to-normal speech conversion. This study also focuses on mapping of spectral features from whisper speech to normal speech like previous methods we discussed.

Initially, the spectral features of both whispered and normal speech is transformed into MFCCs. These MFCCs are the main features of the training model. The SEQ2SEQ framework consists of an encoder and a decoder. The encoder processes the sequence of whispered speech features, summarising them into a fixed-length context vector that encapsulates the entire sequence's information. The decoder then uses this context vector, which generates the normal speech features sequentially.

An important benefit of this architecture is its ability to handle input whisper and target normal speech sequences of different durations without the requirement for frame alignment using DTW. The model's versatility enables it to convert the source whisper feature sequence into a target regular speech feature sequence, even if they have different durations. The encoder and decoder have stacking LSTM networks.

Furthermore, the model employs an auditory attention mechanism to dynamically modify the context vector during the decoding phase to reconstruct the speech. This strategy enhances the model's capacity to establish connections between whispered and regular speech elements. The SEQ2SEQ framework is

trained, and the resulting mapped data is utilised for the prediction of the F0 feature in reconstructed spectral features using two DBLSTM layers.

- **Modified Transformer Network**

Modified transformer network is the successive study of SEQ2SEQ model [62], [63] with the same attention mechanism that we talked about in previous section.

Initially, this study used both whisper and normal speech data and trim the data to match the duration of parallel utterances, and transformed them into MFCCs. The training dataset consists of the wTIMIT corpus [52], [53]. This corpus comprises 450 phonetically balanced phrases extracted from the TIMIT dataset. The sentences are spoken by 48 distinct speakers, resulting in a total of 18,620 parallel speech utterances. After alignment, the dataset amounts to 15 hours of data. In addition, the Speech Ocean dataset [64] consists of 84,000 phonetically balanced phrases, totaling 100 hours of speech data. Both datasets were down-sampled to 16 kHz and DTW was used to time align the dataset [49]. Features such as 80-dimensional MFCCs and 24-dimensional smoothed spectral features were extracted using the Librosa [65] and WORLD vocoder [66] toolkits.

The model utilises a two-layer LSTM network as the encoder to process the whispered speech and normal speech features, summarising them into a context vector that captures the entire sequence's information, which basically encodes the input data.

The decoder also uses the two layers of LSTM, and it decodes the encoded data from the encoder and gives context vectors, which are passed through the WORLD vocoder to reconstruct speech. The authors show better performance than GMM, RBM, DNN, and SEQ2SEQ models in terms of objective and subjective evaluation.

- **Meta-BLSTM**

Meta-BLSTM introduces an advanced method for converting whispered speech into normal speech using a BLSTM network combined with a meta-network [67], [68]. The models used include a standard BLSTM and the proposed meta-BLSTM. The meta-BLSTM differs from the standard BLSTM by generating non-shared weights for the LSTM memory blocks using a meta-network. This meta-network dynamically generates the parameters for the LSTM, thereby reducing model complexity and training time.

The dataset used for training the model had of 337 parallel utterances from the TIMIT database [58]. Each utterance was sampled at 8 kHz and stored with 16-bit PCM. The dataset was divided into a training set with 300 parallel utterances and a test set with 37 parallel utterances. Spectral envelopes were extracted every 40 ms with a 5 ms frameshift and obtained using a 512-point Fast Fourier Transform (FFT). The spectral envelope was then reduced to a 30-dimensional MCC feature vector.

The process begins by extracting the spectral envelope, the aperiodic component, and the F0 of normal speech, as well as the spectral envelope of whispered speech, using the STRAIGHT speech synthesiser as a vocoder [44], [45]. These spectral envelopes are then transformed into MCCs. Since the duration of whispered and normal speech utterances were different, DTW [49] is used to align the speech features temporally before training.

In the training phase, three models are trained to map the whispered MCC features to normal MCCs and then predict F0 for mapped whisper speech. The model shows better results compared to previous LSTM-based architectures.

- **MMSE-GAN**

MMSE-GAN [69]–[71] is being proposed for converting NAM speech to whispered speech using a generative adversarial network (GAN) [72] architecture.

MMSE-GAN used normal adversarial loss that gan uses to minimise the distribution difference between a parallel whisper and NAM speech expressed in 2.1. Also, it has added the second term that uses MSE to reduce the distance between the generated sample and target whisper speech, which can be expressed mathematically in equation 2.2. In equation 2.1 D_{WHSP} denotes the discriminator and equation 2.2 $G_{NAM2WHSP}$ denotes generator.

$$\begin{aligned} \min_D V(D_{WHSP}) = & -\mathbb{E}_{y \sim p(y)} [\log D_{WHSP}(y)] \\ & - \mathbb{E}_{x \sim p(x)} [1 - \log D_{WHSP}(G_{NAM2WHSP}(x))] \end{aligned} \quad (2.1)$$

$$\begin{aligned} \min_G V(G_{NAM2WHSP}) = & -\mathbb{E}_{x \sim p(x)} [\log D_{WHSP}(G_{NAM2WHSP}(x))] \\ & + \frac{1}{2} \mathbb{E}_{y \sim p(y), x \sim p(x)} [\log(y) - \log(G_{NAM2WHSP}(x))]^2 \end{aligned} \quad (2.2)$$

The dataset used for training the models comprises 421 utterances from the CSTR NAM TIMIT Plus corpus [73]–[75], with 350 utterances used for training, 50 utterances used for validation and 21 for testing. MCC feature for 25 channels extracted using AHOCODER [76]–[80] for both whisper and corresponding NAM speech and used DTW [49] to time align data. In the model architecture, the generator has three DNN layers with 512 unit size, and the discriminator has the same architecture and is trained in the GAN framework. The generator learns mapping, and the discriminator distinguishes between real and fake. It showed better performance in both subjective and objective evaluations than models like SEAGAN and pix2pix for this task.

• Disco-GAN

DiscoGAN [69], [81] extends the previously proposed MMSE-GAN framework to better handle the cross-domain relationships between whispered and normal speech. It utilises the same generator and discriminator architecture as MMSE-GAN but incorporates three distinct models and different loss functions during the training process. The first model maps the 40-channel MCC features of whispered speech to normal speech. The second model predicts the F0 for the mapped whispered speech. The third model, a 5-layer Deep Neural Network (DNN), makes Voice (V) and Un-Voice (UV) decisions based on the mapped whispered MCC features. The training process for Disco-GAN is shown in figure 2.1. In that, the X_S is normal speech, and X_W represents whispered speech. And the successor, G_{WS} , does the mapping between whisper and normal speech features, and G_{SW} does the reverse mapping of the first generator. The overall loss function for generator and discriminator are shown mathematically in equation 2.3

$$\begin{aligned}
 L_G &= -\mathbb{E}_{X_S \sim p_s} [\log D_W(G_{SW}(X_S))] \\
 &\quad + \frac{1}{2} \mathbb{E}_{X_w \sim p_w, X_s \sim p_s} [\log(X_W) - \log(G_{SW}(X_S))]^2, \\
 L_D &= -\mathbb{E}_{X_w \sim p_w} [\log D_W(X_W)] \\
 &\quad - \mathbb{E}_{X_s \sim p_s} [\log(1 - D_W(G_{SW}(X_S)))] .
 \end{aligned} \tag{2.3}$$

The adversarial loss helps the generator map the distribution to the target domain, and the regularisation that MSE uses helps in overfitting. Reconstruction loss contributes to learning complex relations between the input whisper data distribution and its corresponding normal speech target domain. The reconstruction loss can be expressed as equation 2.4 and 2.5 for whisper and normal loss.

$$\min_{\theta} L_W = \mathbb{E} \left[(X_{WSW} - X_W)^2 \right] \quad (2.4)$$

$$\min_{\theta} L_S = \mathbb{E} \left[(X_{SWS} - X_S)^2 \right] \quad (2.5)$$

X_{WSW} and X_{SWS} generated data distribution in which the X_{WSW} distribution should be close to X_W . But it is hard to optimise because of MSE so that is the reason why the total loss by G_{WS} is represented as equation 2.6:

$$\mathcal{L}_{G_{WS}} = \mathcal{L}_W + \mathcal{L}_S \quad (2.6)$$

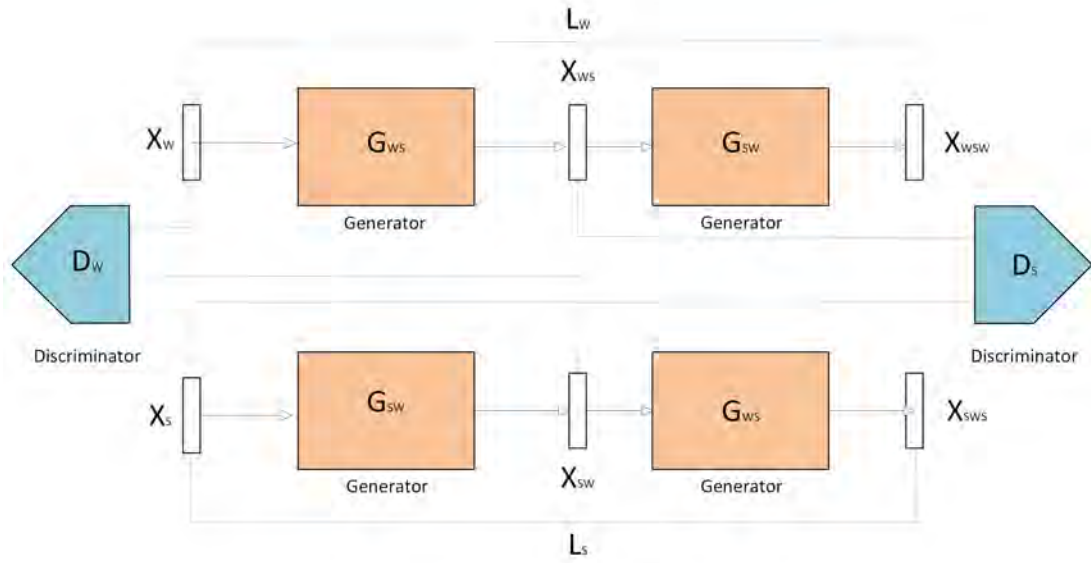


FIGURE 2.1: Disco GAN Training Process

In terms of objective and subjective evaluation, it performed better than MMSE-GAN.

• Cycle-GAN

Cycle-GAN [81] was extending the studies of MMSE-GAN and Disco-GAN. It has similar models and architecture as Disco-GAN, but it has a different objective function. The training of objective function is shown in figure 2.2

This architecture consists of two generators, which are G_{WS} and G_{SW} , which are the mapping between whisper-to-normal speech and normal-to-whisper speech. Also two discriminators which are D_S and D_W .

In cycle-GAN architecture, two types of loss functions are applied, which are adversarial and cycle-consistent loss. In adversarial loss, a least-squares loss is used for a stable training process. The objective of this function is to calculate the loss for mapping between whisper to speech, which is G_{WS} and corresponding discriminator D_S . It can mathematically expressed as equation 2.7.

$$L_{G_{WS}} = \mathbb{E}_{X_s \sim p_s} \left[(\log(D_S(X_s)) - 1)^2 \right] + \mathbb{E}_{X_w \sim p_w} \left[(\log(D_S(X_w)) - 1)^2 \right] \quad (2.7)$$

Cycle-consist loss is similar to the autoencoder objective, where it is trying to reconstruct the input from output, and it is mathematically expressed as equation 2.8. Similarly, we can calculate the G_{SW} same as G_{WS} . And the total loss can be described as equation 2.9.

$$L_{cyc} = \mathbb{E}_{X_w \sim p_w} [\|G_{SW}(G_{WS}(X_w)) - X_w\|_1] + \mathbb{E}_{X_s \sim p_s} [\|G_{WS}(G_{SW}(X_s)) - X_s\|_1] \quad (2.8)$$

$$L_{total} = L_{G_{WS}} + L_{G_{SW}} + \lambda L_{cyc}. \quad (2.9)$$

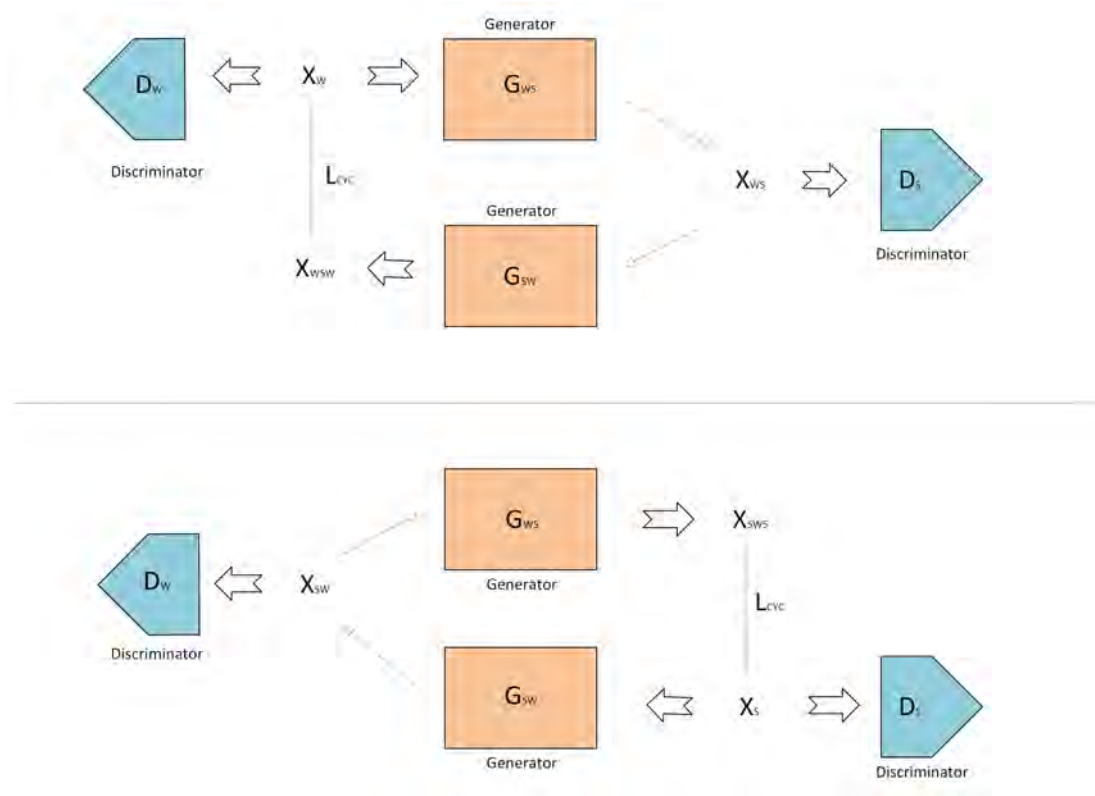


FIGURE 2.2: Cycle GAN Training Process

• CNN-GAN

CNN-GAN [82], [83] utilizes deep convolutional neural networks (DCNNs) and features three models: one for MCC mapping, another for F0 prediction, and a third model for V and UV decision like previous Cycle-GAN, DiscoGAN, and MMSE-GAN. Feature extraction in this model is done using AHOCODER [76]–[80] and for loss function used is a simple adversarial loss as expressed in equation 2.2 of the MMSE-GAN but it only uses a first term which is adversarial loss. It uses the wTIMIT dataset with 1187 parallel utterances for training and testing.

The generator of the MCC mapping model comprises lower and final layers. The initial layers consist of a CNN layer including 128 filters of dimensions 3x3, with a padding of 1. This is then followed by the ReLU activation function. The number of filters is initially increased to 256 and subsequently decreased back to 128, with each step being followed by ReLU activation. The last layers

consist of a solitary CNN layer with 128 filters, which retains its dimensionality as a result of its stride and padding configurations, so preparing the output for the discriminator.

The discriminator in the MCC model consists of CNN layers and pooling layers, which are then followed by fully connected layers. The discriminator starts with a CNN layer consisting of 32 filters with a size of 7x7, a stride of 2, and a padding of 3. This layer is followed by ReLU activations and further CNN layers that gradually increase the size of the filters up to 256. Also, the max-pooling layers are added in between of CNNs.

The authors also note that the training of this model is characterised by instability, since the loss consistently increases after each epoch. Furthermore, the assessment results were inferior to those achieved by MMSE-GAN, DiscoGAN, and Cycle-GAN.

• Inception-GAN

The Inception-GAN [83] architecture builds upon the CNN layers. It consists of three models: one for MCC mapping, another for F0 prediction, and last one for VUV decision, using a similar adversarial loss function as CNN GAN and same training and testing set.

The Inception-GAN model utilises inception and inverse inception modules to analyse MCC features to feed it to an MCC mapping model. The inception module consists of four parallel pathways: a 1x1 CNN that focuses on capturing tiny details, a 1x1 CNN followed by a 3x3 CNN that captures intermediate features, a 1x1 CNN followed by two 3x3 CNNs that capture complex features, and a 3x3 max pooling operation followed by a 1x1 CNN. The outputs are combined.

The inverse inception module copies this arrangement but uses transpose CNNs for the purpose of upsampling. The generator sequentially arranges these

modules to extract features vertically. The discriminator commences with a sizable 7x7 CNN layer. It is then succeeded by inception modules and pooling layers to enhance and improve the extracted features. The model ends with fully linked layers and a dropout layer to mitigate overfitting, utilising a sigmoid activation function for a probability distribution.

In F0 prediction, the generator copies the MCC model and incorporates max pooling layers following each inception module to capture sequential characteristics and improve overfitting. The discriminator uses three DNN layers with ReLU activation and a ending with sigmoid activation for classification.

The VUV decision model has a five-layer DNN module for training purposes, utilising normal speech data. The AHOCODER is used in the same manner as earlier GANs to reconstruct the speech.

The Inception GAN experienced some training challenges, although they were not that severe as the CNN-GAN. Furthermore, it shows superior performance in the assessment, however it was found to be inferior to MMSE, Disco, and Cycle GANs.

• MSpeC-Net

MSpeC-Net is a multi-domain speech conversion architecture that converts NAM speech into both whispered and normal speech [84]. This approach decreases the number of models needed for different speech conversion tasks, enhancing the training efficiency and effectiveness. The CSTR NAM TIMIT Plus corpus [73]–[75] is the dataset utilised for both training and evaluation. It comprises NAM speech and related whispered speech from newspaper texts from the Herald newspaper. In addition, the authors recorded normal speech data where they articulated the same utterance as in the NAM and whisper speech. The data was resampled from a frequency of 96 kHz to 16 kHz, and MCCs were extracted using AHOCODER [76]–[80]. The dataset was split into

82 percent of the data for training , while the remaining portion was used for validation and testing.

The training procedure of MSpeC-Net consists of three domain-specific autoencoders (AE) and a discriminator. The AE captures latent representations of NAM, whispered, and normal speech. The encoder and decoder pair have three DNN layers, each containing 512 units. ReLU activation is applied to each layer, while the output layer uses sigmoid activation. The discriminator uses DNNs with three layers to classify both NAM and whisper speech in order to convert NAM to whisper, as well as to discriminate between whisper and normal speech. The training process converts NAM to whisper speech and whispered speech to normal speech. In this model, the encoder learns features and makes a compressed latent representation, which the decoder uses to generate reconstructed data. The training procedure integrates reconstruction, adversarial, and latent space consistency loss to synchronise the latent spaces across various speech modalities. The MSpeC-Net model was trained on 100 training epochs, with a learning rate of 0.0001, and the model was using the Adam optimiser.

MSpeC-Net was compared to the MMSE-DiscoGAN and showed better performance in NAM-whisper and whisper-normal conversion tasks for both subjective and objective metrics.

- **MelGAN**

MelGAN, originally introduced as a vocoder for speech processing, is a generative model for manipulating and synthesis audio waveforms [85]. This study, modified it to reconstruct whispered speech to normal speech [86].

In MelGAN, the generator takes a mel-spectrogram as input and gradually increases its resolution by passing it through many layers of transposed CNNs.

Each of these layers enhances the depth of feature extraction in the input until it aligns with the intended target's features. This procedure encompasses many layers, which include residual blocks containing dilated CNN). The utilisation of dilated CNNs improves the model's ability to extract features from MCCs, hence enhancing the complexity of feature extraction. The generator uses Leaky-ReLU activation functions for all layers except the final output layer, which utilises a tanh activation function.

The discriminator in MelGAN differs in that it functions at several scales. The audio is processed at three distinct resolutions, enabling a more comprehensive evaluation of the resulting audio's quality. The discriminator block of each scale consists of a sequence of CNNs that analyse the audio's characteristics.

Overall, MelGAN is designed to process whispered input and learn the corresponding target speech spectral properties with complex architecture.

- **VQ-VAE**

Vector-quantised Variational Autoencoders (VQ-VAEs) [86], [87] are models that use variational AEs' strengths with discrete latent representations. Unlike traditional VAEs, which use continuous latent variables, VQ-VAEs use a discrete set of codes, much like a dictionary data structure, to represent the data. In the model, the encoder processes the input data, produces a latent representation, and maps it to the nearest code from a predefined codebook. This means the encoder's output is replaced by the closest code in this dictionary, effectively turning a continuous input into a discrete representation. The decoder then takes this quantised representation and reconstructs the original data from it.

Training a VQ-VAE involves three objective functions. First is the reconstruction loss, which measures how well the decoder's output matches the original input, ensuring the model can accurately reconstruct the input data. Second is the embedding loss, which updates the codebook vectors to better align with the encoder's outputs, ensuring that the dictionary of embeddings learns the data representation. Third is the commitment loss, which encourages the encoder to use the embeddings effectively by penalising large differences between the encoder's outputs and the chosen embeddings.

The VQ-VAE model outputs the reconstructed spectrogram and that can be converted back into an audio using a vocoder WaveGlow [88], or directly through MelGAN.

- **WESPER**

The WESPER [89] model significantly advances in real-time whisper-to-normal speech conversion and shows state-of-the-art results. The core of WESPER's architecture includes a Speech-to-Unit (STU) encoder and a Unit-to-Speech (UTS) decoder.

The STU encoder uses self-supervised pre-training and generates common speech units for both whispered and normal speech. This pre-training phase involves masking parts of the input speech and estimating the corresponding features from the remaining parts, using HuBERT (Hidden-Unit BERT) [90] for learning acoustic features.

For training, WESPER utilises non-parallel and unlabeled normal and whispered speech datasets. The STU encoder learns to produce a consistent latent representation for both types of speech, effectively reducing the differences between whispered and normal utterances.

The UTS decoder, a modified version of FastSpeech2 [91], uses speech units that are generated using STU encoder and converts them into mel-spectrograms

by predicting, which are then converted into audible speech using a vocoder Hifi-GAN [92], which is known for its high-quality speech synthesis capabilities.

The training of the UTS decoder involves using the speech data from specific speakers without needing text labels. This approach allows the model to perform real-time and speaker-independent conversion. However, WESPER has a predictive element because of FastSpeech2 in the decoder, which relies on speech units to predict the words from it instead of phonetically reconstructing speech.

After exploring models in whisper or NAM to speech domain in the following section, we have explored a voice conversion model that translates the speech from source to the target domain.

2.2.2.3 Voice Conversion Based Methods

In this subsection, we will discuss the latest voice conversion that has been used mainly for translating speech from one speaker to another and for translating in different languages.

- **FreeVC**

FreeVC [93] is a voice conversion architecture specifically developed to generate superior-quality speech conversion without needing text labels and data. The VITS framework [94], known in the literature for generating exceptional waveform reconstructions, serves as the foundation for FreeVC. FreeVC proposes novel techniques for improving content information extraction, eliminating the need for text annotations. Removing predictive elements makes it adaptable and effective for many voice conversion applications.

The FreeVC architecture has five components: prior encoder, posterior encoder, decoder, discriminator, and speaker encoder. In FreeVC, the VITS architecture is utilised, with the posterior encoder, decoder, and discriminator adhering to this framework. However, the prior and speaker encoders are distinct and specific to FreeVC, its novel elements compared to the VITS framework. The prior encoder used the WavLM model [95] to extract 1024-dimensional features from the raw audio waveform in a self-supervised way. The SSL features go through a bottleneck extractor to preserve the relevant content information, lowering them to a low-level representation, similar to AE. This creates a bottleneck network in the information flow, forcing the model to discard unrelated speaker information. Next, the reduced representation expands the latent representation and generates output.

FreeVC uses two speaker encoders: a pre-trained speaker encoder and a non-pretrained speaker encoder. The pre-trained encoder, trained on large datasets containing several speakers, is used in various speech conversion and Automated Speech Recognition tasks. The encoder that is not pretrained is a model based on LSTM layers. It is trained together with the rest of the FreeVC components. It aims to learn the absent speaker information and style using the retrieved feature representation. The other thing that FreeVC uses is spectrogram-resize-based data augmentation. This procedure creates mel-spectrograms from the audio waveform, vertically scaling them using bilinear interpolation, slicing it, and reconstructing the waveform from the augmented mel-spectrogram using HifiGAN [92]. This augmentation changes the speaker information while keeping the content unchanged, enabling the model to concentrate on the content characteristics.

- **Diff-HierVC**

Diff-HierVC is an architecture for voice conversion that proposes to address the main problems of imprecise pitch prediction and limited speaker adaption quality in voice conversion tasks. This system uses a hierarchical methodology that incorporates two primary diffusion models: DiffPitch and DiffVoice [96]. The process begins with speech disentanglement, in which the input speech is pre-processed to eliminate unnecessary information. The XLS-R model, which is pre-trained, is used to extract content features from input data. Meanwhile, the style encoder learns style information from the input mel-spectrogram. The F0 is extracted using the YAAPT algorithm [97] for input speech.

The voice conversion process starts with DiffPitch, a technique that utilises a WaveNet-based [98] conditional diffusion model to adjust the F0 in order to align it with the target speech style. This model converts the normalised F0 of the input voice into a pitch representation utilised as a prior. DiffVoice uses features, target F0, and target voice style to generate speech. The system uses a source-filter encoder to partition speech elements into pitch and content, using the reconstructed Mel-spectrogram to improve voice style transmission.

In order to enhance generalisation, Diff-HierVC incorporates a masked prior. This ensures applying a mask to the data before inputting it into DiffVoice, which enables the model to simultaneously learn the procedures of reconstruction and denoising.

The Diff-HierVC uses LibriTTS and VCTK datasets for training. The VC between two different speakers and also translating in different language tasks Diff-HierVC outperformed FreeVC.

- **Hier-speechppVC**

HierSpeechpp [99] is a zero-shot speech synthesis model that utilises a hierarchical framework to improve semantic and acoustic representations of generated speech. This model uses a hierarchical speech synthesiser, a text-to-vec (TTV) system, and a speech super-resolution (SpeechSR) framework to achieve high-quality speech synthesis.

HierSpeech++ uses a hierarchical conditional VAE at its core. This model uses self-supervised speech representation to extract semantic and acoustic information from the input speech.

The hierarchical speech synthesiser includes a dual-audio acoustic encoder that combines representations from both waveform and linear spectrograms. This dual approach improves the information on the spectral features, leading to better feature extraction. Additionally, a source-filter theory-based multi-path semantic encoder is employed to disentangle speech components, further improving the accuracy of the generated speech.

For TTS tasks, HierSpeech++ proposes a TTV model that generates spectral features and F0 from text sequences. This model uses the same self-supervised speech representation to capture linguistic information. The style feature is also handled separately.

To enhance the audio quality, HierSpeech++ uses a speech super-resolution framework that upscales the audio from 16 kHz to 48 kHz. This upsampling improves the information contained in input speech signals.

It demonstrated better performance in literature than Diff-HierVC and FreeVC for all speech conversion tasks.

After discussing different models, we have summarised the information from sub-sections 2.2.2.2 and 2.2.2.3 in table 2.1.

TABLE 2.1: Summary of Deep Learning Algorithms

Model	Dataset	Evaluation Metrics	Vocoder
RBM	TIMIT	MCD	STRAIGHT
Deep Partial Neural Networks	TIMIT	IS distance, LLR, SSNR	STRAIGHT
DNN-MCC-F0	TIMIT	CD , STOI	STRAIGHT
SEQ2SEQ	CSTR-NAM-TIMIT Plus	CD , PESQ, STOI	STRAIGHT
Modified Transformers	wTIMIT, SpeechOcean	WER	WORLD
Meta-BLSTM	TIMIT	CD , PESQ, STOI	STRAIGHT
MMSE-GAN	CSTR-NAM-TIMIT Plus	MCD , RMSE	AHOCODER
Disco-GAN	wTIMIT	MCD , RMSE	AHOCODER
Cycle-GAN	wTIMIT	MCD , RMSE	AHOCODER
CNN-GAN	wTIMIT	MCD , RMSE	AHOCODER
Inception-GAN	wTIMIT	MCD , RMSE	AHOCODER
Mspec-Net	CSTR-NAM-TIMIT Plus, wTIMIT	MCD, RMSE	AHOCODER
MelGAN	wTIMIT	MCD , RMSE	wav2vec
VQ-VAE	wTIMIT	MCD , RMSE	WaveGlow
WESPER	wTIMIT, LibriSpeech	WER, BLEU	Hifi-GAN
FreeVC	VCTK, LibriSpeech	WER, CER, F0-PCC	Hifi-GAN
Diff-HierVC	VCTK, LibriSpeech	CER, EER	Hifi-GAN
Hier-SpeechppVC	wTIMIT, LibriSpeech, EXPRESSO	PESQ, WER	Hifi-GAN, BigVGAN

2.3 Research Gap

In our analysis of the existing literature in previous sections, we identified significant advancements in the field of speech reconstruction using state-of-the-art deep learning algorithms. These algorithms have effectively tackled many of the limitations associated with traditional methods. However, a notable gap remains in their application, as these advancements have primarily concentrated on whispered speech synthesis and have yet to extend their benefits to individuals with laryngectomy.

Some studies, such as those involving deep partial neural networks and RBM,

have hinted at their potential utility for laryngectomy patients. However, practical implementation has been hindered by the lack of appropriate datasets representing laryngectomised speech. They hypothesised that these algorithms could be applied to laryngectomised speech. Most of the existing algorithms have been trained on whispered speech paired with parallel normal speech, not with laryngectomised speech. However, some recent studies suggested that laryngectomy speech has different acoustics characteristics compared to whispered speech [100], [101]. This means these methods might not be efficient on laryngectomised speech, hence, we will investigate them in this thesis.

The second critical gap involves the scarcity of specialised datasets for laryngectomy patients. The research we've reviewed mainly relies on publicly available datasets like wTIMIT and CSTR-NAN-TIMIT, which lack the characteristics required for laryngectomised speech research. However, unlike other studies, we have access to laryngectomised dataset that Unitec researchers have collected in collaboration with North Shore Hospital, Auckland, and Auckland Voice and Swallow Clinic. This dataset includes a diverse range of 22 patients of varying ages and genders, all communicating in English.

To address these gaps, our thesis intends to pioneer the application of state-of-the-art deep learning algorithms to laryngectomised speech datasets. Our unique contribution will involve refining existing algorithms and tailoring them to better suit the distinct requirements of laryngectomy patients. Furthermore, we plan to conduct the analysis of models on laryngectomy patient's data and objective evaluation of this novel architecture using metrics discussed in Chapter 3. By doing so, we aim to bridge the current gaps in research and pioneer the reserach for improving speech synthesis for individual with laryngectomy.

Chapter 3

Analysis and Evaluation of Existing Models

Chapter 3 is divided into three sections. In the first section, we will discuss the publicly available datasets used for training various models and the laryngectomy patient dataset used for testing and evaluating the model results. The laryngectomy dataset has been collected by researchers at Unitec with the assistance of the Auckland Voice and Swallow Clinic.

The second section will introduce the objective measures used to evaluate and analyse the models. These measures are crucial for providing a standardised framework to assess the models' performance.

The third section will analyse, evaluate, and discuss the existing deep learning models. Following this analysis, we will propose our methodology for improving speech conversion for laryngectomy patients in Chapter 4.

3.1 Datasets

In this section, we will discuss the dataset we will be using for the training and testing of models.

3.1.1 wTIMIT

The wTIMIT corpus is frequently used to research automatic recognition of phonemes and for a whisper-to-normal speech conversion [52], [53]. The paired whispered and spoken utterances produced by various speakers are organized systematically and comprise the wTIMIT. The first part of the collection included utterances from 20 Singaporean speakers, while the second phase included utterances from 28 North American speakers. Two subsets that differ only in accent were produced as a result. Each speaker was given a set of 450 prompts to read aloud and whisper. We have used wTIMIT to train the existing models and our proposed models.

3.1.2 Laryngectomy Patients Dataset

The laryngectomy patients dataset has been previously collected by researchers at Unitec with the help of Auckland Voice and Swallow clinic; this dataset comprises 18 male speakers and 4 female speakers. A total of 22 participants with varying degrees of glottis-related impairment and two healthy reference speakers provided the data. Three patients had non-surgical larynx care. Thirteen underwent a total laryngectomy, while four underwent a partial laryngectomy or related surgery (such as a thoracotomy). The third group was under TEP surgery; eight speakers were in this category. The subjects' ages, recorded in an audiology room at North Shore Hospital in Auckland, New Zealand, ranged from 38 to 81. During testing, subjects were seated with a ZOOM H4n recorder around 30 cm from their mouths. Participants were asked to explain their state in recorded sessions before reciting the digits 0 to 9. They next read a series of 15 TIMIT [58] sentences, followed by 11 vowels in carrier words, /hxd/, where "x" is changed with English vowels. Each recitation list was repeated by subjects who could speak clearly and in a whisper, first in their normal voice and then in whispers. All the speakers have at least spoken all

the utterances twice. Mispronounced words or sentences were promptly repeated, and incorrect sentences were cut out before post-processing, leaving roughly 8 hours of recorded speech frames.

TABLE 3.1: Details of Participating Subjects [13]

ID	Bio	G	R	B	A	S	C	I	Notes
Total laryngectomy									
5	72 M	3	2	3	3	1	2	2	Very whispery
6	-M	3	2	3	3	3	3	3	Quite whispers
11	81 M	3	3	2	3	3	3	3	Oesophageal, 8yrs
14	60 M	3	2	3	3	2	3	3	Quite whispers
16	64 M	3	2	3	3	2	3	3	whispery voice
Total laryngectomy with TEP									
7	57 M	3	2	1	1	3	1	1	11yrs
12	76 M	3	3	2	2	2	2	1	
13	-M	3	3	1	2	3	2	2	
15	60s M	3	3	2	1	3	2	1	2yrs
17	79 M	2	3	1	1	2	1	1	
18	80 F	1	2	3	3	1	0	1	Clear whisper 20yrs
19	-M	3	3	2	2	3	2	1	9 and 5yrs
20	73 M	2	3	1	2	2	1	1	9yrs
Partial laryngectomy									
1	60s M	2	3	1	1	2	0	0	3yrs
2	40s F	2	2	0	1	2	0	0	
3	60s M	1	1	1	1	0	0	0	4 wks
4	52 F	2	2	1	1	2	0	0	8 wks, tracheotomy
8	40 M	1	1	1	0	2	0	0	2yrs, botox
9	78 M	2	2	1	2	2	0	1	"many years"
10	69 M	3	3	0	1	2	0	0	7yrs, radiotherapy
Reference									
21	38 M	0	0	0	0	0	0	0	clear voice
22	52 F	0	0	0	0	0	0	0	clear voice

An auditory-perceptual assessment technique called GRBAS is used to measure hoarseness in terms of vocal strain (S), overall grade (G) of dysphonia, breathiness (B), asthenicity (A), and raspiness (R). Together with the time elapsed since the last therapy (measured in months or years) and other information, if available, they also included a subjective evaluation of (C) cadence (speech rate) and intelligibility. The impairment scales for GRBAS, C (cadence), and I (intelligibility) are ranked 0 = normal, 1 = mild, 2 = moderate, and 3 = severe. A subjective GRBAS scale classification of voices is presented in Table

3.1. We have used this dataset presented in the study by McLoughlin et al. [13] to evaluate existing models in Section 3.3 and Chapter 4.

The following section will focus on the objective measures to evaluate the existing models.

3.2 Objective Measures

As stated in Table 2.1, every study uses different evaluation methods to evaluate architecture's performance. For the evaluation of algorithms, there are two main categories of subjective and objective evaluations. For the scope of this thesis, We will focus on the objective evaluation metrics for analysing existing algorithms in Chapter 3.3 and the proposed architecture that we will discuss in Chapter 4.

We will be focusing on six objective evaluation metrics from the literature: Mel Cepstral Distortion (MCD) [49], [102], Short-Time Objective Intelligibility (STOI) [103], Log Likelihood Ratio (LLR) [49], [104], [105], Segmental Signal-to-Noise Ratio (SSNR) [104], [106], Cepstral Distortion (CD) [105] and Perceptual Evaluation of Speech Quality (PESQ) [105], [107].

3.2.1 MCD

The MCD [49], [102] is a measure used in speech and audio signal processing to measure signal distortion. It evaluates how closely two sequence of MFCCs resemble each other. To obtain MFCCs, Mel-frequency scaling, and the discrete cosine transform are applied to the audio signal. Cepstral coefficients, extracted by performing a Fourier transform on the Fourier magnitude's logarithm, measure a signal's spectral information. MCD calculates the difference between a reference signal with a reconstructed signals spectral feature. A lower MCD value indicates a closer spectral similarities between signals.

Mathematically, MCD is defined in equation 3.1.

$$\text{MCD}[\text{dB}] = \frac{10}{\ln 10} \sqrt{2 \sum_{i=1}^n (m_i^t - m_i^c)^2} \quad (3.1)$$

where m_i^t and m_i^c represent the i -th MFCC of the reference and reconstructed signals, respectively, and n is the number of MFCC coefficients.

3.2.2 STOI

The STOI method [103] evaluates the intelligibility of speech signals by analysing them in the time-frequency domain. It segments speech into short, overlapping frames and applies the STFT [108] to capture spectral features. STOI models the human auditory system, considering speech psychoacoustics and critical hearing bands, and includes steps for separating speech from background noise or interference using masking techniques. Modulation filtering [109] is also used to focus on the slower modulations of speech, which is essential for understanding.

To calculate STOI, let $x(n)$ be the clean reference speech signal and $y(n)$ be the reconstructed speech signal. Segment these signals into overlapping frames using a Hamming window and compute the STFT to obtain $X(t, f)$ and $Y(t, f)$, where t is the time frame index and f is the frequency bin index. Obtain the short-time magnitude spectra $|X(t, f)|$ and $|Y(t, f)|$.

Normalise the magnitude spectrum of the reconstructed signal with respect to the reference signal, which is represented in equation 3.2.

$$\hat{Y}(t, f) = \frac{|Y(t, f)|}{|X(t, f)| + \epsilon} \quad (3.2)$$

where ϵ is a small constant to avoid division by zero. Clip the normalized magnitudes to a maximum value of 1 so the range of metric is between 0 and 1. It is calculated as equation 3.3.

$$\tilde{Y}(t, f) = \min(\hat{Y}(t, f), 1) \quad (3.3)$$

Following clipping, we need to calculate the intermediate intelligibility measure for each time frame t and it is illustrated in equation 3.4.

$$d(t) = \frac{1}{F} \sum_f \tilde{Y}(t, f) |X(t, f)| \quad (3.4)$$

where F is the total number of frequency bins. The final STOI score is obtained by averaging these intermediate intelligibility measures across all time frames.

$$\text{STOI} = \frac{1}{T} \sum_t d(t) \quad (3.5)$$

where T is the total number of time frames in equation 3.5. This score quantifies the intelligibility of the reconstructed speech compared to the reference, with higher scores indicating better intelligibility.

3.2.3 LLR

The LLR [49], [104], [105], is used to assess the resemblance between a reference and a reconstructed speech signal. It aids in assessing the probability of the reconstructed speech under two distinct hypotheses. In order to calculate the LLR, the likelihood function is initially created to assess the degree to which a hypothesis can explain the reconstructed speech.

The LLR is calculated by subtracting the log-likelihood calculated using the null hypothesis (H_0) from the log-likelihood calculated using the alternative

hypothesis (H_1). A positive LLR suggests that the reconstructed speech is more consistent with the (H_1), whereas a negative LLR provides evidence in favor of the (H_0). The difference of the LLR values shows the distance between two hypothesis.

In the context of speech processing, $x(n)$ is the clean reference speech signal and $y(n)$ is the reconstructed speech signal. The signals are segmented into overlapping frames using a Hamming window, and the LPC coefficients a_x for the reference signal and a_y for the reconstructed signal are computed. The covariance matrices R_x for the reference signal and R_y for the reconstructed signal are then calculated using the LPC coefficients. The LLR for each frame is computed using the formula in equation 3.6.

$$\text{LLR} = \log \left(\frac{a_y^T R_x a_y}{a_x^T R_x a_x} \right) \quad (3.6)$$

where a_x and a_y are the LPC coefficient vectors for the reference and reconstructed signals, respectively, and R_x is the covariance matrix of the reference signal.

3.2.4 SSNR

SSNR [104], [106] is used to assess the quality of speech signals. The speech signal is divided into short, overlapping segments that capture the signal's features over time. SSNR compares the target speech with the original reference speech to evaluate the denoising effect of models [110].

To calculate SSNR, the energy of the reference speech signal is compared to the energy of the noise or distortion introduced during processing. For each segment, the SSNR is computed as the ratio of the energy of the reference speech to the energy of the difference between the reference and target (distorted)

speech. This ratio is then converted in decibels (dB). Higher SSNR values indicate that the processed speech resembles the clean reference signal, suggesting limited noise or distortion introduced during processing. Lower SSNR values indicate more deterioration or noise in speech signals.

For this thesis, we will use distorted speech as the reference speech and reconstructed speech as a target to evaluate the denoising effect of our models on laryngectomy patients' data. SSNR value around 0 dB indicates that the processed speech is similar to the distorted speech. If the SSNR value is significantly negative, it suggests that the processing has introduced additional distortion. Conversely, if the SSNR value is significantly positive, the processing has overly cleaned the signal, potentially removing important speech features.

Mathematically, SSNR for a single segment is defined in equation 3.7

$$\text{SSNR} = 10 \log_{10} \left(\frac{\sum_{n=1}^N x^2(n)}{\sum_{n=1}^N (x(n) - y(n))^2} \right) \quad (3.7)$$

where $x(n)$ is the reference (distorted) speech signal, $y(n)$ is the target (reconstructed) speech signal, and N is the number of samples in the segment.

3.2.5 CD

CD [105] measures the difference between the cepstral coefficients of two speech signals. Cepstral coefficients are derived from the Fourier transform of the logarithm of the power spectrum of the speech signal. The CD is computed by taking the RMSE between the cepstral coefficients of the reference and reconstructed speech signals. A lower CD value indicates a smaller difference between the reference and the processed speech, indicating better performance.

Mathematically, the CD is defined in equation 3.8.

$$CD = \sqrt{\frac{1}{n} \sum_{i=1}^n (c_i^t - c_i^c)^2} \quad (3.8)$$

where c_i^t and c_i^c represent the i -th cepstral coefficient of the reference and reconstructed signals, respectively, and n is the number of cepstral coefficients.

3.2.6 PESQ

PESQ [105], [107], [111] is a widely used objective metric in speech and signal processing for evaluating the quality of voice signals. PESQ compares the reference and generated speech signals using a perceptual model of human hearing. The reference and generated signals are first aligned in time, and their perceptual representations are then compared using a model that copies the auditory system's response.

PESQ [105], [107], [111], is a commonly used objective measure in speech and signal processing. It is utilised to evaluate the quality of voice signals. PESQ evaluates the reference and reconstructed speech signals by using a perceptual model based on human auditory perception. First, the reference and reconstructed signals are time-aligned, and then their perceptual representations are compared using a model replicating the auditory system's response.

The PESQ method calculates a numerical score ranging from -0.5 to 4.5, where higher values correspond to better speech quality.

The PESQ score is computed using several processes, which involve time-alignment, filtering, and comparison of perceptual representations. Because of that, the exact mathematical formulation is quite complex and hard to interpret.

These objective evaluation metrics will give us a baseline to compare our proposed architecture performance with existing algorithms on laryngectomised

dataset. This will also cover the first and second research questions we mentioned in section 1.3.

3.3 Analysis and Discussion of Existing Models

In this section, we have divided the analysis into three parts to comprehensively cover the different categories of models.

In the first subsection, we analyse five GAN models: MMSE-GAN, DiscoGAN, Cycle-GAN, CNN-GAN, and Inception-GAN. These models are part of a series of studies where each model builds upon the findings of its predecessor.

In the second subsection, we discuss three models in the WSC domain: MelGAN, VQ-VAE, and WESPER. These models are recognised for their advanced architecture and superior performance.

Finally, we evaluate voice conversion models in the third subsection: FreeVC, DiffHierVC, and HierSpeechppVC. These models are specifically designed for translating speech from one language to another, making them highly relevant in multilingual contexts. Our analysis focuses on their methodologies and effectiveness in converting laryngectomised speech to normal speech, evaluating their feasibility in distorted to normal speech reconstruction.

In the testing of this models, we have utilised data from laryngectomy patients, categorised into three groups based on their surgical outcomes. The first group consisted of two patients with total laryngectomy (TL), the second included two patients with total laryngectomy who had also undergone TEP they are under TL-TEP, and the third group comprised patients with partial laryngectomy (PL) . Each patient's data included two selected utterances and one "hvd words" wav file, which contains a sequence of 12 words, each emphasising a different vowel sound, such as HeeD, HaD, HooD, HaRD, HeaRD, HoD,

HiD, HoaRD, HuDD, HeaD, WHo'D, HeYD, HiDe, HoYD. So, there are total of 18 utterances for testing and the sentences that patients have articulated are shown in table 3.2. Also, we have normal parallel data for PL patients and it is still distorted but for total laryngectomy and with TEP we do not have normal data because of that we have taken reference data, which is spoken by normal speech speaker and has same utterances.

TABLE 3.2: Testing Data

ID	File Name	Utterance
Total Laryngectomy		
11	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?
14	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?
Total Laryngectomy with TEP		
7	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?
12	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?
Partial Laryngectomy		
1	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?
2	Sentence01	She had your dark suit in greasy wash water all year.
	Sentence02	Should giraffes be kept in small zoos?

3.3.1 GAN Models

For evaluating the GAN models, we replicated the experiment using consistent data pre-processing and models, training them on the wTIMIT dataset, and for feature extraction training for proposed models in Chapter 4, we will use the same experiment conditions to compare them from the literature. The evaluation used the testing data described in Table 3.2. We used the evaluation metrics discussed in Section 3.2.

In our first analysis, we focused on the spectral distortion of signals, as shown in plots (D) and (E) in Figure 3.2. From these plots, we observed that the MMSE, Disco, Inception, and Cycle GANs performed better for PL patients

compared to the TL and TL-TEP patients categories. The CNN GAN performed poorly, with an MCD value exceeding 30 for TL patients. Similarly, the CD metric shows the same performance trend. It's important to highlight that DNN-based models, such as MMSE, Disco, and Cycle GANs, generally outperformed CNN-based models like CNN and Inception GAN.

Next, we evaluated the speech quality of the reconstructed speech, illustrated in plots (A), (B), and (C). Plot (C) shows that the reconstructed speech for PL patients is closer to normal speech across all models. This is related to spectral distortion, where lower distortion means the speech is closer to normal speech in terms of LLR. However, it's important to note that we only have reference data for TL-TEP and TL patients, not the data from same speaker.

When evaluating plots (A) and (B), we found that CNN-based models performed better in terms of intelligibility. Plot (A) shows that the maximum STOI value for DNN-based models is around 0.3, while it goes up to 0.6 for CNN-based models. Similarly, in plot (B), the PESQ values for CNN-based models are higher than those for MMSE, Disco, and Cycle GANs across three patient categories.

However, when we examined the denoising aspect of the reconstructed signal, we found a value of -10, indicating that the signal contains only noise. Therefore, the higher STOI and PESQ values for CNN-based models are due to the increased noise. Samples are provided in Figure 3.1.

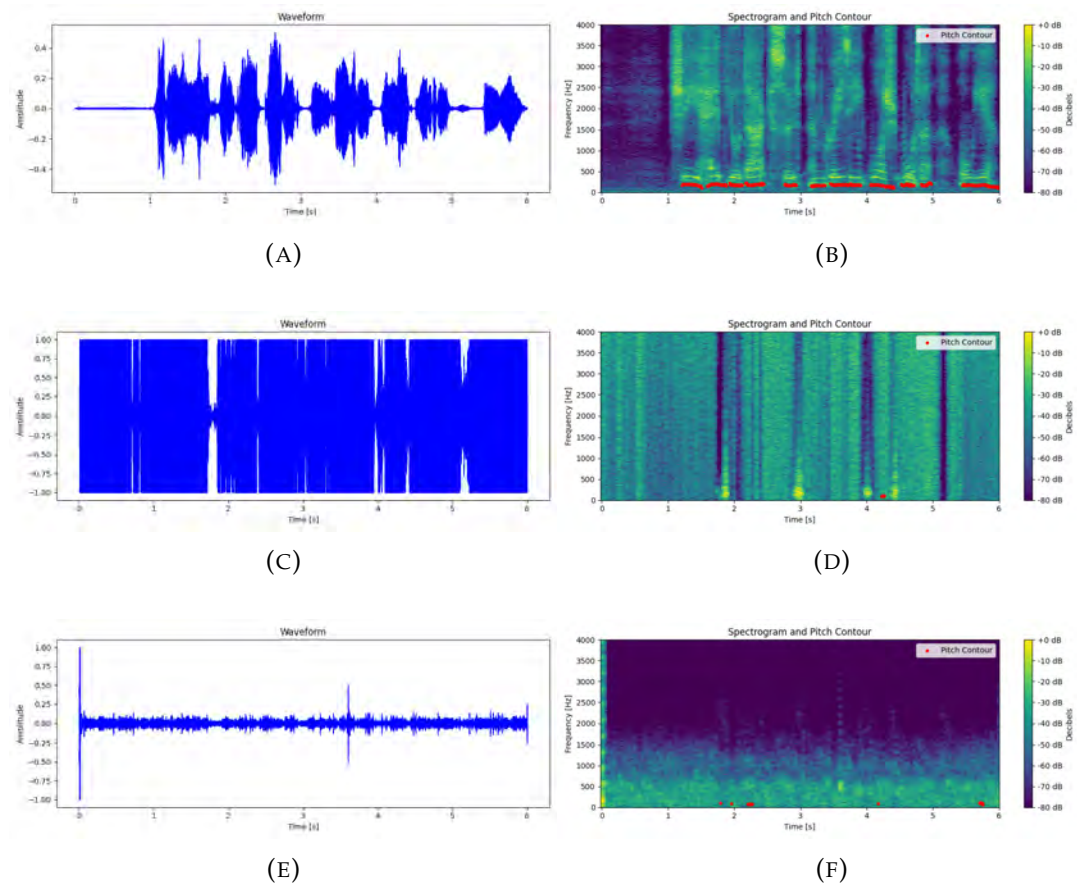


FIGURE 3.1: Waveforms and spectrograms with pitch contours for Normal and Reconstructed speech for PL patient ID 2 articulating "She had your dark suit in greasy wash water all year". (A) Waveform and (B) Spectrogram with pitch contour for Normal speech; (C), (D), (E) and (F) shows the waveforms and spectrograms reconstructed using CNN GAN and Inception GAN, also shows the distortion in reconstructed speech

Figure 3.1 illustrates that the signals generated by both the CNN and Inception GANs are significantly distorted. During the training of these models, we observed unstable behavior and mode collapse for both MCC and F0 models. Despite these challenges, this analysis highlights the potential of these models to phonetically reconstruct speech, addressing our fourth research question in Section 1.3.

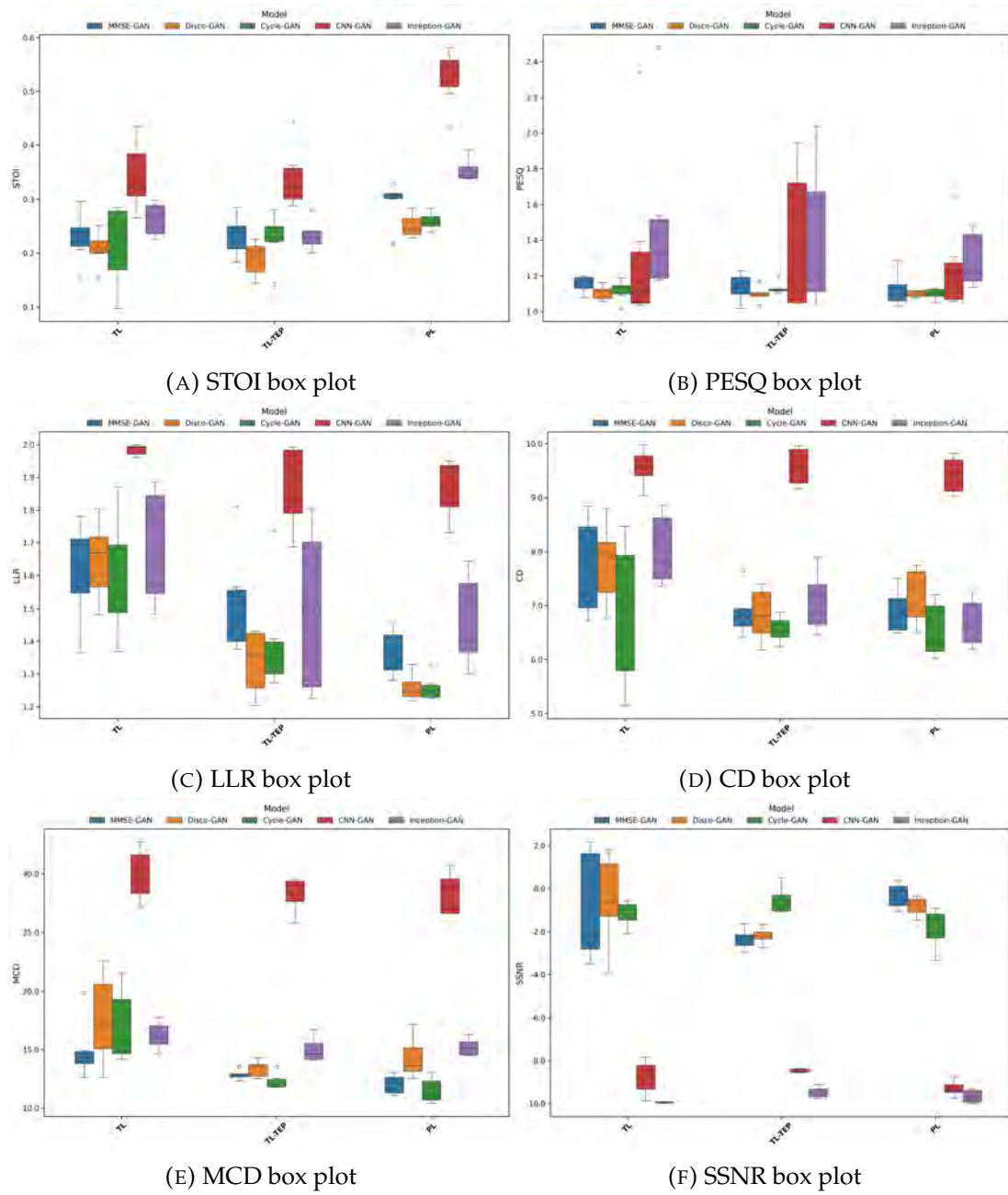


FIGURE 3.2: Objective metrics for Reconstructed data using MMSE-GAN, Disco-GAN, Cycle-GAN, CNN-GAN, and Inception-GAN, depicts the better performance by MMSE, Disco and Cycle GANs and shows the distortion and noise in CNN and Inception GANs

3.3.2 Whisper-Normal Speech Conversion Models

This subsection discusses MelGAN, VQ-VAE, and WESPER for converting whisper-normal speech. Unlike the GANs discussed in Section 3.3.1, we used pre-trained checkpoints provided by the authors of these models. However, we developed our own data pre-processing and inference scripts for these models.

Our analysis of MelGAN, VQ-VAE, and WESPER models focused on various objective metrics discussed in section 3.2, including spectral distortion, speech quality, and intelligibility. As illustrated in Figures 3.5, VQ-VAE consistently shows superior performance across all patient categories—TL, TL-TEP, and PL. It achieved lower values in CD, LLR, and MCD, indicating more accurate speech reconstruction compared to MelGAN and WESPER. WESPER, in particular, shows high variability and poorer performance, especially for PL patients, with CD values exceeding 10. MelGAN showed moderate performance, better than WESPER but less effective than VQ-VAE. The overall lowest MCD value is 5, which is achieved by vq-vae, and it is the best we have seen so far comparing GANs in section 3.3.1.

We then evaluated the speech quality of the reconstructed speech. VQ-VAE again performed better regarding PESQ values. MelGAN followed closely, while WESPER had significantly lower PESQ scores. Also, the STOI value shows quite variable performance for vq-vae, where for some TL patients, the values are 0.45, which is the highest, and for the same category of patients, the lowest value is 0.02, which is unintelligible.

After conducting an objective evaluation, we proceeded to analyse the reconstructed data through listening tests because we observed a large variance in speech quality metric values. We listened to all the reconstructed speech data from the three models—MelGAN, VQ-VAE, and WESPER—and found some

surprising results.

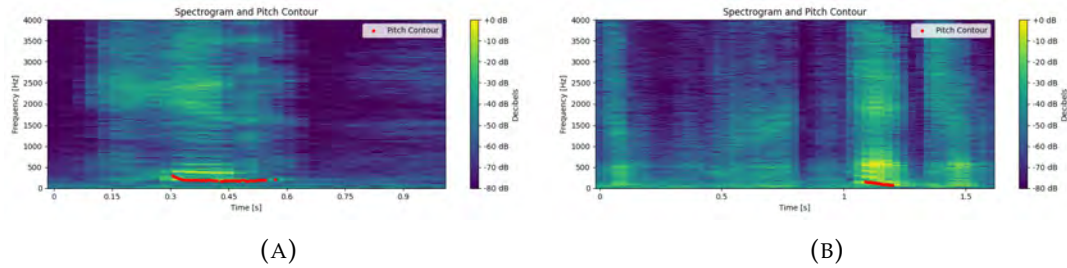


FIGURE 3.3: Spectrograms with pitch contour for (A) normal and (B) reconstructed HvD word for PL patient ID 1 articulating HYeD using WESPER shows the totally different linguistic compared to normal speech

Firstly, we examined WESPER and identified that the reconstructed speech changed the linguistics of the distorted speech used as testing data. This indicates that WESPER is not phonetically reconstructing speech accurately. Instead, it tries to predict words based on spectral features, converting these to speech units and reconstructing them using Fast2Speech. This approach leads to inaccuracies, as demonstrated in Figure 3.3. This guesswork is particularly problematic in the TL patient category which does not have much power in speech signal combined with noise.

Secondly, we listened to Melgan, and it was not inserting pitch, and the signal was noisy for TL patients and passed as it is. We have illustrated one example in figure 3.4 plot (E) and (F).

Lastly, we evaluated the reconstructed speech from VQ-VAE and found it to be notably noisy, with pitch levels exceeding 500 Hz, resulting in a shrill and unpleasant sound. The use of AEs in the VQ-VAE model contributed to this issue, as the model attempted to learn the data distribution, resulting in additional noise to each frame of speech. Consequently, the speech sounded artificial and monotonous. An example of this issue is illustrated in Figure 3.4 plot (G) and (H).

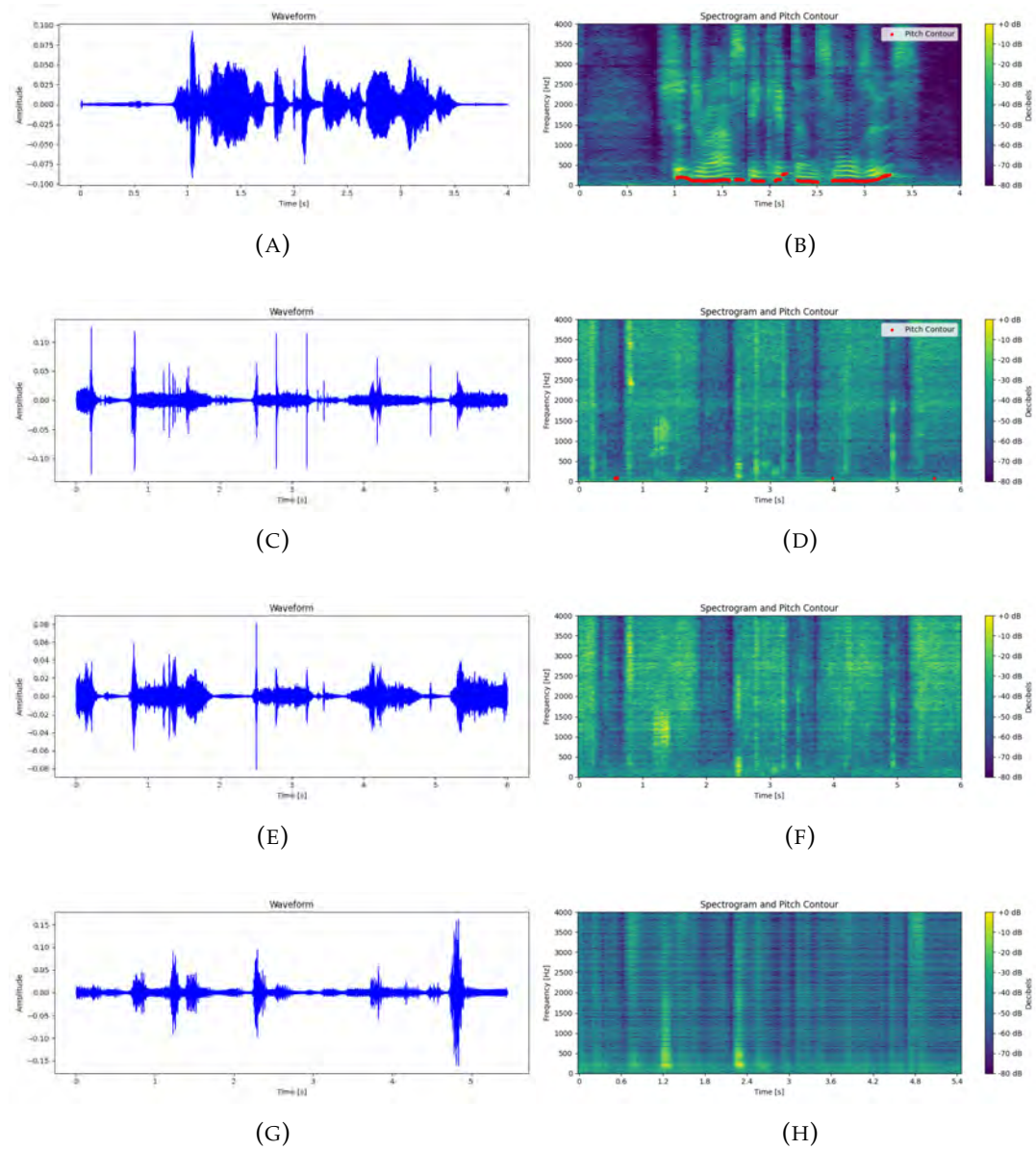


FIGURE 3.4: Waveforms and spectrograms with pitch contours for Normal and Reconstructed speech for TL patient ID 11 articulating "She had your dark suit in greasy wash water all year". (A), (B), (C), and (D) shows Waveforms and spectrograms with pitch contour for Reference and Distorted speech, respectively. (E) and (F) shows the waveform and spectrogram using MelGAN and shows the added noise in reconstructed speech without adding pitch. (G) and (F) shows reconstructed signal using VQ-VAE and shows noisy signal

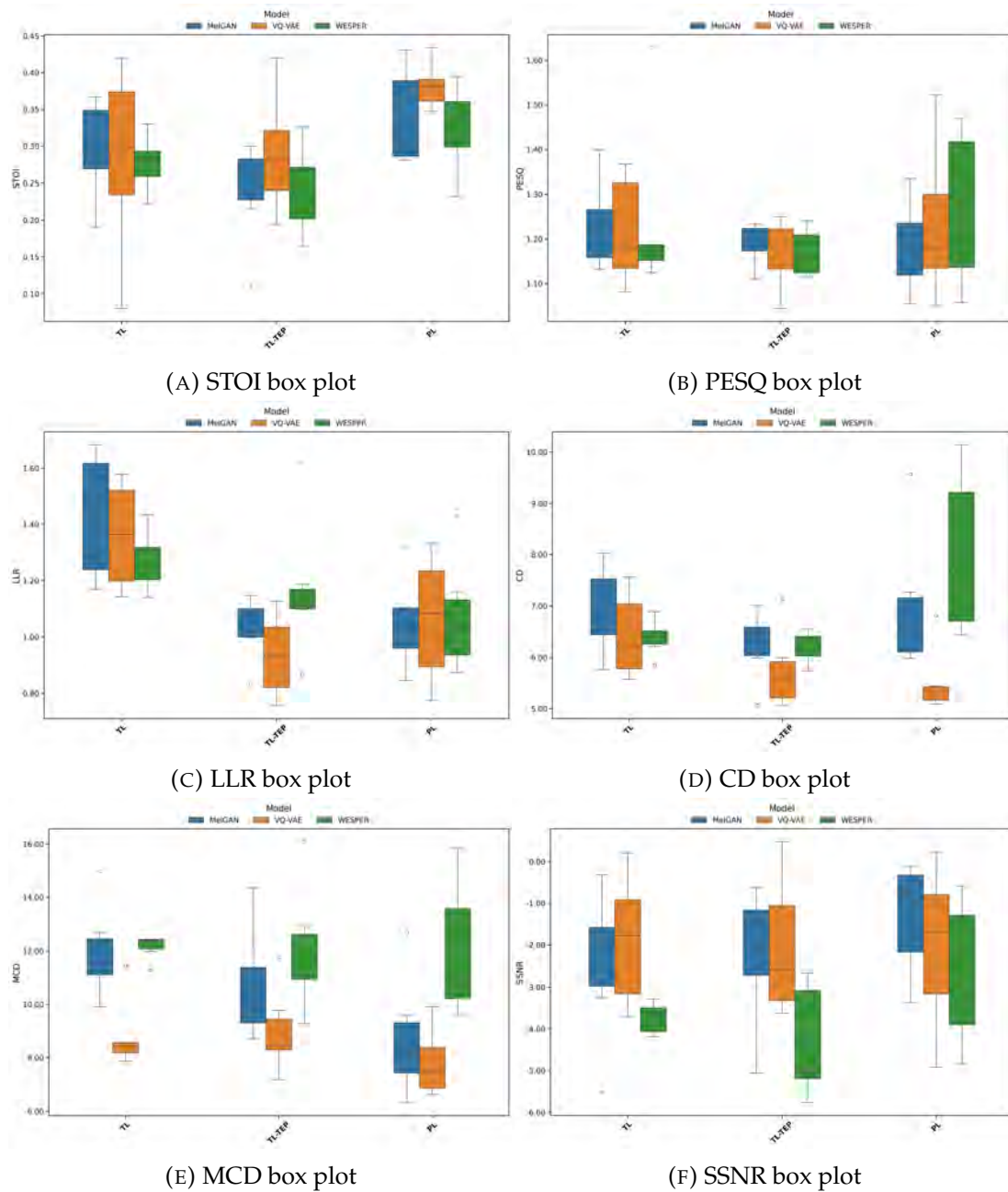


FIGURE 3.5: Objective metrics for Reconstructed data using MelGAN, VQ-VAE and WESPER. Illustrates high volatility in speech quality metrics in plots (A) and (B)

3.3.3 Voice Conversion Models

In this section, similar to Section 3.3.2, we utilised pre-trained checkpoints for FreeVC, DiffHierVC, and HierSpeechppVC. We developed our own data pre-processing and inference scripts to test these models on our dataset. We have used the same metrics as mentioned in section 3.2.

First, we examined the spectral distortion of signals, as illustrated in Figures 3.7. From these plots, we observed that DiffHierVC consistently performed the best across all patient categories—TL, TL-TEP, and PL—with lower CD, LLR, and MCD values compared to FreeVC and HierSpeechppVC. Specifically, DiffHierVC shows the lowest CD values, indicating better reconstruction of spectral features. FreeVC showed moderate performance, generally better than HierSpeechppVC but not as effective as DiffHierVC. HierSpeechppVC displayed higher variability and poorer performance, particularly for PL patients, where its CD values exceeded those of the other models. MCD values for DiffHierVC go as low as 5 for PL, which indicates that the models outperform all the models in evaluation in terms of MCD and CD.

Next, we evaluated the speech quality of the reconstructed speech. The PESQ values indicated that DiffHierVC showed the highest perceived speech quality with significantly higher scores, especially for TL-TEP patients. FreeVC also performed well but was slightly behind DiffHierVC. HierSpeechppVC, on the other hand, showed lower PESQ scores, suggesting a lower perceived quality of the reconstructed speech. However, the values of STOI were not that high in any of the models, which is going around 0.35 for all the models suggest low intelligibility.

When examining the SSNR values, we found that FreeVC really high positive values around 6 and in contrast of the DiffHierVC and HierSpeechppVC had negative values around -6. Positive values in FreeVC suggest that the model is cleansing the signal completely, and there might not be any pitch or energy added to the reconstructed speech signal. Also, the negative values in the other two models suggest that when reconstructing speech, it added noise, which is a common problem in diffusion-based architecture because it tries to add the noise and predict where it has been added. To check this, we have listened to the speech and provided the sample in figure 3.6

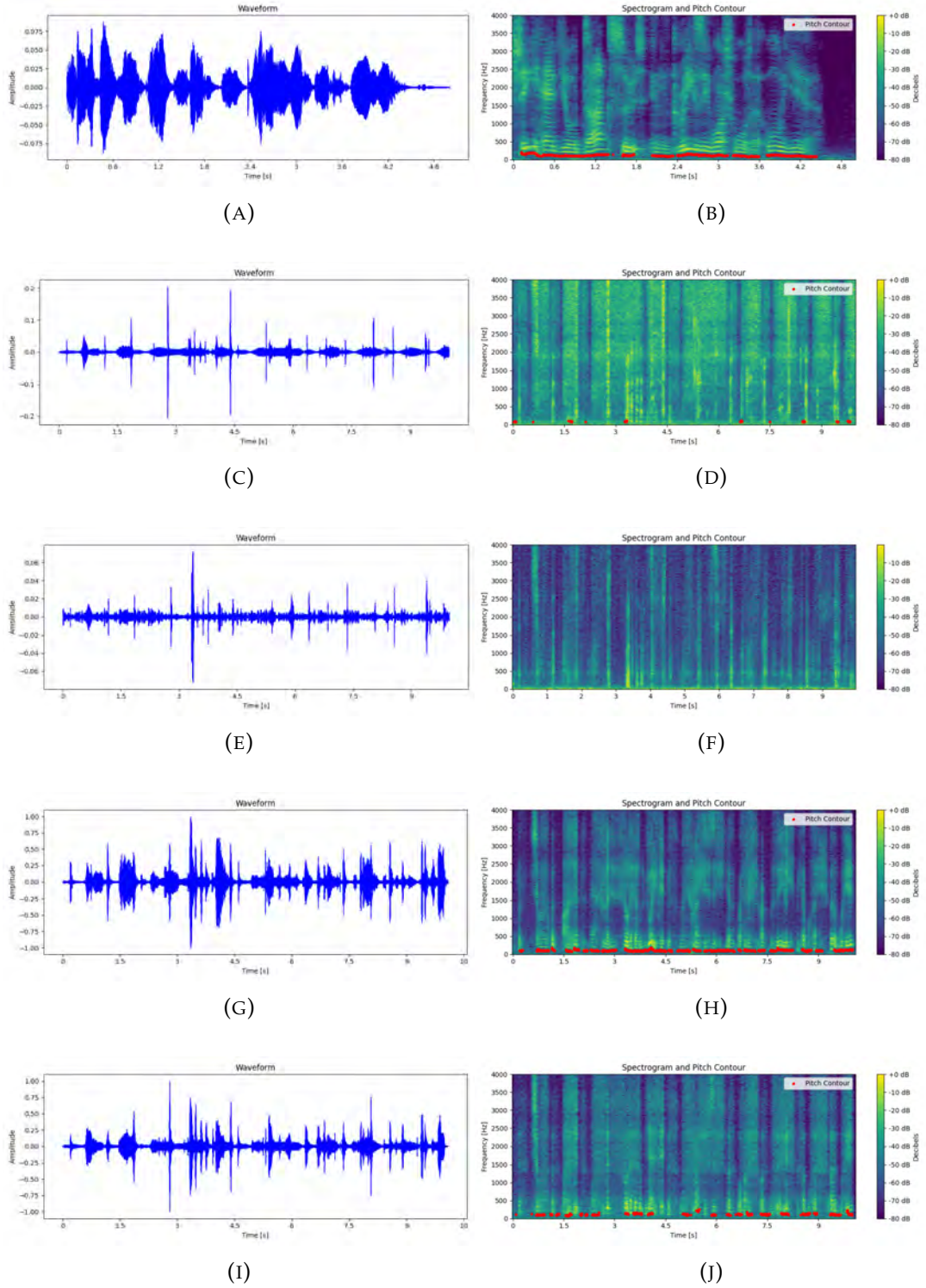


FIGURE 3.6: Waveforms and spectrograms with pitch contours for Normal and Reconstructed speech for TL patient ID 11 articulating "She had your dark suit in greasy wash water all year." (A) Waveform and (B) Spectrogram with pitch contour for Normal speech; (C) Waveform and (D) Spectrogram with pitch contour for Distorted speech; (E) Waveform and (F) Spectrogram for Reconstructed speech using FreeVC, which shows no pitch insertion; (G) Waveform and (H) Spectrogram with pitch contour for Reconstructed speech using DiffHierVC shows random pitch insertion with adding noise in signal; (I) Waveform and (J) Spectrogram with pitch contour for Reconstructed speech using HierSpeechppVC shows slightly better pitch insertion than DiffHierVC but added more noise in signal.

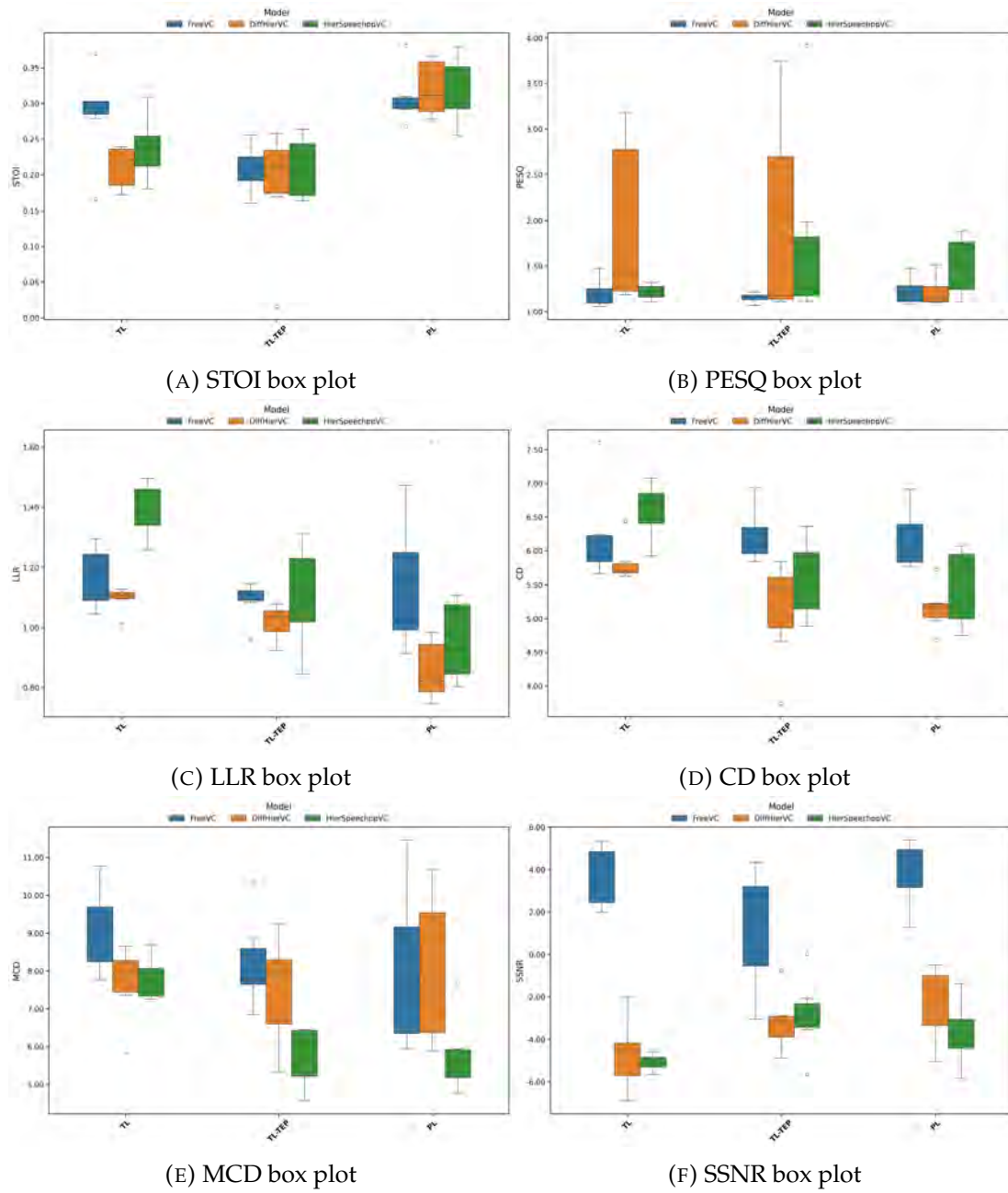


FIGURE 3.7: Objective metrics for Reconstructed data using FreeVC, DiffHierVC and HierSpeechppVC. DiffHierVC shows better performance in spectral distortion metrics (C), (D), and (E) than HierSpeechpp and FreeVC. FreeVC shows the removing data from signal in plot (F)

After conducting the analysis in Section 3.3, we identified the limitations of the models and analysed their data requirements and training times to determine which models are best suited for reconstructing distorted speech. We chose to focus on the GANs discussed in Section 3.3.1 because they can be effectively

trained on just 1 or 2 hours of data. Additionally, these GANs align with our fourth research question in Section 1.3 by reconstructing speech phonetically without introducing additional noise, unlike diffusion models. They also do not require large amounts of data like MelGAN and VQ-VAE, nor do they predict words based on spectral features like WESPER. So, We are proposing the modified GANs in Chapter 4. Additionally, we have summarised the limitations of the models in table 3.3, which answers our 3rd research question.

TABLE 3.3: Summary of Analysis

Model	Limitations
WESPER	Predicting Words
MelGAN	Large amount of parallel Data Needed
VQ-VAE	Large amount of parallel Data Needed and Noise
FreeVC	Not Adding Pitch and Power to Speech
Diff-HierVC	Adding Noise
HierSpeechppVC	Adding Noise
CNN GAN	Unstable Training Process
Inception GAN	Unstable Training Process

Chapter 4

Proposed Modified GANs

This section is divided into four parts. The first part discusses the proposed methodology in which we will discuss the feature extraction process, training, validation, and testing sets, as well as the training phase and inference phase. In the second part, we present the proposed models: MMSE-GAN, Disco-GAN, Cycle-GAN, CNN-GAN, and Inception-GAN, detailing the modifications to their architectures and loss functions. Finally, the last part covers the objective evaluation of results and a comparative analysis with previous results from the same models using measures discussed in section 3.2.

4.1 Proposed Methodology

This section will discuss the proposed methodology and the workflow of feature extraction, training, and inference. It will discuss all the phases in subsequent sub-sections and the block diagram of this proposed methodology represented in figure 4.1.

4.1.1 Feature Extraction

For extracting features, we have used AHOCODER [76]–[80] which is a high-quality vocoder that can extract the features from the input wav files and also synthesise the speech from these features. The difficult part was that the

AHOCODER is in binary format, which only supports Linux-based distributions, so to run that, we have configured vs code with WSL remote server [112] to get ubuntu Linux distribution for running AHOCODER. Following that, AHOCODER only accepts the input wav file in 16KHz mono channel format, so we have used SoX [113], [114] to convert the file into a desirable extension and sampling rate. SoX is a command line utility that we can use to modify WAV files without degrading or corrupting the file's contents at the frame level.

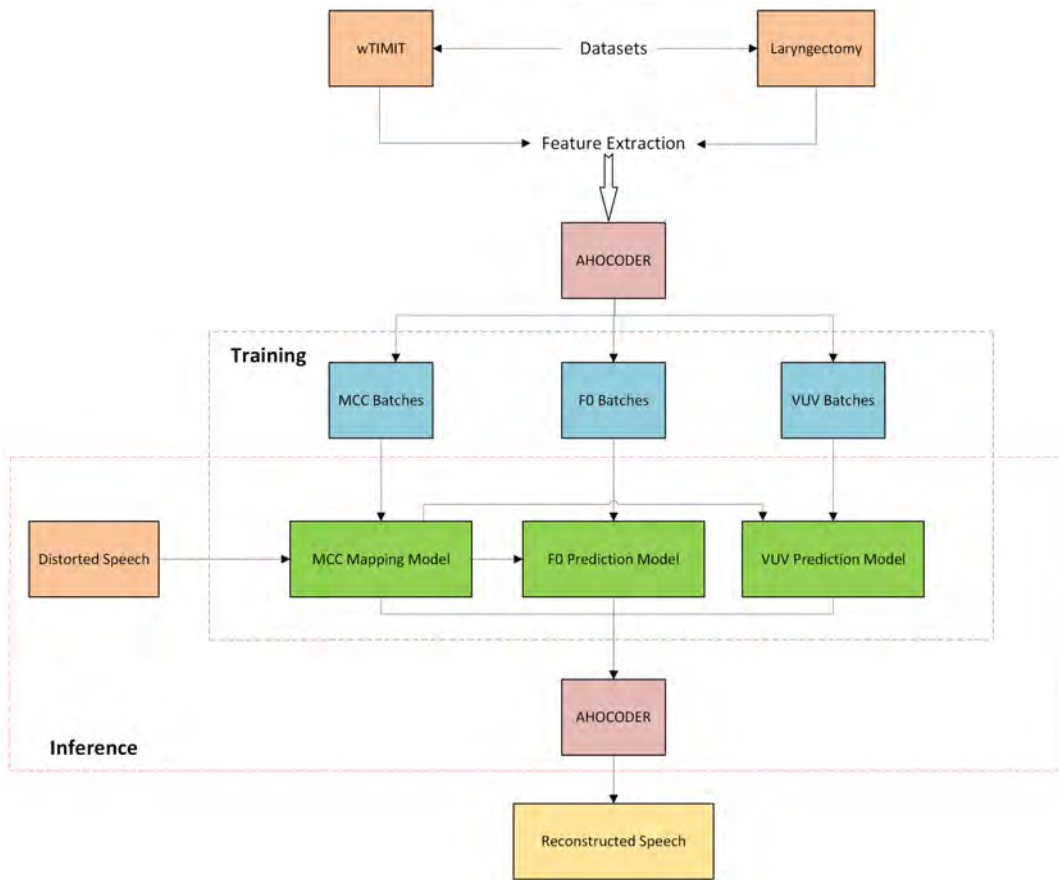


FIGURE 4.1: Block Diagram of Proposed Methodology

We used the wTIMIT dataset and our patient data to train these models, excluding testing data. Specifically, we used 1444 parallel utterances for training and 300 for validation from the wTIMIT dataset from six speakers. Additionally, we incorporated data from laryngectomy patients, using 250 utterances

from all three categories of patients: PL, TL, and TL-TEP. For validation, we used 50 utterances.

We had access to parallel normal speech data for the PL patients, whereas, for TL and TL-TEP patients, we used reference data spoken by different individuals in normal speech. In total, our training set comprised 1694 parallel utterances and 550 utterances for validation, with an additional 18 utterances set aside for testing, which we have discussed in table 3.2.

We have extracted MCC, F0, and Fv for normal and whisper speech data from the wTIMIT dataset and reference speech and distorted speech data from the patient's dataset. We have extracted features with a 25 ms window size and a 5 ms humming window. We have extracted 40 mel channel MCC data and 1 channel f0 data through AHOCODER. After feature extraction, the files were in binary format, so we used MATLAB to read these files and batch the features using it. However, whisper or distorted speech and parallel normal speech have different time alignments when we looked at data in MATLAB. So, we have used the DTW algorithm to time-align MCC features of whisper or distorted and normal speech and batch the data into 1000,40 shape MCC batches for both normal and whisper or distorted speech. Then we batched f0, which is the most challenging part because it extracts f0 in log base e format, so the pitch becomes negative infinity when the voice is not present. When we trained the model with that, it was skewed and did not add any pitch. Then, we skipped that data and only took voiced frames, batched them with the corresponding 40-channel MCC feature, and made an f0 feature batch of 1000,1 shape with 1000,40 shape corresponding MCC feature, which has voiced frames. For our data, if f0 is greater than 0, then the voice is present and taken as 1 and batched with the corresponding normal MCC feature and taken 0 otherwise and batch them with corresponding 40 channel MCC features. This was the whole feature extraction and pre-processing part we did on data to feed it to GANs. In

the following subsection, we will discuss the training phase.

4.1.2 Training Phase

In figure 4.1, we have shown the training phase by black dotted lines. In the previous section, we discussed the data that goes into a model for training.

As shown in block diagram 4.1, we use 3 independent models in the training phase. The first model is the MCC mapping model, which is the core model that takes in a whisper or distorted data and parallels normal speech data simultaneously to map the whisper or distorted speech to normal speech. For this mapping of the MCC feature, we have used 5 GANs, namely MMSE-GAN, Disco-GAN, Cycle-GAN, CNN-GAN, and Inception-GAN. We have discussed the modified GANs for MCC mapping models in section 4.2.

The second model is the F0 prediction model, which is being trained on F0 and their corresponding 40-channel MCC feature from normal or reference speech because whisper or distorted speech has no pitch. We used the same GAN architectures as in the previous sub-section but modified the architecture loss functions and tuned the hyperparameters and the details discussed in section 4.3.

The last Model is the VUV prediction model, which takes 40 channels of MCC data with a corresponding label of 0 if the frame is unvoiced or 1 if the frame is voiced. We have adjusted the simple DNN module to accommodate laryngectomy patients, and the adjustments are discussed in section 4.4.

Following the training process, in next subsection, we will look into the inference phase to get reconstructed speech.

4.1.3 Inference Phase

This phase has been shown in figure 4.1 as red dotted lines where we have trained the models and can be used for inference. We first need to pass the whisper or distorted speech to the MCC mapping model for inference. However, we cannot directly pass the speech to the model for that. We extracted testing data with the same humming window of 5 ms and window size of 25 ms and used AHOCODER to extract features from distorted speech.

When it passes through the MCC mapping model, the distorted speech is mapped phonetically, where each frame is being considered. After mapping, it has the MCC features nearing normal speech and passes through the F0 prediction model to predict pitch based on MCC data.

After going through the F0 prediction model, it will have pitch data for each frame, but it does not know which frame should be voiced or unvoiced, and the VUV model is making that decision. The mapped MCC data will be passed through this model, and the VUV prediction will be obtained after this prediction. It will be combined with the F0 prediction output, and which frame has 0 for that frame F0 will also become 0, and which frame has 1 will retain the pitch. This mapped MCC data, and combined F0 and VUV data passes through AHOCODER, which will synthesising the speech.

This is the generalised process for the proposed models. We will swap the MCC mapping and F0 prediction models but will follow this process.

4.2 Proposed MCC Mapping Models

This section will discuss the adjustments we have made in models to adapt for laryngectomy patients for MCC features. We have modified all five GANs and provided details about adjustments in the sub-sections. Additionally, we

have discussed the stability of GANs and the adjustments made to the models during experiments based on observations from the generators and discriminators. These adjustments are crucial as they directly impact the reconstructed speech quality for laryngectomy patients.

- **MMSE-GAN**

In MMSE-GAN, when we were experimenting with adding patient data that we discussed in the previous section, the learning speed for the generator was slowing down. Discriminator loss decreased each epoch, so based on this analysis from the current model, we added two dropout layers with a 0.3 dropout rate in the discriminator after the first two DNN layers. We have a regularised discriminator, and the second modification we have done is label smoothing.

Label smoothing is a regularisation technique that helps prevent the model from becoming overconfident in its predictions [115], [116]. Instead of assigning a probability of 1 to the correct class and 0 to all others, label smoothing assigns a slightly lower probability to the correct class like 0.9. It distributes the remaining probability mass among the other classes. This technique reduces the model's sensitivity to noise in the training data and helps it generalise unseen data better.

We have used this approach, modified the labels, and put 0.9 for real and 0.1 for fake during the training and validation process.

- **Disco-GAN**

In Disco GAN, similar to MMSE-GAN, we did not make any changes in the generator, but in the discriminator, we added two dropout layers after the first two DNN layers with a 0.4 dropout rate. Additionally, we have applied label smoothing and set 0.9 for real and 0.1 for fake in training.

- **Cycle-GAN**

Compared to MMSE-GAN and Disco-GAN, the training process for Cycle-GAN was unstable, and the discriminator quickly distinguished between real and fake speech. So, we have regularised the discriminator by adding three dropout layers after each DNN layer and increased the dropout rate to 0.5 for the last layer and 0.4, respectively, for the first and second layers. Then we have also implored label smoothing same as previous GANs.

- **CNN-GAN**

After modifying the MCC models based on DNN layers, we transitioned to CNN-based models, which initially proved to be unstable and performed poorly in evaluations. This necessitated significant modifications. During experiments, we observed that the generator had an exploding gradient problem and the discriminator was overfitting. Given that our CNN GAN model already had a sufficient number of layers, our primary focus shifted to addressing these issues, along with improving the loss function and optimiser.

To address the issues with the layers, we first identified that the generator was experiencing exploding gradient problems. To mitigate this, we added batch normalisation after each layer transitioning from 128 to 256 and back to 128 channels.

For the discriminator, we implemented dropout layers to improve regularisation. Specifically, we added dropout layers with a 0.3 dropout rate after the 32 and 64-channel layers. Following the max pooling layer, we applied a dropout rate of 0.5. Additionally, we maintained a 0.5 dropout rate for layers transitioning from 128 to 256 channels. After the final pooling layer, we incorporated a dropout rate of 0.3.

Despite these architectural changes, the training remained unstable and inefficient. To address the second issue, we switched to the AdamW optimiser [117] with a weight decay of 0.0001 to enhance model convergence, and it is proven

to be working well with the exploding gradients problem [118], also used label smoothing of factor 0.1.

Aside from these changes, we have changed the loss function for this model. Initially, the CNN GAN utilised only adversarial loss to calculate the generator loss. We introduced MSE loss as an identity loss which is used in image-to-image translation tasks to minimise the distance between pixel values for original and generated samples [119]. For our task, it will try to close the distance between the generated MCC or F0 feature with the reference feature to make it closer to normal speech. This combination of adversarial and MSE losses allowed for better training, as mathematically expressed in equations 4.1 and 4.2. The schematic diagram for this proposed loss function is illustrated in figure 4.2

$$\begin{aligned} \min_D V(D_{\text{NORMAL}}) = & -\mathbb{E}_{y \sim p(y)} [\log D_{\text{NORMAL}}(y)] \\ & - \mathbb{E}_{x \sim p(x)} [\log(1 - D_{\text{DIS}}(G_{\text{DIST2NORM}}(x)))] \end{aligned} \quad (4.1)$$

$$\begin{aligned} \min_G V(G_{\text{DIST2NORM}}) = & -\mathbb{E}_{x \sim p(x)} [\log D_{\text{DIS}}(G_{\text{DIST2NORM}}(x))] \\ & + \frac{1}{2} \mathbb{E}_{y \sim p(y), x \sim p(x)} [\log(y) - \log(G_{\text{DIST2NORM}}(x))]^2 \end{aligned} \quad (4.2)$$

In the equation 4.1, D_{Normal} is used to discriminate between real and fake samples based on probability distribution. And second equation 4.2, first term where $G_{\text{DIST2NORM}}$ is the generator to generate the samples from input x normal speech and it passes through D_{DIS} to calculate the adversarial loss. We

have added our novel second term that calculates how far the generated sample is from the original sample y using MSE.

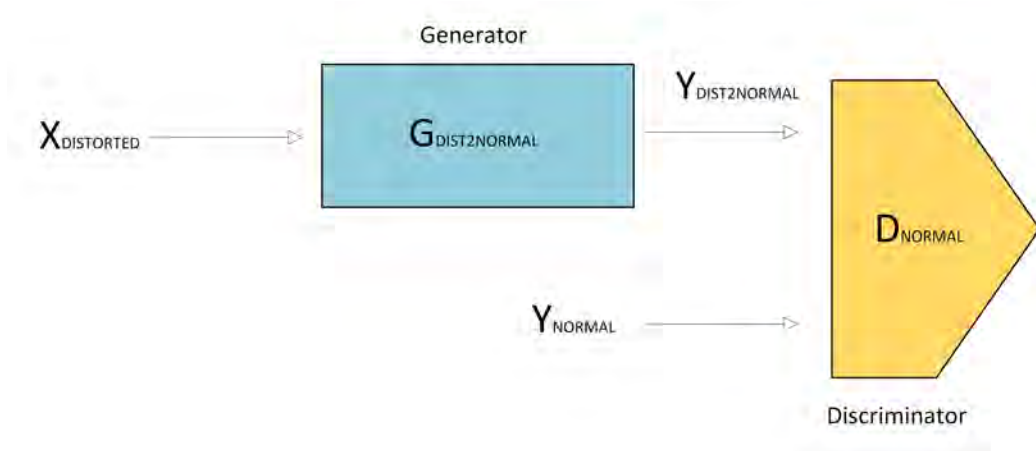


FIGURE 4.2: Schematic Representation of Proposed Loss Function

• Inception-GAN

The Inception GAN had problems similar to those of the CNN GAN, particularly with stability and performance during training. To address these challenges, we used the AdamW optimiser and implemented label smoothing with values of 0.8 for real labels and 0.2 for fake labels.

In terms of architecture, the model was already quite complex and had enough layers to extract the features, so we did not add layers. Instead, we have added dropout layers after every layer in the inception module with a dropout rate of 0.3 and did not change the inverse inception module, which stabilises the gradients from the initial inception module in both the generator and discriminator.

Additionally, we have modified the loss function to make the training stable. We have used a similar loss function that we have proposed in figure 4.2 and equations 4.1 and 4.2. But we have added hyperparameters λ_1 and λ_2 in a generator for both adversarial and MSE loss to give a relative importance to the model's optimised loss. The modified loss function for generator is represented mathematically in equation 4.3.

$$\begin{aligned}
\min_G V(G_{\text{DIST2NORM}}) = & -\lambda_1 * \left(\mathbb{E}_{x \sim p(x)} [\log D_{\text{DIS}}(G_{\text{DIST2NORM}}(x))] \right) \\
& + \lambda_2 * \left(\frac{1}{2} \mathbb{E}_{y \sim p(y), x \sim p(x)} [\log(y) - \log(G_{\text{DIST2NORM}}(x))]^2 \right)
\end{aligned}
\tag{4.3}$$

For this model, we have set λ_1 as 1 and λ_2 as 2 to give more importance to reducing loss between generated speech and normal speech because we are using the wTIMIT dataset, which has parallel data in our training.

After making changes in the MCC mapping model, we need to improve the prediction of F0 for the mapped MCC feature. In the subsequent section, we will discuss the modifications we have done on models and validation loss during training for models to discuss the stability of models with modification.

4.3 Proposed F0 Prediction Models

In this section, we try to improve the F0 model to more accurately predict pitch for distorted speech. Improving this model presented several challenges. The discriminator began to overfit, and the generator struggled to predict F0 for the mapped MCC accurately. We observed the F0 models' training closely because of these problems. We plotted the validation loss for the discriminator and the generator at each epoch in order to better understand these problems. Plots like this made it easier to monitor model performance and make necessary modifications to stabilise the training process.

- **MMSE-GAN for F0 Prediction Model**

In this model, we have not changed the architecture of the generator, but in the discriminator, we have added two dropout layers with a 0.4 dropout rate after the first two DNN layers and done label smoothing of 0.1 factor.

From figure 4.3, we can observe that the generator is getting better at each epoch from plot (A), and the discriminator loss being stable around 0.7 from plot (B) suggests stable training of models.

• Disco-GAN for F0 Prediction Model

For the Disco-GAN, we implemented changes similar to those made for the MMSE-GAN. However, instead of a 0.4 dropout rate, we used a 0.5 dropout rate. Additionally, for label smoothing, we assigned 0.8 as the real label and 0.2 as the fake label.

• Cycle-GAN for F0 Prediction Model

The most difficult model to train was the CycleGAN model because, as we observed after a few initial epochs, the discriminator easily distinguished between real and generated samples. Because of that, the generator did not have a chance to learn. To mitigate this issue, the first thing we have done is to tune the learning rate. We have increased the learning rate of the generator to 0.0002, and for the discriminator, we have set the 0.00001, and we have used AdamW as an optimiser with a weight decay of 0.00001. Second thing, we have done label smoothing of 0.9 for real and 0.1 for fake.

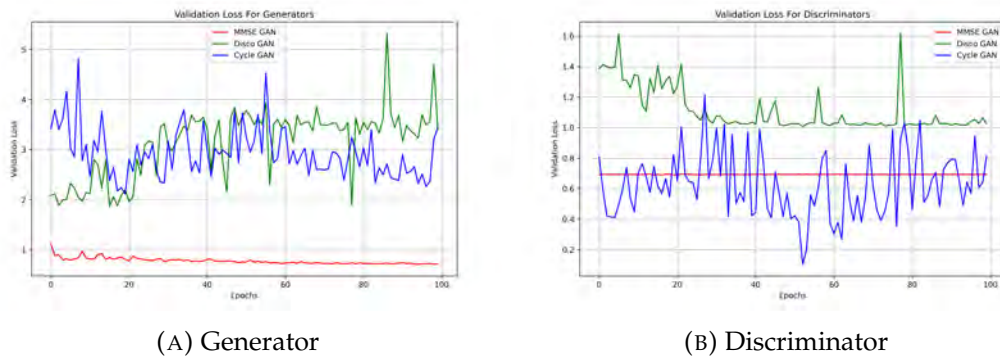


FIGURE 4.3: Validation loss of G and D for Proposed MMSE-GAN, Disco-GAN and Cycle-GAN for F0 Prediction Models

For the last modification, we tuned the discriminator and did not change the generator. In the discriminator in all three DNN layers, we have used Batch

Normalisation, and after the first two layers, we have added a dropout of 0.5. After making this modification, the validation loss stabilises, and they are represented in 4.3.

• CNN-GAN for F0 Prediction Model

When we moved toward modifying CNN models, it was hard to deal with because we could not add more layers due to complexity and computation power, the second thing is that make it a stable training process. We have smoothed the label for factor 0.1. In terms of architecture, we did not make any changes to the generator because it was working well from the current model.

In the discriminator, we have developed a totally new architecture, which is comprised of 4 DNN layers with 512 node size and with 4 dropout layers after every DNN layer with 0.5 dropout rate and Sigmoid activation function for the last layer.

Apart from that, we have used the novel loss function that we have used in section 4.2. The discriminator uses the same objective function as represented in 4.1 and for the generator we have used the objective function represented in equation 4.3 but in this function, we have taken λ_1 as 1 and λ_2 as 2 and the resultant validation loss is illustrated in figure 4.4.

• Inception-GAN for F0 Prediction Model

Lastly, we have made modifications to the model for the F0 feature, which was the most unstable and computation-heavy model. For the generator, we have made some changes in the inception module and added the dropout layers with a 0.5 dropout rate after every layer. We did not change the inverse inception module and used the same layers.

But to make the training efficient, we have used the loss function for discriminator represented in 4.1 and the generator represented in 4.3 but for hyperparameter, we have set 0.5 for λ_1 and 2 for λ_2 and the resultant stable training is illustrated in 4.4.

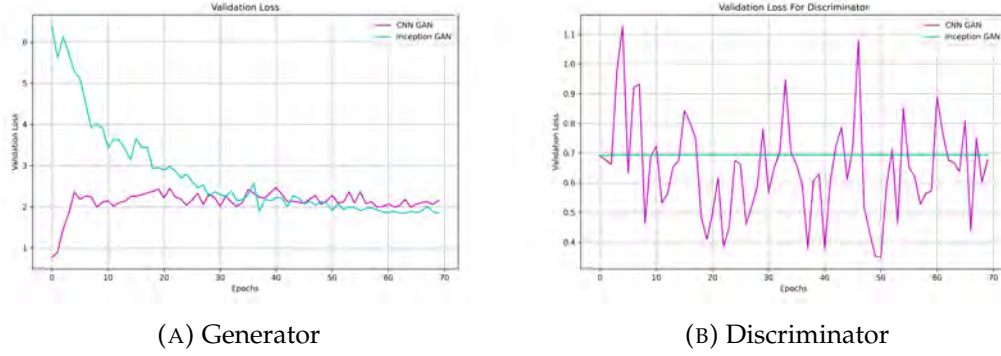


FIGURE 4.4: Validation loss of G and D for Proposed CNN-GAN and Inception-GAN for F0 Prediction Models

4.4 Proposed VUV Prediction Model

The VUV feature model is used for determining whether a frame in data should be voiced or unvoiced, which is particularly challenging for laryngectomy patients because we do not have normal speech data from the same speaker. To improve the performance of the VUV model and to better accommodate laryngectomy patient's data, we introduced new layers and increased the model's depth to extract better features.

Previously, the model comprised five DNN layers, which proved insufficient. To address this, we added three more DNN layers, resulting in a total of eight layers. The sizes of the nodes in the subsequent layers were set to 128, 256, 512, 512, 256, 128, 64, and 32. For the activation function, we replaced ReLU with the PELU [120]. PELU is similar to ReLU, and it is a non-linear activation function, but it works better with data of varying shapes, and it seemed a good choice for our requirements.

However, increasing the number of layers increased the depth of the model, but we have observed the overfitting in validation loss. To mitigate this, we applied batch normalisation followed by a dropout layer with a dropout rate of 0.5 for each layer to make it regularised after each layer. This approach helps regularise the DNN module and improves the generalisation of the validation set.

We have added sample codes for generators and discriminators of proposed models in Appendix B.

After proposing this model in the following section, we will evaluate this model and give a discussion.

4.5 Evaluation of Proposed Models

This section will provide an evaluation of proposed models, a discussion on the effectiveness of laryngectomy patient data, and a comparative analysis with previous GANs from section 3.3.1.

After developing the proposed models, we have made them cross-compatible. We can train the MCC mapping model with MMSE-GAN and the F0 model in Disco-GAN to use as a predictor of F0 for mapped MCC by the MMSE-GAN model. We have a total of 25 models after this process, and we have done 450 evaluations with these models. This section will provide an in-depth discussion of five selected models. For a more detailed analysis, including the complete evaluation tables, please refer to the link provided in Appendix (A). This supplementary material is hosted on Google Drive and includes 36 tables with all the evaluations performed, including the analysis of models in Chapter 3.

The five models we will discuss are MMSE-CNN, Disco-Inception, Cycle-CNN,

CNN-Inception, and Inception-Inception GANs. In each model, we use a specific GAN architecture for MCC feature mapping and another for F0 prediction. For example, the MMSE-CNN model employs MMSE GAN for MCC feature mapping and CNN GAN for F0 prediction, and it is similar to the remaining four models. Also, we have taken five different GANs in the MCC mapping model to do a comparative analysis with existing GANs.

We have started our evaluation by analysing the PL patient category. First, we observed the spectral distortion by examining plots (C),(D), and (E) from figure 4.8. The PL patient category exhibited the lowest spectral distortion in terms of LLR, MCD, and CD compared to the TL-TEP and TL patient categories. Disco-Inception performed the best among all proposed algorithms, achieving the lowest MCD value of 7, while Inception-Inception GAN had the highest MCD value compared to the other four algorithms. Similar results were observed for CD values, with Disco-Inception reaching the lowest value of around 5.

When comparing these proposed models with the current models from figure 3.2, the proposed models outperformed all the existing ones. Notably, the CNN-Inception model showed the most significant improvement. In the current models, the CNN GAN had an MCD value of 35, which was reduced to around 11 after modifying the model. The MCD for MMSE-CNN GAN reduced from 13 to around 10; for Inception-Inception GAN, it decreased from 15 to 11.70. Similarly, for CD, the value from CNN-Inception GAN reduced from 9.50 to 6.50, which is a drastic change in all other proposed models. It has shown a similar reduction in spectral distortion. In addition, the LLR value shows that for PL patients, the reconstructed speech is closer to normal speech compared to the models in section 3.3.1. Again, the CNN-Inception GAN value came down from 1.9 to 1.2. also, for all the other models, it at least reduced by 0.2.

When we evaluated speech quality measures, as illustrated in plot (A) and plot

(B) in figure 4.8, we observed improvements in STOI for MMSE-CNN, Disco-Inception, and Cycle-CNN GANs. Their maximum STOI value, previously around 0.3, improved to 0.4, indicating more intelligible speech. However, the values for CNN-Inception and Inception-Inception GANs decreased. Initially, noise in the speech resulted in artificially high STOI and PESQ values. As the models made the speech more intelligible, these values dropped. An example of this effect is provided in Figure 4.5.

Figure 4.5 shows that the proposed model demonstrates superior performance. Additionally, the CNN-Inception and Inception-Inception GANs show significant improvement in signal denoising. This is further supported by plot (F) in figure 4.8, which illustrates better performance for all the proposed models. Inception-Inception GAN showed the best denoising and pitch insertion in all the proposed models.

Next, we evaluated TL-TEP patient data regarding spectral distortion metrics, as shown in figure 4.8. The TL-TEP patients performed better than TL patients but slightly worse than PL patients. The proposed models showed significant improvement across all metrics compared to previous models. Notably, the CNN-Inception and Inception-Inception models demonstrated a drastic decrease in MCD values, reducing from 27 to around 12 and from 15 to 11.9, respectively. In terms of CD, it shows a similar trend of decrement, specifically for CNN-based models.

However, LLR values of DNN-based proposed models are closer to the previous model. This is because we do not have parallel normal speech data from the same speaker for TL-TEP and TL patients, and we have taken a reference, as discussed previously in section 4.1.

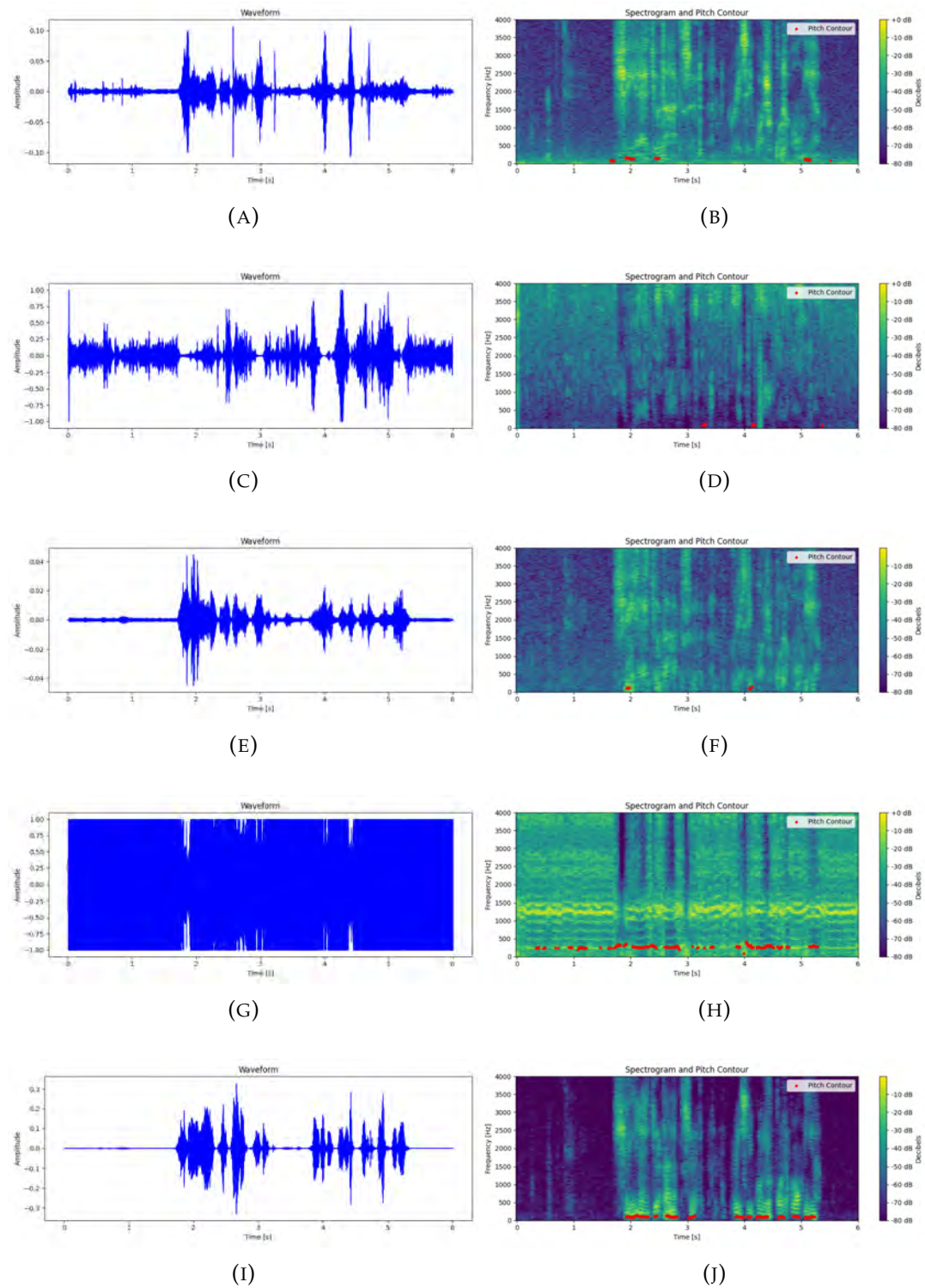


FIGURE 4.5: Waveforms and spectrograms with pitch contours for Distorted and Reconstructed speech for PL patient ID 1 articulating "She had your dark suit in greasy wash water all year". (A) Waveform and (B) Spectrogram with pitch contour for Distorted speech; (C), (D), (G) and (H) waveforms and Spectrograms for Reconstructed speech using existing CNN and Inception GANs showing distortion in signal; (E) and (F) waveform and spectrogram of proposed CNN-Inception GAN; (I) and (J) waveform and spectrogram of proposed Inception-Inception GAN showing accurate pitch insertion and denoising effect.

Due to the lack of parallel data when we observed the speech quality measures exhibit slight improvement in terms of STOI and PESQ values for MMSE-CNN, Disco-Inception, and Cycle-CNN, but for CNN-Inception and Inception-Inception, the values decreased. As discussed, this decrease was influenced by the presence of noise. Still, the speech quality was better than previous models, and pitch insertion is far superior, and the example has been provided in figure 4.6. The proposed model also shows superior performance in denoising, and it is shown as an example in figure 4.6 and in particular for Inception-Inception GAN.

Lastly, we evaluated TL patients. Due to the lack of parallel data for TL patients, their performance in terms of MCD, CD, and LLR showed improvement compared to previous models, as shown in figure 4.8 plots (C), (D), and (E). However, these metrics do not provide much insight into improvement due to the absence of parallel data. Therefore, we focused more on speech quality measures and signal denoising. From plot (B), we can see that the overall PESQ values for DNN-based models increased to an average of around 0.1, with some outliers indicating that the proposed models performed better on certain data. The real difference is observed in plot (A), where the STOI values improved significantly. For MMSE-CNN, the average STOI value increased from 0.25 to 0.35. Disco-Inception, which performed best in this category, saw its value rise from 0.22 to 0.37. Similarly, for Cycle-Inception GAN, the value went up from 0.23 to 0.34. We have also provided the sample in figure 4.7.

From figure 4.7, we can observe that the Disco-Inception model added more pitch, while the Inception-Inception model exhibited the best denoising effect among all the proposed models.

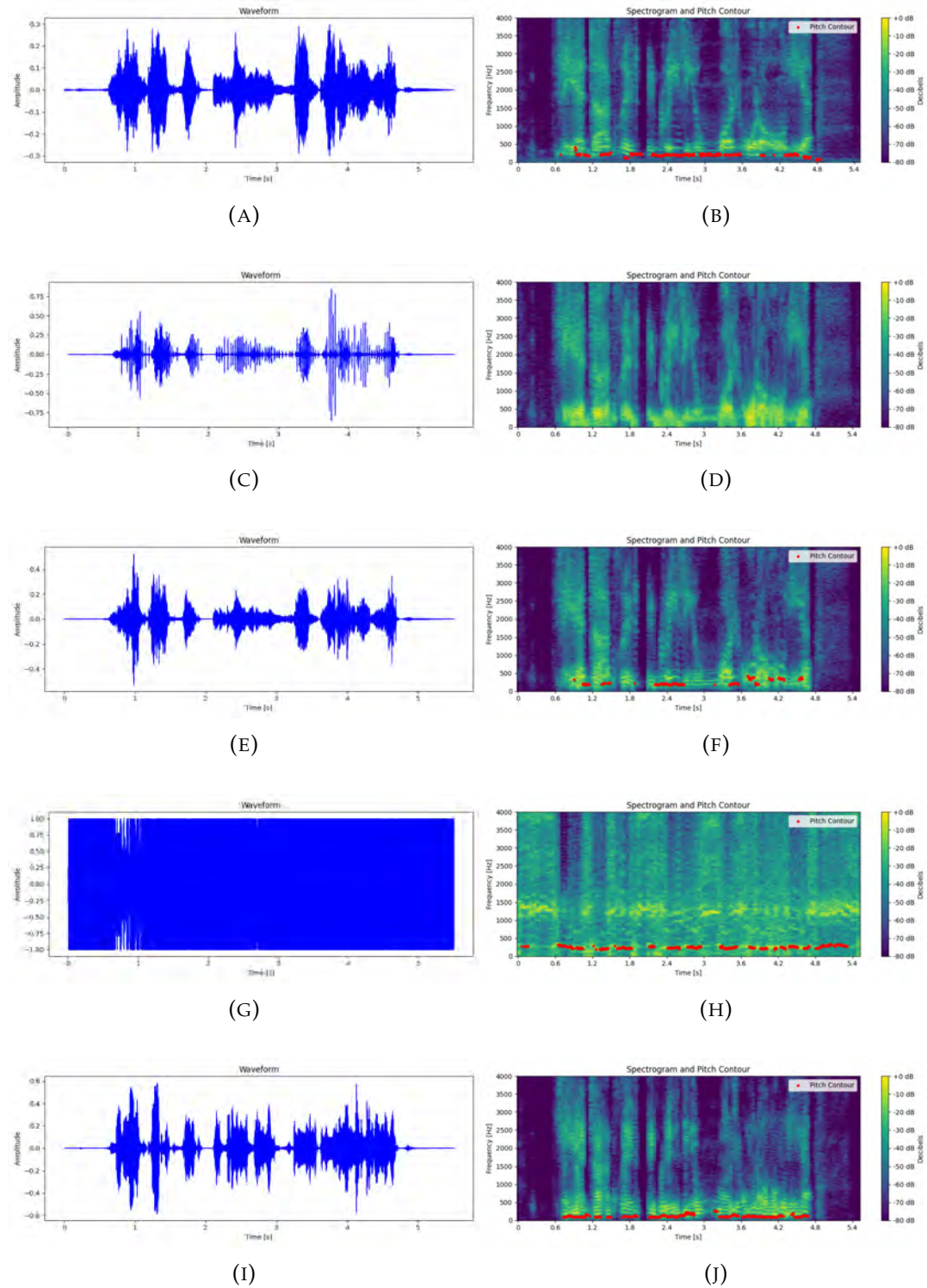


FIGURE 4.6: Waveforms and spectrograms with pitch contours for Distorted and Reconstructed speech for PL patient ID 12 articulating "She had your dark suit in greasy wash water all year." (A) Waveform and (B) Spectrogram with pitch contour for Distorted speech; (C), (D), (G), and (H) Waveforms and Spectrogram with pitch contour for Reconstructed speech using existing Cycle and Inception GANs and cycle GAN shows better spectral features but did not insert and pitch; (E), (F), (I) and (J) shows Waveforms and Spectrograms with pitch contours for proposed Cycle-CNN and Inception-Inception GANs depicting effective pitch insertion with denoising effect.

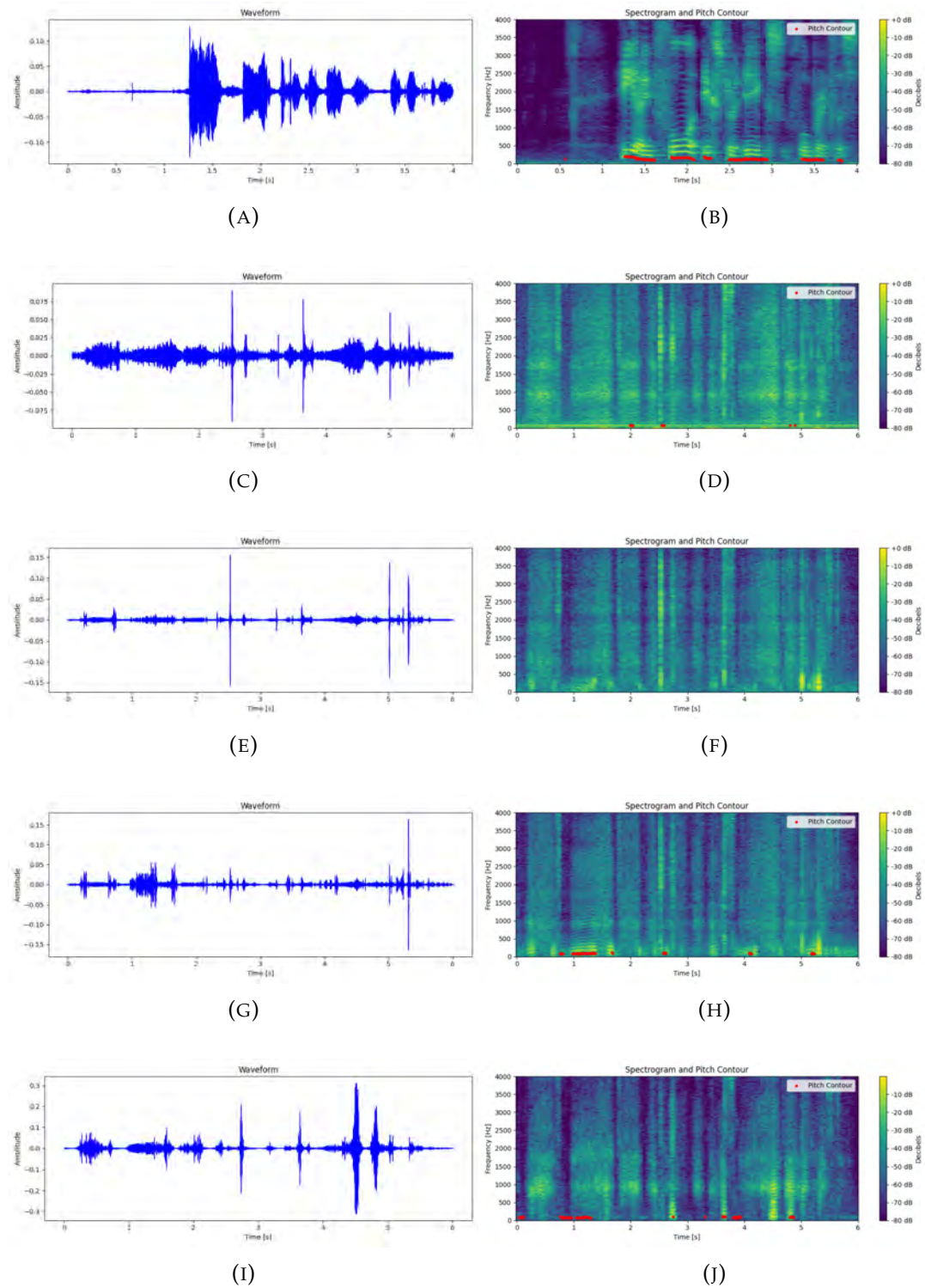


FIGURE 4.7: Waveforms and spectrograms with pitch contours for Distorted and Reconstructed speech for PL patient ID 14 articulating "Should giraffes be kept in small zoos?". (A), (B), (C), and (D) Waveforms and (B) Spectrograms with pitch contour for Normal and Distorted speech; (E) and (F) Waveform and Spectrogram using existing Disco-GAN shows slight denoising effect but not pitch insertion; (G), (H), (I), and (J) Waveforms and Spectrograms using proposed Disco-Inception and Inception-Inception GANs shows better denoising and pitch insertion

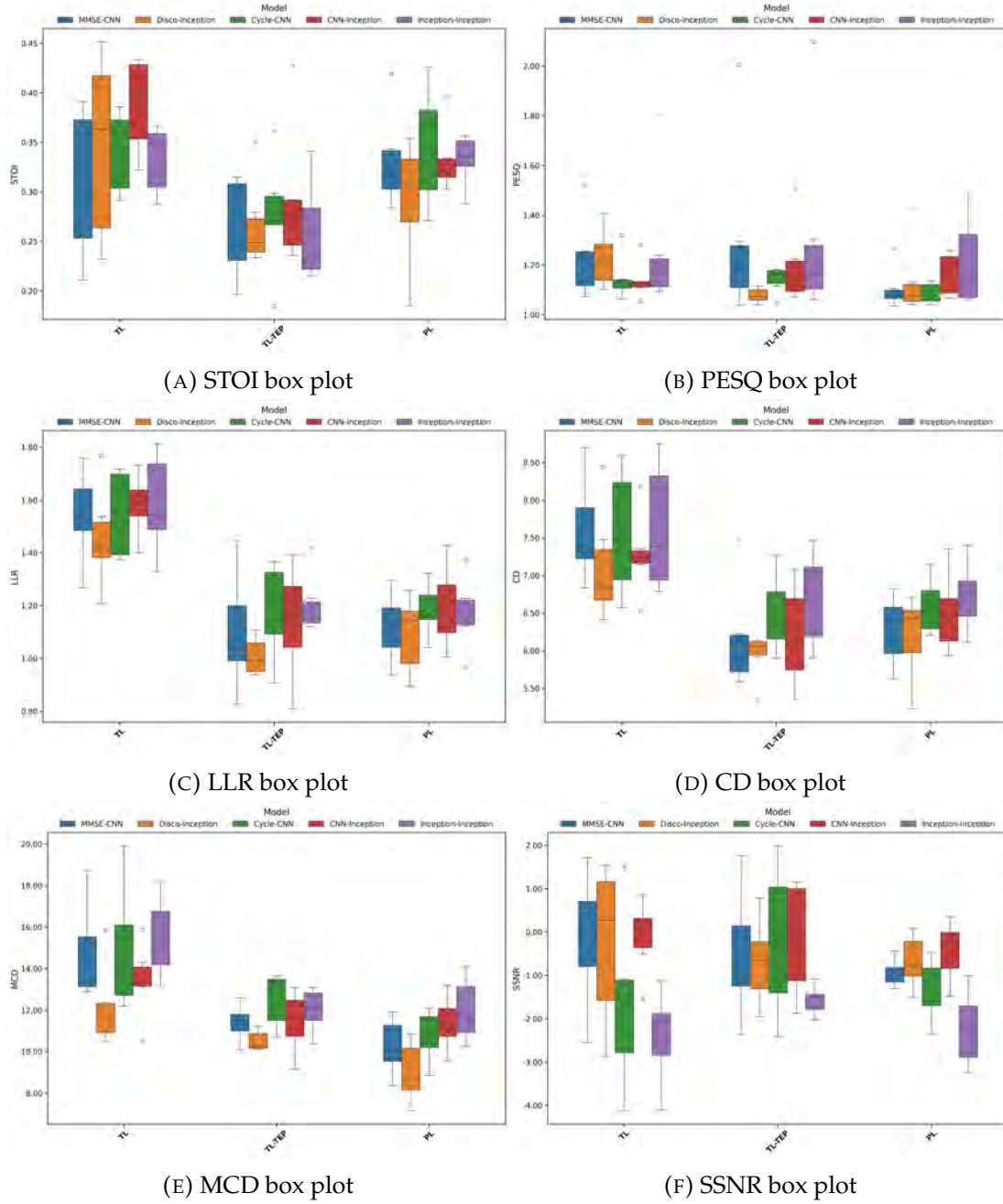


FIGURE 4.8: Objective metrics for Reconstructed data using proposed MMSE-CNN, Disco-Inception, Cycle-CNN, CNN-Inception and Inception-Inception GANs shows Disco-Inception performed best on spectral metrics in plots (C), (D) and (E), In terms of speech quality CNN-Inception and Inception-CNN performed better than three DNN based models.

After discussing the proposed models, we found that in terms of spectral features and speech quality, the proposed models outperformed previous GANs for PL and TL+TEP patient data. For TL patients, although the speech was less

intelligible, the models effectively denoised the speech signals. Following this analysis, we will provide the conclusion for our thesis in the next chapter.

Chapter 5

Conclusion

In this thesis, we explored how to improve speech for laryngectomy patients using state-of-the-art DL techniques for the first time in literature. We began by reviewing existing methods and identifying significant challenges in achieving natural and clear speech for these patients, which answers our first three research questions discussed in section 1.3. This led us to evaluate and refine various DL models.

We first examined consecutive proposed GANs MMSE-GAN, Disco-GAN, CycleGAN, CNN-GAN, and Inception-GAN. These models showed potential but had issues. They often struggled with training stability, leading to inconsistent results. For example, the Inception GAN and CNN GAN could insert pitch incorrectly and distort the speech signal, resulting in unnatural and noisy speech. However, these models were reconstructing speech phonetically without predicting what the sentence or next word would be.

Next, we tested the WESPER model to convert whisper-to-normal speech in real time. Although it represented a significant improvement, it still had problems. The WESPER model often struggled to reconstruct words accurately, especially for TL patients, producing incorrect vowel sounds or random words.

This was because it relied on predictions rather than direct phonetic reconstruction. It also depended heavily on FastSpeech2, which introduced prediction errors, particularly for TL patients. Also, for TL-TEP and PL, we have heard and observed the linguistic changes.

We also examined MelGAN and VQ-VAE. When evaluating MelGAN, we observed that for TL patients, it merely passed the speech signal without adding any pitch data and introduced noise in the reconstructed speech. Additionally, MelGAN trained on the entire wTIMIT dataset (45,000 parallel utterances) for 3000 epochs, which required significant computational power. For VQ-VAE, using an AE architecture led to noise in the signal when learning the latent space, producing reconstructed speech with pitches over 600 Hz, resulting in a harsh sound. It was trained on the same amount of data as MelGAN.

We then explored voice conversion models such as FreeVC, Diff-HierVC, and HierSpeechppVC. While these models showed some promising results, they also had significant drawbacks. FreeVC had trouble inserting pitch correctly, leading to unnatural-sounding speech, and for TL patients, it totally removed data from input speech, and the generated speech did not have any pitch or power in the reconstructed speech. Diff-HierVC and HierSpeechpp diffusion-based models introduced a lot of noise into the speech, masking the intended signals and reducing clarity. In addition to this, these models randomly inserted a pitch for TL patients, which resulted in noisy and meaningless reconstructed speech. Overall, we have done 198 evaluations to finish evaluating existing state-of-the-art models.

After the analysis, we developed our own cross-compatible GANs from the MMSE, Disco, Cycle, CNN, and Inception GANs. We made three major contributions to these current GANs, which answer the fourth research question discussed in section 1.3.

Firstly, we enhanced the discriminators for DNN-based MCC mapping models in the MMSE, Disco, and Cycle GANs by adding additional dropout layers, which improved overall performance. For F0 prediction models, while the MMSE and Disco GANs retained the same discriminator as the MCC mapping models, we introduced batch normalisation followed by dropout layers after all three DNN layers in the Cycle GAN's discriminator. Additionally, in the CNN GAN, we added batch normalisation after each CNN layer in the generator and included dropout layers after each CNN layer in the discriminator for the MCC model. For the F0 prediction model, we employed the same CNN GAN generator but introduced a new discriminator architecture with five DNN layers, each followed by dropout layers. In the Inception GAN, we incorporated dropout layers within the inception module after every CNN layer for both the generator and discriminator.

Secondly, to address the instability observed in the CNN and Inception GANs after modifying the model layers, we introduced a novel loss function combining identity loss with adversarial loss. This approach helped measure the distance between the generated and original samples, using MSE as the identity loss in the CNN GAN model. We used the same loss function for the Inception GAN but included an additional hyperparameter to balance the importance of identity and adversarial losses in the generator. These modifications stabilised the training process, even including laryngectomy patient data in the wTIMIT dataset.

Lastly, we achieved cross-compatibility between models, allowing MCC mapping models to utilize MMSE-GAN, while Disco GAN handled F0 prediction for the mapped MCC features. This integration proposed 25 novel GANs.

After evaluating the models, we observed the following results for the different patient categories:

The Disco-Inception GAN model achieved the lowest MCD value of 9 for the

PL patient category across all proposed models. Additionally, the MMSE-CNN GAN achieved a CD value of 6.5. Regarding speech quality, the Inception-Inception GAN had the highest STOI value of 0.35 for this category.

For TL-TEP patients, the Disco-Inception GAN had the lowest LLR value of 0.92 across all categories, along with an MCD value of 10 and a CD value of 6, both of which are the lowest for this category.

For TL patients, the CNN-Inception model achieved STOI value of 0.4, which is the highest in this category and across all other categories.

However, the objective metrics did not significantly improve because of the lack of normal speech data from the same patient for TL and TL-TEP. However, they were still in the desirable range and showed improvement compared to previous models.

In conclusion, this thesis has pioneered the study of improving speech for laryngectomy patients by DL models. Our proposed GANs have shown promising results, particularly for PL and TL-TEP patients. However, this research has provided a foundation for future research in speech reconstruction for laryngectomy patients. In the future, we plan to use transformer-based GANs with multiheaded attention mechanisms to encode the time-varying sequence of input speech more effectively, helping in better feature learning and training for GANs.

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Appendix A

Evaluation Tables

We have provided 36 tables with all the evaluations in this Appendices A.

A.1 WESPER

TABLE A.1: Wesper Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.27	1.147	1.321	6.21	12.37	-3.439
	Sentence01	0.255	1.124	1.586	7.01	12.08	-1.757
	Sentence02	0.33	1.628	1.434	6.55	11.97	-3.95
14	HvD	0.223	1.165	1.305	6.62	12.40	-0.29
	Sentence01	0.298	1.39	1.355	6.74	11.06	-1.986
	Sentence02	0.293	1.187	1.308	6.69	12.43	-3.669
Total Laryngectomy with TEP							
7	HvD	0.165	1.116	1.038	5.99	10.72	0.573
	Sentence01	0.222	1.122	1.127	6.16	9.29	1.764
	Sentence02	0.198	1.106	1.033	5.78	9.37	-0.985
12	HvD	0.23	1.241	0.868	5.74	11.75	-1.353
	Sentence01	0.326	1.216	1.063	6.55	16.08	-1.254
	Sentence02	0.285	1.181	1.117	6.58	12.91	-0.678
Partial Laryngectomy							
1	HvD	0.298	1.263	0.905	5.62	9.57	-0.584
	Sentence01	0.395	1.054	0.873	5.82	10.74	-1.97
	Sentence02	0.365	1.057	0.91	6.05	10.14	-3.83
2	HvD	0.232	1.252	1.16	6.54	11.40	-0.237
	Sentence01	0.300	1.468	1.025	6.36	13.20	-2.08
	Sentence02	0.395	1.469	1.31	7.19	15.83	-2.93

A.2 MelGAN

TABLE A.2: Melgan Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.367	1.132	1.432	6.86	14.98	-0.319
	Sentence01	0.365	1.154	1.683	8.03	12.37	-2.126
	Sentence02	0.300	1.339	1.557	7.16	12.04	-1.40
14	HvD	0.191	1.169	0.999	5.76	9.90	-2.157
	Sentence01	0.265	1.287	1.19	6.74	11.76	-2.691
	Sentence02	0.294	1.261	1.174	6.65	11.00	-3.248
Total Laryngectomy with TEP							
7	HvD	0.215	1.232	0.98	5.93	9.19	-1.624
	Sentence01	0.266	1.234	0.893	5.40	9.74	0.355
	Sentence02	0.288	1.19	0.83	5.06	8.70	0.506
12	HvD	0.265	1.233	1.005	6.41	14.98	-1.00
	Sentence01	0.111	1.101	1.032	7.01	14.38	0.514
	Sentence02	0.300	1.079	0.97	6.35	11.35	-0.872
Partial Laryngectomy							
1	HvD	0.284	1.329	0.977	5.98	6.91	-0.574
	Sentence01	0.292	1.054	0.954	6.03	8.52	-2.48
	Sentence02	0.431	1.095	1.02	6.13	10.36	-1.87
2	HvD	0.417	1.334	0.845	5.36	6.32	0.238
	Sentence02	0.281	1.09	0.841	5.72	8.29	-1.967
	Sentence02	0.306	1.223	1.319	7.27	10.31	-0.87

A.3 VQ-VAE

TABLE A.3: Vq-Vae Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.387	1.081	1.541	6.61	11.43	0.203
	Sentence01	0.334	1.124	1.463	7.18	8.14	-0.81
	Sentence02	0.184	1.196	1.278	5.48	8.83	-0.82
14	HvD	0.26	3.567	1.142	5.66	7.87	1.157
	Sentence01	0.08	1.479	1.167	5.59	9.23	-1.705
	Sentence02	0.226	1.163	1.175	5.57	8.32	0.838
Total Laryngectomy with TEP							
7	HvD	0.249	1.168	1.052	5.69	8.39	-0.636
	Sentence01	0.237	1.184	1.052	5.54	8.33	1.614
	Sentence02	0.194	1.224	0.894	4.76	8.33	2.418
12	HvD	0.313	1.249	0.756	5.11	8.78	-0.176
	Sentence01	0.324	1.044	0.799	5.51	11.77	1.323
	Sentence02	0.42	1.128	0.881	5.69	9.97	1.014
Partial Laryngectomy							
1	HvD	0.39	1.201	0.800	5.09	6.62	0.879
	Sentence01	0.357	1.048	0.88	5.54	8.69	-1.866
	Sentence02	0.391	1.06	0.775	5.32	7.82	-0.385
2	HvD	0.347	1.157	0.832	5.11	6.88	1.164
	Sentence01	0.434	1.333	0.883	5.46	9.27	-1.919
	Sentence02	0.374	1.522	1.284	6.82	9.90	-1.358

A.4 FreeVC

TABLE A.4: FreeVC Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.279	1.104	1.261	6.24	10.77	1.996
	Sentence01	0.37	1.473	1.294	6.20	7.96	2.299
	Sentence02	0.302	1.056	1.045	5.73	8.07	2.33
14	HvD	0.303	1.088	1.159	6.17	7.77	5.342
	Sentence01	0.167	1.511	1.292	6.49	9.64	3.995
	Sentence02	0.302	1.283	1.067	5.66	9.69	3.995
Total Laryngectomy with TEP							
7	HvD	0.256	1.14	1.145	6.34	7.76	3.438
	Sentence01	0.185	1.173	1.109	6.33	7.83	4.093
	Sentence02	0.205	1.386	0.963	5.68	7.68	2.342
12	HvD	0.217	1.162	0.912	5.91	8.85	0.462
	Sentence01	0.161	1.068	1.112	6.90	12.00	0.305
	Sentence02	0.228	1.182	1.081	6.68	10.12	0.208
Partial Laryngectomy							
1	HvD	0.295	1.231	0.957	5.76	6.32	4.984
	Sentence01	0.381	1.077	0.913	5.87	9.19	2.928
	Sentence02	0.292	1.133	0.992	6.14	8.83	1.8
2	HvD	0.302	1.202	1.00	5.93	5.94	5.16
	Sentence01	0.310	1.473	1.088	6.39	10.00	0.648
	Sentence02	0.268	1.104	1.255	6.91	11.46	1.287

A.5 DiffHier-VC

TABLE A.5: DiffHier-VC Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.239	3.176	1.012	5.67	7.69	-4.159
	Sentence01	0.227	1.574	1.105	5.62	8.66	-2.005
	Sentence02	0.239	3.165	1.21	5.23	7.98	-3.475
14	HvD	0.214	1.12	1.094	5.72	7.36	-4.821
	wave01	0.173	1.218	1.025	5.44	7.67	-7.618
	wave02	0.177	1.267	1.28	6.44	8.30	-6.889
Total Laryngectomy with TEP							
7	HvD	0.235	3.93	1.078	5.66	7.62	-0.745
	Sentence01	0.106	1.149	0.974	5.34	7.22	3.280
	Sentence02	0.17	1.131	1.058	5.24	8.32	-3.008
12	HvD	0.231	3.746	0.707	4.66	6.26	-2.509
	Sentence01	0.258	1.089	0.923	5.47	9.28	-1.998
	Sentence02	0.189	2.058	1.023	5.83	6.64	-1.313
Partial Laryngectomy							
1	HvD	0.278	1.114	0.824	4.97	5.89	-0.878
	Sentence01	0.367	1.098	0.821	5.23	8.32	-3.838
	Sentence02	0.391	1.091	0.801	5.14	7.12	-1.938
2	HvD	0.288	1.313	0.746	4.68	5.43	-0.391
	Sentence01	0.332	1.514	1.096	5.93	9.19	-6.135
	Sentence02	0.367	1.162	0.983	5.72	9.94	-5.485

A.6 HierSpeechppVC

TABLE A.6: HierSpeech-VC Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.239	1.106	1.258	5.92	8.70	-4.58
	Sentence01	0.309	1.224	1.496	7.08	8.14	-4.28
	Sentence02	0.193	1.248	1.389	6.86	8.46	-2.26
14	HvD	0.203	1.194	1.355	6.72	7.26	-5.34
	Sentence01	0.181	1.157	1.342	6.55	7.72	-6.16
	Sentence02	0.259	1.32	1.399	6.30	7.84	-5.21
Total Laryngectomy with TEP							
7	HvD	0.264	3.914	1.263	6.36	7.80	0.022
	Sentence01	0.171	1.137	1.027	5.71	6.43	1.1
	Sentence02	0.174	1.985	1.314	5.78	6.45	-3.538
12	HvD	0.237	1.111	0.845	4.98	6.23	-3.007
	Sentence01	0.164	1.182	0.986	5.64	9.20	-1.66
	Sentence02	0.246	1.224	1.086	6.02	6.42	-1.457
Partial Laryngectomy							
1	HvD	0.291	1.287	0.839	4.77	4.78	-0.85
	Sentence01	0.369	1.227	0.862	5.19	7.67	-2.87
	Sentence02	0.380	1.073	0.904	5.51	6.70	-3.42
2	HvD	0.298	1.782	0.803	4.75	5.22	-1.38
	Sentence01	0.255	1.886	1.077	5.86	8.85	-5.41
	Sentence02	0.299	1.106	0.99	5.56	8.58	-3.99

A.7 Existing GANs

These are the models from the literature that we have done analysis on.

TABLE A.7: MMSE GAN Voiced Region Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.296	1.164	1.684	8.190	18.944	-2.233
	Sentence01	0.236	1.077	1.781	8.837	14.548	-3.012
	Sentence02	0.233	1.109	1.721	8.541	11.956	-4.948
14	HvD	0.251	1.197	1.566	7.52	12.91	0.110
	Sentence01	0.153	1.119	1.367	5.752	10.744	2.158
	Sentence02	0.206	1.170	1.542	6.732	12.583	0.793
Total Laryngectomy with TEP							
7	HvD	0.201	1.120	1.529	7.647	12.847	-2.749
	Sentence01	0.285	1.088	1.404	6.893	11.846	-0.300
	Sentence02	0.183	1.196	1.339	6.643	11.564	-1.636
12	HvD	0.228	1.229	1.374	6.402	10.939	-1.211
	Sentence01	0.251	1.117	1.556	6.402	12.910	-1.211
	Sentence02	0.247	1.106	1.879	5.944	12.301	0.007
Partial Laryngectomy							
1	HvD	0.216	1.078	1.384	6.625	9.510	-0.251
	Sentence01	0.227	1.228	1.256	6.976	11.350	0.376
	Sentence02	0.311	1.254	1.100	6.617	11.059	0.139
2	HvD	0.309	1.684	1.281	6.488	11.245	-0.331
	Sentence01	0.305	1.117	1.348	7.179	13.032	-1.070
	Sentence02	0.301	1.161	1.431	7.499	12.974	-0.950

TABLE A.8: Disco GAN Voiced Region Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.209	1.112	1.720	7.826	22.547	-1.446
	Sentence01	0.199	1.056	1.804	8.874	21.165	-0.814
	Sentence02	0.227	1.060	1.710	8.280	18.783	-3.977
14	HvD	0.251	1.262	1.629	8.035	14.861	-0.465
	Sentence01	0.153	1.070	1.482	6.752	14.125	0.560
	Sentence02	0.204	1.127	1.547	7.053	15.632	1.683
Total Laryngectomy with TEP							
7	HvD	0.214	1.168	1.430	7.398	13.710	-2.360
	Sentence01	0.201	1.104	1.431	7.047	14.039	-0.598
	Sentence02	0.207	1.089	1.408	6.934	13.210	-2.297
12	HvD	0.144	1.105	1.306	6.427	12.487	-1.995
	Sentence01	0.153	1.101	1.204	6.738	14.173	-1.135
	Sentence02	0.226	1.030	1.242	6.669	14.272	-1.672
Partial Laryngectomy							
1	HvD	0.228	1.075	1.330	7.208	12.526	-0.354
	Sentence01	0.278	1.047	1.250	6.918	14.223	-0.951
	Sentence02	0.235	1.116	1.235	6.665	13.643	-1.167
2	HvD	0.248	1.112	1.285	6.540	13.541	-1.475
	Sentence01	0.283	1.116	1.401	7.431	17.128	-1.010
	Sentence02	0.235	1.085	1.513	7.752	15.642	-0.497

TABLE A.9: Cycle GAN Voiced Region Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.282	1.123	1.565	7.469	21.459	-2.098
	Sentence01	0.185	1.09	1.736	8.469	19.504	-0.599
	Sentence02	0.204	1.095	1.644	8.075	15.611	-3.511
14	HvD	0.265	1.189	1.369	6.389	12.180	-1.222
	Sentence01	0.220	1.088	1.277	5.993	12.605	1.346
	Sentence02	0.097	1.144	1.492	7.182	14.755	-0.571
Total Laryngectomy with TEP							
7	HvD	0.220	1.814	1.312	6.873	12.524	-1.082
	Sentence01	0.280	1.074	1.291	6.438	13.646	1.036
	Sentence02	0.142	1.126	1.273	6.493	13.468	0.515
12	HvD	0.226	1.189	0.917	5.763	11.169	-0.929
	Sentence01	0.256	1.147	0.855	5.728	11.920	0.379
	Sentence02	0.228	1.204	0.738	5.235	11.772	-0.222
Partial Laryngectomy							
1	HvD	0.261	1.090	1.042	6.023	10.745	-0.923
	Sentence01	0.283	1.048	1.180	6.507	10.874	-1.609
	Sentence02	0.238	1.083	1.025	6.142	10.627	-2.581
2	HvD	0.233	1.125	1.225	6.715	10.380	-1.283
	Sentence01	0.248	1.151	1.227	6.707	12.293	-2.531
	Sentence02	0.256	1.083	1.325	7.202	12.754	-3.346

TABLE A.10: CNN Voiced Region Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.434	1.150	1.961	9.634	29.122	-8.217
	Sentence01	0.399	2.342	1.964	9.816	28.026	-7.836
	Sentence02	0.337	1.159	1.943	9.693	21.179	-8.220
14	HvD	0.309	1.034	1.994	9.976	32.720	-9.881
	Sentence01	0.308	1.056	1.985	9.938	31.033	-9.800
	Sentence02	0.265	1.078	2.000	9.996	29.021	-9.824
Total Laryngectomy with TEP							
7	HvD	0.362	1.047	1.991	9.956	32.540	-8.477
	Sentence01	0.307	1.051	1.921	9.653	31.920	-5.885
	Sentence02	0.444	1.041	1.960	9.822	32.489	-6.693
12	HvD	0.298	1.056	1.814	9.176	25.805	-8.567
	Sentence01	0.288	1.041	1.782	8.974	29.371	-7.023
	Sentence02	0.339	1.037	1.687	8.702	27.760	-5.867
Partial Laryngectomy							
1	HvD	0.432	1.056	1.950	9.818	26.613	-9.741
	Sentence01	0.559	1.063	1.787	9.148	29.805	-9.184
	Sentence02	0.581	1.066	1.731	9.023	26.563	-9.071
2	HvD	0.553	1.169	1.937	9.733	30.728	-9.423
	Sentence01	0.495	1.050	1.906	9.578	29.978	-9.438
	Sentence02	0.548	1.067	1.921	9.585	27.326	-8.751

TABLE A.11: Inception Voiced Region Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.398	1.539	1.800	7.992	17.379	-9.906
	Sentence01	0.276	2.479	1.886	8.836	17.750	-9.911
	Sentence02	0.227	1.444	1.839	8.555	17.152	-9.970
14	HvD	0.293	1.221	1.731	7.611	14.577	-9.959
	Sentence01	0.276	4.173	1.765	7.351	15.924	-9.974
	Sentence02	0.361	1.176	1.845	7.452	15.288	-9.954
Total Laryngectomy with TEP							
7	HvD	0.240	1.156	1.676	7.408	14.090	-9.759
	Sentence01	0.280	1.368	1.804	7.891	16.699	-9.116
	Sentence02	0.241	2.039	1.710	7.302	15.668	-9.258
12	HvD	0.200	1.097	1.243	6.483	12.163	-9.730
	Sentence01	0.216	1.257	1.325	7.236	11.418	-9.859
	Sentence02	0.219	1.040	1.316	7.138	12.992	-9.514
Partial Laryngectomy							
1	HvD	0.343	1.179	1.644	6.622	12.564	-9.953
	Sentence01	0.320	1.079	1.489	6.939	15.203	-9.887
	Sentence02	0.337	2.261	1.483	7.173	13.446	-10.000
2	HvD	0.365	1.268	1.759	7.412	15.167	-8.765
	Sentence01	0.391	1.135	1.719	8.060	16.266	-9.239
	Sentence02	0.337	1.168	1.624	7.262	15.649	-9.922

A.8 Proposed GANs

This section will provide proposed GANs evaluations.

TABLE A.12: MMSE - MMSE Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.385	1.244	1.555	7.270	18.861	-1.729
	Sentence01	0.264	1.094	1.767	8.676	14.901	-0.390
	Sentence02	0.377	1.088	1.591	7.804	13.713	-2.303
14	HvD	0.280	1.314	1.416	7.058	12.889	0.562
	Sentence01	0.390	1.463	1.358	7.196	12.956	1.459
	Sentence02	0.240	1.168	1.463	7.010	15.734	1.276
Total Laryngectomy with TEP							
7	HvD	0.202	2.004	1.501	7.659	11.421	-1.144
	Sentence01	0.372	1.113	1.135	5.865	11.819	-0.092
	Sentence02	0.299	1.098	1.103	5.862	11.933	0.266
12	HvD	0.228	1.261	1.012	6.193	10.027	-2.084
	Sentence01	0.241	1.049	1.068	6.116	11.463	-0.625
	Sentence02	0.272	1.439	0.972	6.127	10.855	-0.332
Partial Laryngectomy							
1	HvD	0.235	1.185	1.189	6.576	8.326	-0.389
	Sentence01	0.267	1.036	1.196	6.615	10.318	-1.142
	Sentence02	0.277	1.056	1.025	6.121	9.526	-0.442
2	HvD	0.316	1.323	0.976	5.494	9.686	-0.212
	Sentence01	0.303	1.398	1.198	6.628	11.902	-1.045
	Sentence02	0.334	1.149	1.274	6.788	11.555	-0.833

TABLE A.13: MMSE - Disco Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.424	1.117	1.573	7.449	18.844	-1.323
	Sentence01	0.211	1.088	1.704	8.482	14.918	0.005
	Sentence02	0.355	1.072	1.594	7.865	13.674	-2.205
14	HvD	0.224	1.327	1.457	7.133	12.935	0.738
	Sentence01	0.379	1.250	1.255	6.803	13.001	0.714
	Sentence02	0.225	1.170	1.463	7.006	15.758	1.287
Total Laryngectomy with TEP							
7	HvD	0.184	1.107	1.333	6.912	11.373	-1.781
	Sentence01	0.346	1.124	1.252	6.355	12.588	-0.707
	Sentence02	0.308	1.101	1.074	5.746	11.946	-1.687
12	HvD	0.230	1.057	0.871	5.712	10.062	-1.924
	Sentence01	0.334	1.046	1.072	6.210	11.483	0.024
	Sentence02	0.326	1.112	0.959	6.032	10.817	-1.472
Partial Laryngectomy							
1	HvD	0.184	1.136	1.241	6.901	8.379	-0.338
	Sentence01	0.284	1.300	1.157	6.566	10.356	-0.587
	Sentence02	0.326	1.068	0.988	6.033	9.518	0.069
2	HvD	0.305	1.066	1.097	6.048	9.702	-0.787
	Sentence01	0.373	1.055	1.204	6.606	11.917	0.379
	Sentence02	0.342	1.080	1.281	6.821	11.591	-1.301

TABLE A.14: MMSE - Cycle Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.346	1.110	1.548	7.263	18.458	-1.580
	Sentence01	0.250	1.071	1.717	8.541	14.910	-0.269
	Sentence02	0.302	1.064	1.666	8.053	13.612	-2.526
14	HvD	0.230	1.300	1.499	7.375	12.925	1.053
	Sentence01	0.345	1.541	1.372	7.266	12.999	-0.483
	Sentence02	0.237	1.345	1.440	7.210	15.749	0.485
Total Laryngectomy with TEP							
7	HvD	0.236	1.301	1.234	6.473	11.431	-1.231
	Sentence01	0.263	1.074	1.193	6.098	11.818	-0.255
	Sentence02	0.274	1.066	1.047	5.630	12.018	0.584
12	HvD	0.270	2.976	0.866	5.641	9.928	-1.879
	Sentence01	0.294	1.048	1.052	6.142	11.534	1.479
	Sentence02	0.326	1.035	0.949	6.120	10.535	-0.409
Partial Laryngectomy							
1	HvD	0.260	1.109	1.216	6.738	8.405	-0.681
	Sentence01	0.274	1.323	1.210	6.752	10.360	-0.872
	Sentence02	0.320	1.063	0.981	5.956	9.535	-0.395
2	HvD	0.321	1.079	1.019	5.683	9.771	-0.708
	Sentence01	0.340	1.735	1.288	6.819	11.958	-0.838
	Sentence02	0.299	2.583	1.446	7.366	11.588	-0.664

TABLE A.15: MMSE - CNN Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.391	1.217	1.572	7.345	18.718	-0.858
	Sentence01	0.324	1.073	1.759	8.698	14.887	-0.618
	Sentence02	0.330	1.083	1.664	8.074	13.676	-2.549
14	HvD	0.230	1.255	1.482	7.180	12.897	0.790
	Sentence01	0.387	1.521	1.267	6.835	12.938	1.708
	Sentence02	0.211	1.245	1.490	7.376	15.741	0.441
Total Laryngectomy with TEP							
7	HvD	0.196	1.295	1.446	7.470	11.444	-1.361
	Sentence01	0.312	1.224	1.250	6.227	12.596	-0.911
	Sentence02	0.297	1.097	1.022	5.584	11.909	0.455
12	HvD	0.258	1.144	0.981	6.024	10.083	-2.366
	Sentence01	0.222	1.038	1.047	6.139	11.462	-0.803
	Sentence02	0.315	2.005	0.826	5.617	10.850	1.756
Partial Laryngectomy							
1	HvD	0.321	1.106	1.121	6.283	8.368	-1.225
	Sentence01	0.284	1.034	1.143	6.474	10.358	-0.847
	Sentence02	0.419	1.064	0.936	5.854	9.497	-1.307
2	HvD	0.335	1.077	1.017	5.626	9.643	-0.439
	Sentence01	0.297	1.267	1.208	6.613	11.908	-0.943
	Sentence02	0.344	1.068	1.296	6.820	11.567	-0.813

TABLE A.16: MMSE - Inception Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.397	1.223	1.563	7.295	18.748	-1.269
	Sentence01	0.358	1.088	1.689	8.385	14.875	-0.515
	Sentence02	0.359	1.082	1.678	8.161	13.700	-2.607
14	HvD	0.279	1.236	1.439	7.104	12.912	0.969
	Sentence01	0.394	1.247	1.358	7.258	12.898	1.450
	Sentence02	0.223	1.299	1.523	7.345	15.721	1.232
Total Laryngectomy with TEP							
7	HvD	0.222	1.178	1.393	7.160	11.329	-1.744
	Sentence01	0.263	1.066	1.088	5.852	11.832	-0.388
	Sentence02	0.235	1.077	1.141	6.015	11.885	-1.063
12	HvD	0.213	2.914	0.939	5.981	10.074	-1.619
	Sentence01	0.305	1.099	1.046	6.117	11.460	-0.195
	Sentence02	0.256	1.084	0.851	5.814	10.836	0.849
Partial Laryngectomy							
1	HvD	0.215	1.102	1.236	6.837	8.323	-0.824
	Sentence01	0.269	1.032	1.168	6.559	10.340	-0.643
	Sentence02	0.368	1.061	0.946	5.886	9.495	0.612
2	HvD	0.255	1.066	1.083	5.936	9.662	-0.578
	Sentence01	0.368	1.366	1.174	6.555	11.895	-0.979
	Sentence02	0.341	1.077	1.276	6.756	11.563	-0.781

TABLE A.17: Disco - MMSE Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.383	1.103	1.505	7.029	15.827	-0.986
	Sentence01	0.311	1.094	1.760	8.398	11.303	-0.344
	Sentence02	0.323	1.114	1.530	7.253	12.133	-2.435
14	HvD	0.231	1.315	1.409	7.049	10.468	1.201
	Sentence01	0.319	1.106	1.142	6.289	10.461	-0.038
	Sentence02	0.235	1.409	1.354	6.291	12.365	1.153
Total Laryngectomy with TEP							
7	HvD	0.218	1.928	1.233	6.623	10.222	-1.307
	Sentence01	0.303	1.088	0.976	5.625	10.117	-0.520
	Sentence02	0.263	1.083	0.993	5.472	10.284	0.090
12	HvD	0.212	1.059	0.979	6.131	10.106	-0.967
	Sentence01	0.273	1.052	0.974	5.876	11.077	1.442
	Sentence02	0.144	1.129	0.976	6.244	11.141	0.176
Partial Laryngectomy							
1	HvD	0.199	1.094	1.202	6.682	7.485	0.066
	Sentence01	0.312	1.037	1.141	6.468	9.023	-1.156
	Sentence02	0.270	1.302	0.912	5.719	8.265	-0.292
2	HvD	0.277	1.101	0.946	5.351	8.119	-0.240
	Sentence01	0.411	1.062	1.168	6.437	10.846	-1.169
	Sentence02	0.275	1.102	1.270	6.784	10.533	-0.587

TABLE A.18: Disco - Disco Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.352	1.090	1.476	6.893	16.272	-1.434
	Sentence01	0.310	1.088	1.708	8.204	11.318	-0.131
	Sentence02	0.340	1.094	1.606	7.495	12.138	-1.989
14	HvD	0.217	1.297	1.454	7.257	10.514	1.105
	Sentence01	0.378	1.047	1.131	6.311	10.749	0.769
	Sentence02	0.296	1.395	1.338	6.181	12.415	1.328
Total Laryngectomy with TEP							
7	HvD	0.249	2.106	1.284	6.798	10.462	-1.871
	Sentence01	0.315	1.094	0.962	5.507	10.687	-0.304
	Sentence02	0.219	1.105	1.087	5.954	10.351	0.003
12	HvD	0.247	1.111	0.904	5.833	10.067	-1.634
	Sentence01	0.292	1.055	1.082	6.295	11.161	0.534
	Sentence02	0.282	1.051	0.997	6.322	10.907	-0.918
Partial Laryngectomy							
1	HvD	0.196	2.296	1.130	6.315	7.493	0.109
	Sentence01	0.290	1.060	1.033	6.132	8.979	0.584
	Sentence02	0.250	1.049	0.920	5.837	8.274	-1.002
2	HvD	0.218	1.094	0.966	5.422	8.113	-0.202
	Sentence01	0.381	1.125	1.145	6.359	10.860	-0.628
	Sentence02	0.254	1.084	1.274	6.798	10.488	-0.216

TABLE A.19: Disco - Cycle Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.349	1.099	1.511	7.132	16.223	-1.434
	Sentence01	0.196	1.094	1.589	8.102	11.397	-0.131
	Sentence02	0.297	1.165	1.529	7.335	12.079	-1.989
14	HvD	0.338	1.246	1.343	6.793	10.567	1.105
	Sentence01	0.378	1.510	1.181	6.454	10.723	0.769
	Sentence02	0.263	1.686	1.348	6.808	12.366	1.328
Total Laryngectomy with TEP							
7	HvD	0.187	1.903	1.229	6.576	10.284	-1.871
	Sentence01	0.294	1.071	1.084	6.108	10.245	-0.304
	Sentence02	0.284	1.073	0.949	5.278	10.142	0.003
12	HvD	0.220	1.054	0.906	5.862	9.987	-1.634
	Sentence01	0.251	1.071	1.077	6.216	11.001	0.534
	Sentence02	0.290	1.075	0.744	5.394	10.928	-0.918
Partial Laryngectomy							
1	HvD	0.343	1.115	1.061	6.123	7.491	0.109
	Sentence01	0.287	1.082	1.083	6.354	9.005	0.584
	Sentence02	0.271	1.178	0.960	5.869	8.320	-1.002
2	HvD	0.351	1.132	1.243	6.753	8.128	-0.202
	Sentence01	0.341	1.122	1.188	6.477	10.869	-0.628
	Sentence02	0.358	1.289	1.235	6.671	10.550	-0.216

TABLE A.20: Disco - CNN Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.322	1.247	1.446	6.794	15.831	-1.001
	Sentence01	0.305	1.102	1.763	8.397	11.338	-1.460
	Sentence02	0.309	1.118	1.596	7.513	12.169	-2.602
14	HvD	0.240	1.300	1.385	6.961	10.468	1.025
	Sentence01	0.329	1.819	1.229	6.613	10.766	1.810
	Sentence02	0.210	1.779	1.389	6.432	12.365	1.888
Total Laryngectomy with TEP							
7	HvD	0.205	1.204	1.287	6.842	10.256	-2.148
	Sentence01	0.258	1.060	1.062	6.127	10.124	0.699
	Sentence02	0.214	1.067	0.954	5.380	10.254	-0.475
12	HvD	0.201	1.077	1.083	6.564	10.112	-1.528
	Sentence01	0.315	1.140	0.927	5.676	11.102	0.533
	Sentence02	0.234	1.057	0.932	6.048	11.178	0.749
Partial Laryngectomy							
1	HvD	0.232	1.118	1.072	6.113	7.146	-0.044
	Sentence01	0.279	1.040	1.037	6.075	9.010	0.621
	Sentence02	0.237	1.228	0.916	5.777	8.267	-0.809
2	HvD	0.329	1.143	0.908	5.329	8.128	-0.511
	Sentence01	0.324	1.157	1.207	6.507	10.854	-1.501
	Sentence02	0.328	1.084	1.240	6.606	10.537	-0.424

TABLE A.21: Disco - Inception Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.342	1.257	1.450	6.746	15.831	-1.949
	Sentence01	0.384	1.130	1.768	8.442	11.321	-0.468
	Sentence02	0.429	1.099	1.537	7.478	12.122	-2.880
14	HvD	0.232	1.292	1.384	6.947	10.468	1.012
	Sentence01	0.451	1.165	1.208	6.645	10.769	1.532
	Sentence02	0.237	1.406	1.381	6.410	12.366	1.203
Total Laryngectomy with TEP							
7	HvD	0.244	1.115	1.108	6.123	10.237	-1.960
	Sentence01	0.253	1.073	1.068	5.924	10.145	-0.448
	Sentence02	0.233	1.072	0.960	5.345	10.246	-0.151
12	HvD	0.237	1.054	0.947	6.014	10.112	-1.462
	Sentence01	0.349	1.109	1.029	6.097	11.084	0.783
	Sentence02	0.280	1.040	0.939	6.136	11.216	-0.850
Partial Laryngectomy							
1	HvD	0.185	1.430	1.130	6.395	7.152	0.081
	Sentence01	0.302	1.040	1.157	6.555	9.047	-0.687
	Sentence02	0.259	1.051	0.931	5.832	8.246	-0.057
2	HvD	0.354	1.092	0.894	5.226	8.117	-1.071
	Sentence01	0.338	1.058	1.186	6.474	10.851	-1.513
	Sentence02	0.317	1.131	1.258	6.703	10.535	-0.858

TABLE A.22: Cycle - MMSE Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.302	1.110	1.679	8.065	20.130	-2.648
	Sentence01	0.351	1.083	1.783	8.861	16.727	-1.831
	Sentence02	0.273	1.122	1.716	8.362	13.945	-3.693
14	HvD	0.252	1.156	1.429	6.893	12.059	-0.748
	Sentence01	0.389	1.086	1.338	6.941	12.276	2.302
	Sentence02	0.396	1.117	1.430	6.769	14.449	-0.650
Total Laryngectomy with TEP							
7	HvD	0.233	1.245	1.395	7.045	12.981	-1.981
	Sentence01	0.249	1.117	1.433	6.851	13.622	-0.033
	Sentence02	0.199	1.195	1.315	6.369	12.899	-0.018
12	HvD	0.232	2.865	1.273	7.230	10.685	-1.835
	Sentence01	0.362	1.064	1.411	7.308	13.611	0.547
	Sentence02	0.304	1.217	0.946	6.073	10.968	-1.270
Partial Laryngectomy							
1	HvD	0.311	1.143	1.188	6.646	8.845	-0.944
	Sentence01	0.339	1.049	1.140	6.514	11.021	-1.365
	Sentence02	0.343	1.059	1.024	6.127	10.313	-1.333
2	HvD	0.175	1.097	1.173	6.172	10.159	-1.091
	Sentence01	0.314	1.088	1.204	6.598	11.908	-2.652
	Sentence02	0.355	1.088	1.354	7.221	12.124	-1.575

TABLE A.23: Cycle - Disco Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.334	1.140	1.688	8.045	20.284	-2.860
	Sentence01	0.316	1.107	1.820	8.990	16.675	-2.189
	Sentence02	0.306	1.116	1.720	8.478	13.942	-3.584
14	HvD	0.292	1.215	1.437	7.015	12.178	-0.532
	Sentence01	0.365	1.296	1.315	6.870	12.338	1.227
	Sentence02	0.363	1.118	1.403	6.283	14.571	-1.024
Total Laryngectomy with TEP							
7	HvD	0.209	1.348	1.570	8.064	12.853	-1.988
	Sentence01	0.256	1.160	1.291	6.388	13.673	-1.222
	Sentence02	0.272	1.175	1.231	6.015	12.776	0.138
12	HvD	0.274	1.179	1.069	6.357	10.684	-2.894
	Sentence01	0.279	1.056	1.410	7.458	13.571	0.970
	Sentence02	0.323	1.080	0.992	6.244	10.945	-0.058
Partial Laryngectomy							
1	HvD	0.300	1.123	1.148	6.388	8.875	-0.737
	Sentence01	0.316	1.046	1.142	6.443	11.045	-1.290
	Sentence02	0.384	1.054	0.994	5.977	10.341	-1.506
2	HvD	0.232	1.117	1.129	6.116	10.175	-1.560
	Sentence01	0.331	1.100	1.211	6.695	12.033	-1.763
	Sentence02	0.398	1.079	1.377	7.400	12.085	-0.748

TABLE A.24: Cycle - Cycle Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.373	1.135	1.617	7.778	20.137	-2.420
	Sentence01	0.282	1.081	1.778	8.868	16.705	-1.239
	Sentence02	0.329	1.136	1.741	8.603	13.883	-3.462
14	HvD	0.282	1.164	1.387	6.850	11.988	-1.698
	Sentence01	0.418	1.121	1.201	6.186	12.101	0.337
	Sentence02	0.311	1.204	1.429	6.732	13.656	-0.154
Total Laryngectomy with TEP							
7	HvD	0.276	1.160	1.350	6.901	12.643	-2.372
	Sentence01	0.301	1.128	1.243	6.332	13.140	0.694
	Sentence02	0.136	1.096	1.237	6.195	12.519	0.650
12	HvD	0.272	1.123	0.859	5.546	10.572	-0.932
	Sentence01	0.285	1.096	1.335	7.081	13.379	1.441
	Sentence02	0.321	1.372	0.782	5.372	10.730	0.979
Partial Laryngectomy							
1	HvD	0.294	1.132	1.131	6.339	8.696	-0.908
	Sentence01	0.305	1.084	1.139	6.493	10.658	-1.712
	Sentence02	0.298	1.179	1.102	6.421	9.659	-0.509
2	HvD	0.414	1.185	1.093	5.976	10.136	-1.345
	Sentence01	0.327	2.103	1.168	6.508	11.920	-1.517
	Sentence02	0.252	1.248	1.566	8.154	12.122	-0.168

TABLE A.25: Cycle - CNN Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.292	1.141	1.704	8.195	19.906	-3.053
	Sentence01	0.376	1.133	1.717	8.594	16.640	-1.985
	Sentence02	0.328	1.127	1.677	8.248	13.939	-4.125
14	HvD	0.296	1.319	1.375	6.863	12.193	-1.090
	Sentence01	0.363	1.061	1.374	7.175	12.293	1.463
	Sentence02	0.386	1.100	1.447	6.571	14.477	-1.198
Total Laryngectomy with TEP							
7	HvD	0.268	1.182	1.352	6.922	13.137	-1.475
	Sentence01	0.185	1.114	1.231	6.346	13.600	0.638
	Sentence02	0.299	1.158	1.248	6.107	13.121	-1.191
12	HvD	0.267	1.167	1.046	6.291	10.704	-2.407
	Sentence01	0.361	1.044	1.366	7.265	13.640	1.165
	Sentence02	0.286	1.182	0.908	5.897	10.967	1.984
Partial Laryngectomy							
1	HvD	0.294	1.136	1.165	6.570	8.867	-0.754
	Sentence01	0.329	1.040	1.159	6.504	11.018	-1.767
	Sentence02	0.400	1.053	1.041	6.202	10.255	-1.499
2	HvD	0.271	1.130	1.144	6.218	10.176	-1.070
	Sentence01	0.426	1.062	1.265	6.870	11.901	-2.349
	Sentence02	0.326	1.091	1.323	7.144	12.107	-0.471

TABLE A.26: Cycle - Inception Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.384	1.133	1.644	7.895	20.299	-2.651
	Sentence01	0.374	1.097	1.834	9.101	16.690	-0.752
	Sentence02	0.325	1.145	1.721	8.449	13.927	-4.031
14	HvD	0.278	1.205	1.421	6.964	12.128	-0.463
	Sentence01	0.390	1.044	1.382	7.191	12.303	2.207
	Sentence02	0.377	1.153	1.446	6.632	14.506	-0.571
Total Laryngectomy with TEP							
7	HvD	0.220	1.143	1.498	7.513	13.204	-2.363
	Sentence01	0.234	1.148	1.243	6.246	13.747	0.246
	Sentence02	0.266	1.146	1.260	6.236	13.174	-0.532
12	HvD	0.250	1.962	1.185	6.845	10.730	-1.574
	Sentence01	0.289	1.045	1.397	7.409	13.750	-0.925
	Sentence02	0.343	1.232	1.206	6.993	10.966	0.234
Partial Laryngectomy							
1	HvD	0.270	1.131	1.202	6.712	8.882	-0.669
	Sentence01	0.349	1.043	1.155	6.515	11.128	-1.468
	Sentence02	0.390	1.060	0.942	5.818	10.267	-1.479
2	HvD	0.317	1.108	1.109	6.009	10.191	-1.469
	Sentence01	0.358	1.056	1.267	6.949	12.019	-2.322
	Sentence02	0.336	1.069	1.380	7.371	12.074	-0.613

TABLE A.27: Inception - MMSE Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.359	1.125	1.575	7.598	18.144	-3.089
	Sentence01	0.327	1.092	1.834	9.066	15.785	-1.959
	Sentence02	0.392	1.094	1.778	8.473	13.092	-4.006
14	HvD	0.271	2.906	1.528	7.345	14.202	-2.212
	Sentence01	0.359	1.798	1.329	6.876	14.401	-1.127
	Sentence02	0.329	1.385	1.492	7.180	17.114	-2.070
Total Laryngectomy with TEP							
7	HvD	0.191	1.164	1.339	7.017	12.290	-2.369
	Sentence01	0.260	1.114	1.316	6.827	13.104	-1.313
	Sentence02	0.276	1.130	1.266	6.425	11.916	0.442
12	HvD	0.190	3.125	1.008	6.282	10.352	-2.284
	Sentence01	0.220	1.079	1.192	6.666	11.458	-0.603
	Sentence02	0.267	1.081	0.986	6.283	12.941	-0.841
Partial Laryngectomy							
1	HvD	0.344	1.273	1.252	7.021	10.233	-2.039
	Sentence01	0.354	1.063	1.024	6.293	11.792	-2.901
	Sentence02	0.302	1.437	1.057	6.497	10.754	-2.157
2	HvD	0.301	1.182	1.216	6.725	11.575	-2.201
	Sentence01	0.318	1.504	1.267	7.001	13.539	-3.423
	Sentence02	0.278	1.343	1.489	7.936	14.059	-2.000

TABLE A.28: Inception - Disco Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.319	1.102	1.627	7.920	18.212	-3.094
	Sentence01	0.277	1.078	1.811	8.754	15.784	-2.049
	Sentence02	0.314	1.106	1.782	8.509	13.161	-3.267
14	HvD	0.318	1.143	1.548	7.472	14.220	-2.477
	Sentence01	0.341	1.096	1.450	7.450	14.537	-1.524
	Sentence02	0.383	1.110	1.529	6.928	17.155	-1.654
Total Laryngectomy with TEP							
7	HvD	0.257	1.193	1.335	7.041	12.448	-2.678
	Sentence01	0.254	1.090	1.383	7.002	13.177	-1.266
	Sentence02	0.314	1.127	1.132	5.826	12.039	0.401
12	HvD	0.260	1.549	1.098	6.560	10.425	-1.780
	Sentence01	0.221	1.056	1.185	6.700	11.512	0.417
	Sentence02	0.295	1.049	0.923	6.054	12.804	0.492
Partial Laryngectomy							
1	HvD	0.336	1.170	1.195	6.765	10.269	-1.502
	Sentence01	0.279	1.064	1.063	6.411	11.779	-3.309
	Sentence02	0.280	1.113	1.011	6.234	10.771	-1.068
2	HvD	0.215	1.164	1.340	7.233	11.479	-2.596
	Sentence01	0.370	1.077	1.229	6.829	13.452	-2.871
	Sentence02	0.225	1.503	1.454	7.729	14.028	-2.136

TABLE A.29: Inception - Cycle Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.342	1.165	1.557	7.520	18.184	-2.738
	Sentence01	0.279	1.103	1.820	8.896	15.851	-1.757
	Sentence02	0.438	1.129	1.777	8.476	13.088	-3.587
14	HvD	0.301	1.242	1.456	7.036	14.216	-2.198
	Sentence01	0.413	1.112	1.326	6.792	14.805	-0.448
	Sentence02	0.229	1.210	1.573	6.844	17.033	-1.362
Total Laryngectomy with TEP							
7	HvD	0.241	1.162	1.369	7.128	12.346	-1.222
	Sentence01	0.255	1.129	1.315	6.650	13.004	0.702
	Sentence02	0.315	1.093	1.216	6.195	11.767	0.170
12	HvD	0.182	1.080	0.959	6.026	10.334	-1.124
	Sentence01	0.260	1.103	1.128	6.347	11.221	1.465
	Sentence02	0.286	1.080	0.854	5.750	12.549	0.771
Partial Laryngectomy							
1	HvD	0.304	1.171	1.200	6.774	10.151	-1.059
	Sentence01	0.354	1.259	1.114	6.575	11.734	-2.151
	Sentence02	0.334	1.059	1.067	6.561	10.719	-2.595
2	HvD	0.301	2.214	1.054	5.993	11.589	-0.987
	Sentence01	0.344	1.195	1.205	6.912	13.425	-3.126
	Sentence02	0.397	1.075	1.377	7.428	14.012	-1.650

TABLE A.30: Inception - CNN Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.320	1.134	1.589	7.682	18.144	-2.611
	Sentence01	0.322	1.096	1.811	8.716	15.841	-1.835
	Sentence02	0.423	1.118	1.750	8.488	13.140	-4.151
14	HvD	0.246	1.205	1.589	7.779	14.204	-2.206
	Sentence01	0.322	2.070	1.332	6.883	14.384	-0.974
	Sentence02	0.252	1.123	1.552	6.912	17.089	-2.839
Total Laryngectomy with TEP							
7	HvD	0.262	1.192	1.398	7.315	12.256	-2.471
	Sentence01	0.300	1.165	1.280	6.668	13.045	-0.352
	Sentence02	0.264	1.163	1.193	6.104	11.971	-0.666
12	HvD	0.256	1.169	1.007	6.209	10.370	-1.644
	Sentence01	0.187	1.046	1.195	6.755	11.393	-1.182
	Sentence02	0.223	1.091	0.991	6.383	12.941	-1.775
Partial Laryngectomy							
1	HvD	0.268	1.223	1.169	6.687	10.239	-1.667
	Sentence01	0.357	1.387	1.084	6.520	11.821	-2.053
	Sentence02	0.315	1.614	1.069	6.448	10.745	-0.857
2	HvD	0.290	1.153	1.245	6.784	11.580	-1.402
	Sentence01	0.433	1.055	1.148	6.633	13.563	-3.802
	Sentence02	0.250	1.088	1.441	7.778	14.113	-2.466

TABLE A.31: Inception - Inception Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.362	1.167	1.572	7.592	18.149	-2.564
	Sentence01	0.349	1.095	1.812	8.746	15.861	-1.825
	Sentence02	0.366	1.239	1.790	8.564	13.144	-4.105
14	HvD	0.290	1.805	1.481	7.172	14.130	-2.040
	Sentence01	0.348	1.179	1.329	6.863	14.367	-1.128
	Sentence02	0.276	1.198	1.553	6.912	17.094	-2.838
Total Laryngectomy with TEP							
7	HvD	0.262	1.192	1.398	7.315	12.256	-2.471
	Sentence01	0.300	1.165	1.280	6.668	13.045	-0.352
	Sentence02	0.264	1.163	1.193	6.104	11.971	-0.666
12	HvD	0.256	1.169	1.007	6.209	10.370	-1.644
	Sentence01	0.187	1.046	1.195	6.755	11.393	-1.182
	Sentence02	0.223	1.091	0.991	6.383	12.941	-1.775
Partial Laryngectomy							
1	HvD	0.268	1.223	1.169	6.687	10.239	-1.667
	Sentence01	0.357	1.387	1.084	6.520	11.821	-2.053
	Sentence02	0.315	1.614	1.069	6.448	10.745	-0.857
2	HvD	0.290	1.153	1.245	6.784	11.580	-1.402
	Sentence01	0.433	1.055	1.148	6.633	13.563	-3.802
	Sentence02	0.250	1.088	1.441	7.778	14.113	-2.466

TABLE A.32: CNN - MMSE Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.428	1.122	1.615	7.233	15.872	-0.430
	Sentence01	0.262	1.098	1.723	7.977	13.407	-0.711
	Sentence02	0.350	1.179	1.624	7.123	10.534	-1.548
14	HvD	0.344	1.257	1.566	7.413	13.300	0.815
	Sentence01	0.386	2.676	1.475	7.284	13.131	0.218
	Sentence02	0.347	1.352	1.537	6.589	14.320	0.824
Total Laryngectomy with TEP							
7	HvD	0.315	1.292	1.348	6.885	12.540	-1.771
	Sentence01	0.232	1.128	1.007	5.392	12.361	1.017
	Sentence02	0.255	1.351	1.101	5.514	13.115	0.303
12	HvD	0.246	1.142	0.921	5.847	9.147	-2.265
	Sentence01	0.394	1.062	1.350	7.177	12.207	-0.468
	Sentence02	0.253	1.185	0.982	5.987	10.225	1.085
Partial Laryngectomy							
1	HvD	0.333	1.201	1.192	6.503	9.571	0.720
	Sentence01	0.364	1.137	1.195	6.637	11.373	-0.244
	Sentence02	0.361	1.049	1.076	6.309	10.825	-0.760
2	HvD	0.327	1.063	1.072	5.935	10.982	-0.252
	Sentence01	0.339	1.161	1.319	6.765	13.291	0.370
	Sentence02	0.364	1.232	1.409	7.212	12.312	-0.209

TABLE A.33: CNN - Disco Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.435	1.170	1.613	7.239	15.855	-0.447
	Sentence01	0.281	1.132	1.740	8.071	13.429	-0.577
	Sentence02	0.285	1.130	1.640	7.146	10.563	-0.746
14	HvD	0.359	1.264	1.562	7.387	13.320	0.860
	Sentence01	0.376	1.954	1.414	6.652	13.122	0.942
	Sentence02	0.349	1.536	1.518	6.577	14.351	0.837
Total Laryngectomy with TEP							
7	HvD	0.291	1.153	1.352	6.880	12.594	-1.357
	Sentence01	0.201	1.141	1.114	5.828	12.477	0.316
	Sentence02	0.288	1.164	1.208	6.019	13.235	0.257
12	HvD	0.196	1.116	1.058	6.388	9.145	-1.688
	Sentence01	0.336	1.077	1.230	6.636	12.274	0.990
	Sentence02	0.188	1.132	0.942	5.934	10.218	1.702
Partial Laryngectomy							
1	HvD	0.381	1.129	1.186	6.525	9.830	0.726
	Sentence01	0.320	1.092	1.170	6.592	11.380	-0.117
	Sentence02	0.331	1.077	1.133	6.471	10.893	-0.531
2	HvD	0.339	1.060	1.047	5.808	10.982	-0.421
	Sentence01	0.338	1.215	1.336	6.820	13.286	0.828
	Sentence02	0.369	1.143	1.410	7.200	12.227	0.013

TABLE A.34: CNN - Cycle Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.432	1.130	1.635	7.280	16.018	-0.664
	Sentence01	0.241	1.146	1.751	8.089	13.469	-0.012
	Sentence02	0.377	1.205	1.622	7.342	10.510	-1.401
14	HvD	0.337	1.180	1.573	7.350	13.298	0.745
	Sentence01	0.447	1.097	1.398	7.138	13.193	0.769
	Sentence02	0.368	1.189	1.501	6.451	14.319	0.764
Total Laryngectomy with TEP							
7	HvD	0.196	1.178	1.354	6.896	12.530	-1.878
	Sentence01	0.334	1.164	1.011	5.372	12.422	-0.196
	Sentence02	0.348	1.140	1.047	5.301	13.206	0.737
12	HvD	0.249	1.184	0.903	5.737	9.083	-1.016
	Sentence01	0.344	1.096	1.096	6.248	12.374	1.535
	Sentence02	0.255	1.333	0.933	5.866	10.227	0.620
Partial Laryngectomy							
1	HvD	0.341	1.115	1.099	6.142	9.965	0.954
	Sentence01	0.328	1.386	1.189	6.608	11.434	-0.848
	Sentence02	0.313	1.129	0.998	6.030	10.687	-0.722
2	HvD	0.310	1.234	1.103	6.062	11.007	-0.188
	Sentence01	0.342	1.237	1.299	6.790	13.292	-0.593
	Sentence02	0.378	1.359	1.413	7.225	12.313	0.164

TABLE A.35: CNN - CNN Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.453	1.126	1.607	7.210	15.878	-0.357
	Sentence01	0.350	1.103	1.738	8.142	13.416	0.051
	Sentence02	0.342	1.149	1.629	7.168	10.536	-0.833
14	HvD	0.335	1.277	1.565	7.374	13.295	0.732
	Sentence01	0.439	1.118	1.392	7.143	13.125	0.234
	Sentence02	0.365	1.370	1.525	6.554	14.314	0.857
Total Laryngectomy with TEP							
7	HvD	0.253	1.259	1.377	7.108	12.529	-1.518
	Sentence01	0.279	1.140	1.168	5.835	12.370	0.307
	Sentence02	0.292	1.052	1.145	5.715	13.080	0.272
12	HvD	0.273	1.154	0.919	5.812	9.166	-1.471
	Sentence01	0.322	1.131	1.194	6.564	12.203	-0.770
	Sentence02	0.231	1.704	0.980	5.986	10.234	-1.198
Partial Laryngectomy							
1	HvD	0.298	1.294	1.215	6.596	9.554	0.447
	Sentence01	0.316	1.048	1.134	6.463	11.390	0.012
	Sentence02	0.292	1.075	1.020	6.080	10.647	-0.788
2	HvD	0.350	1.063	1.028	5.769	10.983	-0.266
	Sentence01	0.339	1.148	1.310	6.747	13.205	-1.516
	Sentence02	0.364	1.098	1.404	7.191	12.302	0.026

TABLE A.36: CNN - Inception Final Experiment Evaluation

ID	File Name	STOI	PESQ	LLR	CD	MCD	SSNR
Total Laryngectomy							
11	HvD	0.433	1.133	1.618	7.249	15.863	-0.505
	Sentence01	0.427	1.120	1.732	8.186	13.397	0.065
	Sentence02	0.374	1.054	1.644	7.215	10.532	-1.534
14	HvD	0.322	1.281	1.571	7.353	13.289	0.354
	Sentence01	0.429	1.109	1.400	7.142	13.122	0.201
	Sentence02	0.347	1.133	1.528	6.532	14.303	0.852
Total Laryngectomy with TEP							
7	HvD	0.292	1.189	1.392	7.076	12.506	-1.260
	Sentence01	0.291	1.106	1.086	5.629	12.377	-0.714
	Sentence02	0.236	1.224	1.226	6.093	13.087	0.578
12	HvD	0.255	1.069	0.809	5.346	9.155	-1.872
	Sentence01	0.427	1.089	1.288	6.883	12.197	1.136
	Sentence02	0.243	1.508	1.027	6.119	10.256	1.146
Partial Laryngectomy							
1	HvD	0.327	1.233	1.186	6.423	9.564	0.350
	Sentence01	0.310	1.233	1.160	6.516	11.374	-0.213
	Sentence02	0.303	1.068	1.005	6.028	10.652	-1.034
2	HvD	0.330	1.065	1.077	5.936	10.983	-0.256
	Sentence01	0.334	1.145	1.309	6.749	13.200	-1.480
	Sentence02	0.395	1.258	1.429	7.353	12.303	0.052

Appendix B

Code Samples

B.1 Generators

```
1 class dnn_generator(nn.Module):
2
3     # Weight Initialization
4     def weight_init(self):
5         nn.init.xavier_uniform_(self.fc1.weight)
6         nn.init.xavier_uniform_(self.fc2.weight)
7         nn.init.xavier_uniform_(self.fc3.weight)
8         nn.init.xavier_uniform_(self.out.weight)
9
10        nn.init.xavier_uniform_(self.fc1.bias)
11        nn.init.xavier_uniform_(self.fc2.bias)
12        nn.init.xavier_uniform_(self.fc3.bias)
13        nn.init.xavier_uniform_(self.out.bias)
14
15    def __init__(self, G_in, G_out, w1, w2, w3):
16        super(dnn_generator, self).__init__()
17
18        self.fc1= nn.Linear(G_in, w1)
19        self.fc2= nn.Linear(w1, w2)
20        self.fc3= nn.Linear(w2, w3)
21        self.out= nn.Linear(w3, G_out)
```

```

22
23     self.weight_init()
24
25     # Deep neural network
26     def forward(self, x):
27
28         x = F.relu(self.fc1(x))
29         x = F.relu(self.fc2(x))
30         x = F.relu(self.fc3(x))
31         x = self.out(x)
32     return x

```

LISTING B.1: Generator for MMSE, Disco, and Cycle GAN for
MCC and F0 feature

```

1 class cnn_mcc_generator(nn.Module):
2
3     def __init__(self):
4         super(cnn_mcc_generator, self).__init__()
5
6         lower_layers = []
7         self.lower_layers = nn.Sequential(*lower_layers)
8
9         inception_layers = []
10        inception_layers += [nn.Conv2d(1, 128, 3, padding=1)]
11        inception_layers += [nn.ReLU(True)]
12        inception_layers += [nn.BatchNorm2d(128)]
13        inception_layers += [nn.Conv2d(128, 256, 3, padding=1)]
14        inception_layers += [nn.ReLU(True)]
15        inception_layers += [nn.BatchNorm2d(256)]
16        inception_layers += [nn.Conv2d(256, 128, 3, padding=1)]
17        inception_layers += [nn.ReLU(True)]
18        inception_layers += [nn.BatchNorm2d(128)]
19        self.inception_layers = nn.Sequential(*inception_layers)
20
21        final_layers = []

```

```

22         final_layers += [nn.Conv2d(128, 1, 3, stride=1, padding
23         =1)]
24
25     def forward(self, x):
26         h1 = self.lower_layers(x)
27         h2 = self.inception_layers(h1)
28         return self.final_layers(h2)

```

LISTING B.2: CNN GAN for MCC feature Generator

```

1 class cnn_f0_generator(nn.Module):
2
3
4     def __init__(self):
5         super(cnn_f0_generator, self).__init__()
6
7         lower_layers = []
8         self.lower_layers = nn.Sequential(*lower_layers)
9
10        inception_layers = []
11        inception_layers += [nn.Conv2d(1, 128, 3, padding=1)]
12        inception_layers += [nn.ReLU(True)]
13        inception_layers += [nn.MaxPool2d((1,7),(1,2))]
14        inception_layers += [nn.Conv2d(128, 256, 3, padding=1)]
15        inception_layers += [nn.ReLU(True)]
16        inception_layers += [nn.MaxPool2d((1,7),(1,2))]
17        inception_layers += [nn.Conv2d(256, 128, 3, padding=1)]
18        inception_layers += [nn.ReLU(True)]
19        inception_layers += [nn.MaxPool2d((1,6),(1,2))]
20        self.inception_layers = nn.Sequential(*inception_layers)
21
22        final_layers = []
23        final_layers += [nn.Conv2d(128, 1, 3, stride=1, padding
24        =1)]
25        self.final_layers = nn.Sequential(*final_layers)

```

```
25
26
27     def forward(self, x):
28
29         h1 = self.lower_layers(x)
30         h2 = self.inception_layers(h1)
31         return self.final_layers(h2)
```

LISTING B.3: CNN GAN for F0 feature Generator

```
1 class inception(nn.Module):
2
3     def __init__(self, inp, n1, n3r, n3, n5r, n5, mxp):
4         super(inception, self).__init__()
5
6         layers = []
7         layers += [nn.Conv2d(inp, n1, 1)]
8         layers += [nn.ReLU(True)]
9         nn.init.xavier_uniform_(layers[0].weight)
10
11         self.one = nn.Sequential(*layers)
12
13         layers = []
14         layers += [nn.Conv2d(inp, n3r, 1)]
15         layers += [nn.ReLU(True)]
16         layers += [nn.Conv2d(n3r, n3, 3, padding=1)]
17         layers += [nn.ReLU(True)]
18         nn.init.xavier_uniform_(layers[0].weight)
19         nn.init.xavier_uniform_(layers[2].weight)
20
21         self.three = nn.Sequential(*layers)
22
23         layers = []
24         layers += [nn.Conv2d(inp, n5r, 1)]
25         layers += [nn.ReLU(True)]
26         layers += [nn.Dropout(0.3)]
```

```

27     layers += [nn.Conv2d(n5r, n5, 3, padding=1)]
28     layers += [nn.ReLU(True)]
29     layers += [nn.Dropout(0.3)]
30     layers += [nn.Conv2d(n5, n5, 3, padding=1)]
31     layers += [nn.ReLU(True)]
32     layers += [nn.Dropout(0.3)]
33     nn.init.xavier_uniform_(layers[0].weight)
34     nn.init.xavier_uniform_(layers[2].weight)
35     nn.init.xavier_uniform_(layers[4].weight)
36
37     self.five = nn.Sequential(*layers)
38
39     layers = []
40
41     layers += [nn.MaxPool2d(3, 1, 1)]
42     layers += [nn.Conv2d(inp, mxp, 1)]
43     layers += [nn.ReLU(True)]
44     layers += [nn.Dropout(0.5)]
45     nn.init.xavier_uniform_(layers[1].weight)
46     self.maxp = nn.Sequential(*layers)
47
48     def forward(self, x):
49         h1 = self.one(x)
50         h2 = self.three(x)
51         h3 = self.five(x)
52         h4 = self.maxp(x)
53         h = torch.cat([h1, h2, h3, h4], 1)
54         return h

```

LISTING B.4: Inception Module

```

1 class inv_inception(nn.Module):
2
3     def __init__(self, inp, n1, n3r, n3, n5r, n5, mxp):
4         super(inv_inception, self).__init__()
5

```

```
6         layers = []
7         layers += [nn.ConvTranspose2d(inp, n1, 1, stride=1,
padding=0, output_padding=0)]
8         layers += [nn.ReLU(True)]
9         nn.init.xavier_uniform_(layers[0].weight)
10
11        self.one = nn.Sequential(*layers)
12
13        layers = []
14        layers += [nn.ConvTranspose2d(inp, n3r, 1, stride=1,
padding=0, output_padding=0)]
15        layers += [nn.ReLU(True)]
16        layers += [nn.ConvTranspose2d(n3r, n3, 3, stride=1,
padding=1, output_padding=0)]
17        layers += [nn.ReLU(True)]
18        nn.init.xavier_uniform_(layers[0].weight)
19        nn.init.xavier_uniform_(layers[2].weight)
20
21        self.three = nn.Sequential(*layers)
22
23        layers = []
24        layers += [nn.ConvTranspose2d(inp, n5r, 1, stride=1,
padding=0, output_padding=0)]
25        layers += [nn.ReLU(True)]
26        layers += [nn.ConvTranspose2d(n5r, n5, 3, stride=1,
padding=1, output_padding=0)]
27        layers += [nn.ReLU(True)]
28        layers += [nn.ConvTranspose2d(n5, n5, 3, stride=1,
padding=1, output_padding=0)]
29        layers += [nn.ReLU(True)]
30        nn.init.xavier_uniform_(layers[0].weight)
31        nn.init.xavier_uniform_(layers[2].weight)
32        nn.init.xavier_uniform_(layers[4].weight)
33
34        self.five = nn.Sequential(*layers)
```



```

35
36     layers = []
37     layers += [nn.MaxPool2d(3, 1, 1)]
38     layers += [nn.ConvTranspose2d(inp, mxp, 1, stride=1,
padding=0, output_padding=0)]
39     layers += [nn.ReLU(True)]
40     nn.init.xavier_uniform_(layers[1].weight)
41
42     self.maxp = nn.Sequential(*layers)
43
44     def forward(self, x):
45         h1 = self.one(x)
46         h2 = self.three(x)
47         h3 = self.five(x)
48         h4 = self.maxp(x)
49
50         h = torch.cat([h1, h2, h3, h4], 1)
51
52     return h

```

LISTING B.5: Inverse Inception Module

```

1 class inception_mcc_generator(nn.Module):
2
3
4     def __init__(self):
5         super(inception_mcc_generator, self).__init__()
6
7         lower_layers = []
8         self.lower_layers = nn.Sequential(*lower_layers)
9
10        inception_layers = []
11        inception_layers += [inception(1, 32, 50, 64, 16, 32, 16)
]
12        inception_layers += [inception(144, 64, 64, 128, 32, 64,
32)]

```

```

13         inception_layers += [inv_inception(288, 64, 64, 128, 32,
14         64, 32)]
15         inception_layers += [inv_inception(288, 32, 128, 64, 64,
16         32, 16)]
17         self.inception_layers = nn.Sequential(*inception_layers)
18
19         final_layers = []
20         final_layers += [nn.Conv2d(144, 1, 3, stride=1, padding
21         =1)] # Out: 1000x40x1
22         self.final_layers = nn.Sequential(*final_layers)
23
24     def forward(self, x):
25
26         h1 = self.lower_layers(x)
27         h2 = self.inception_layers(h1)
28         return self.final_layers(h2)

```

LISTING B.6: Inception GAN for MCC feature Generator

```

1 class inception_f0_generator(nn.Module):
2
3
4     def __init__(self):
5         super(inception_f0_generator, self).__init__()
6
7         lower_layers = []
8         self.lower_layers = nn.Sequential(*lower_layers)
9
10        inception_layers = []
11        inception_layers += [inception(1, 32, 50, 64, 16, 32, 16)
12        ]
13        inception_layers += [nn.MaxPool2d((1,7),(1,2))]
14        inception_layers += [inception(144, 64, 64, 128, 32, 64,
15        32)]
16        inception_layers += [nn.MaxPool2d((1,7),(1,2))]

```

```

15         inception_layers += [inception(288, 64, 64, 128, 32, 64,
16         32)]
17         inception_layers += [nn.MaxPool2d((1,6),(1,2))]
18         inception_layers += [inception(288, 32, 128, 64, 64, 32,
19         16)]
20         self.inception_layers = nn.Sequential(*inception_layers)
21
22         final_layers = []
23         final_layers += [nn.Conv2d(144, 1, 3, stride=1, padding
24         =1)] # Out: 1000x40x1
25         self.final_layers = nn.Sequential(*final_layers)
26
27     def forward(self, x):
28
29         h1 = self.lower_layers(x)
30         h2 = self.inception_layers(h1)
31         return self.final_layers(h2)

```

LISTING B.7: Inception GAN for F0 feature Generator

B.2 Discriminators

```

1 class dnn_discriminator(nn.Module):
2     def __init__(self, D_in, D_out, w1, w2, w3,rate):
3         super(dnn_discriminator, self).__init__()
4         self.fc1 = nn.Linear(D_in, w1)
5         # self.bn1 = nn.BatchNorm1d(w1) # Cycle
6         self.fc2 = nn.Linear(w1, w2)
7         # self.bn2 = nn.BatchNorm1d(w2) # Cycle
8         self.fc3 = nn.Linear(w2, w3)
9         # self.bn3 = nn.BatchNorm1d(w3) # Cycle
10        self.out = nn.Linear(w3, D_out)
11        self.dropout = nn.Dropout(rate)

```

```

12
13     def forward(self, y):
14         y = F.relu(self.fc1(y))
15         y = self.dropout(y)
16         y = F.relu(self.fc2(y))
17         y = self.dropout(y)
18         y = F.relu(self.fc3(y))
19         # uncomment for Cycle GAN
20         # y = self.dropout(y)
21         y = torch.sigmoid(self.out(y))
22     return y

```

LISTING B.8: Discriminator for MMSE, Disco, and Cycle GAN
for MCC and F0 feature

```

1 class cnn_f0_discriminator(nn.Module):
2     def __init__(self, D_in, D_out, w1, w2, w3, w4):
3         super(cnn_f0_discriminator, self).__init__()
4         self.fc1 = nn.Linear(D_in, w1)
5         self.fc2 = nn.Linear(w1, w2)
6         self.fc3 = nn.Linear(w2, w3)
7         self.fc4 = nn.Linear(w3, w4)
8         self.out = nn.Linear(w4, D_out)
9         self.dropout = nn.Dropout(0.5)
10
11     def forward(self, y):
12         y = F.relu(self.fc1(y))
13         y = self.dropout(y)
14         y = F.relu(self.fc2(y))
15         y = self.dropout(y)
16         y = F.relu(self.fc3(y))
17         y = self.dropout(y)
18         y = F.relu(self.fc4(y))
19         y = self.dropout(y)
20         y = torch.sigmoid(self.out(y))

```

```
21         return y
```

LISTING B.9: CNN and Inception for F0 feature Discriminators

```
1 class cnn_mcc_discriminator(nn.Module):
2
3     def __init__(self):
4         super(cnn_mcc_discriminator, self).__init__()
5
6         lower_layers = []
7         lower_layers += [nn.Conv2d(1, 32, 7, 2, 3)]
8         lower_layers += [nn.ReLU(True)]
9         lower_layers += [nn.Dropout(0.3)]
10        lower_layers += [nn.Conv2d(32, 64, 3, 1, 1)]
11        lower_layers += [nn.ReLU(True)]
12        lower_layers += [nn.Dropout(0.3)]
13        lower_layers += [nn.MaxPool2d(3, (3,2), 1)]
14        lower_layers += [nn.Conv2d(64, 128, 3, 1, 1)]
15        lower_layers += [nn.ReLU(True)]
16        lower_layers += [nn.Dropout(0.5)]
17        lower_layers += [nn.MaxPool2d(3, (4,2), 1)]
18        lower_layers += [nn.Conv2d(128, 256, 3, 1, 1)]
19        lower_layers += [nn.ReLU(True)]
20        lower_layers += [nn.Dropout(0.5)]
21        lower_layers += [nn.MaxPool2d(3, (3,2), 1)]
22        lower_layers += [nn.Conv2d(256, 256, 3, 1, 1)]
23        lower_layers += [nn.ReLU(True)]
24        lower_layers += [nn.Dropout(0.3)]
25
26
27        self.lower_layers = nn.Sequential(*lower_layers)
28
29
30        final_layers = []
31
32        final_layers += [nn.Linear(3*14*256, 1028)]
```

```

33         final_layers += [nn.ReLU(True)]
34         final_layers += [nn.Dropout(0.5)]
35         final_layers += [nn.Linear(1028, 1)]
36         final_layers += [nn.Sigmoid()]
37
38         nn.init.xavier_uniform_(final_layers[0].weight)
39         nn.init.xavier_uniform_(final_layers[3].weight)
40
41         self.final_layers = nn.Sequential(*final_layers)
42
43     def forward(self, x):
44         h1 = self.lower_layers(x)
45         h2 = h1.view(h1.size(0), -1)
46         return self.final_layers(h2)

```

LISTING B.10: CNN GAN for MCC feature Discriminator

```

1 class inception_mcc_discriminator(nn.Module):
2
3     def __init__(self):
4         super(inception_mcc_discriminator, self).__init__()
5
6         lower_layers = []
7         lower_layers += [nn.Conv2d(1, 32, 7, 2, 3)] # Out: 500
8         lower_layers += [nn.ReLU(True)]
9         lower_layers += [nn.Dropout(0.5)]
10
11         self.lower_layers = nn.Sequential(*lower_layers)
12
13         inception_layers = []
14
15         inception_layers += [inception(32, 32, 50, 64, 16, 32,
16         16)]
17         inception_layers += [nn.AvgPool2d((25, 2), (5, 1))]

```

```

18         inception_layers += [inception(144, 64, 64, 128, 32, 64,
19         32)]
20
21         inception_layers += [nn.MaxPool2d(3, 2, 1)]
22
23         inception_layers += [inception(288, 128, 128, 256, 32,
24         64, 48)]
25
26         inception_layers += [nn.AvgPool2d((10,2),(6,6))]
27
28         self.inception_layers = nn.Sequential(*inception_layers)
29
30         final_layers = []
31
32         final_layers += [nn.Linear(7*2*496, 1028)]
33         final_layers += [nn.ReLU(True)]
34         final_layers += [nn.Dropout(0.5)]
35         final_layers += [nn.Linear(1028, 1)]
36         final_layers += [nn.Sigmoid()]
37
38         nn.init.xavier_uniform_(final_layers[0].weight)
39         nn.init.xavier_uniform_(final_layers[3].weight)
40
41         self.final_layers = nn.Sequential(*final_layers)
42
43     def forward(self, x):
44         h1 = self.lower_layers(x)
45         h2 = self.inception_layers(h1)
46         h2 = h2.view(h2.size(0), -1)
47         return self.final_layers(h2)

```

LISTING B.11: Inception GAN for MCC feature Discriminator

B.3 DNN Module for VUV feature Model

```

1 class dnn(nn.Module):

```

```
2
3     def weight_init(self):
4         for m in self.modules():
5             if isinstance(m, nn.Linear):
6                 nn.init.xavier_uniform_(m.weight)
7
8     def __init__(self, G_in, G_out, w1, w2, w3, w4, w5, w6, w7,
9 w8, dropout_rate=0.3):
10
11         super(dnn, self).__init__()
12
13         self.fc1 = nn.Linear(G_in, w1)
14         self.prelu1 = nn.PReLU()
15         self.bn1 = nn.BatchNorm1d(w1)
16
17         self.fc2 = nn.Linear(w1, w2)
18         self.prelu2 = nn.PReLU()
19         self.bn2 = nn.BatchNorm1d(w2)
20
21         self.fc3 = nn.Linear(w2, w3)
22         self.prelu3 = nn.PReLU()
23         self.bn3 = nn.BatchNorm1d(w3)
24
25         self.fc4 = nn.Linear(w3, w4)
26         self.prelu4 = nn.PReLU()
27         self.bn4 = nn.BatchNorm1d(w4)
28
29         self.fc5 = nn.Linear(w4, w5)
30         self.prelu5 = nn.PReLU()
31         self.bn5 = nn.BatchNorm1d(w5)
32
33         self.fc6 = nn.Linear(w5, w6)
34         self.prelu6 = nn.PReLU()
35         self.bn6 = nn.BatchNorm1d(w6)
36
37         self.fc7 = nn.Linear(w6, w7)
```



```
36         self.prelu7 = nn.PReLU()
37         self.bn7 = nn.BatchNorm1d(w7)
38
39         self.fc8 = nn.Linear(w7, w8)
40         self.prelu8 = nn.PReLU()
41         self.bn8 = nn.BatchNorm1d(w8)
42
43         self.out = nn.Linear(w8, G_out)
44         self.dropout = nn.Dropout(dropout_rate)
45
46         self.weight_init()
47
48     def forward(self, x):
49         x = self.prelu1(self.bn1(self.fc1(x)))
50         x = self.dropout(x)
51         x = self.prelu2(self.bn2(self.fc2(x)))
52         x = self.dropout(x)
53         x = self.prelu3(self.bn3(self.fc3(x)))
54         x = self.dropout(x)
55         x = self.prelu4(self.bn4(self.fc4(x)))
56         x = self.dropout(x)
57         x = self.prelu5(self.bn5(self.fc5(x)))
58         x = self.dropout(x)
59         x = self.prelu6(self.bn6(self.fc6(x)))
60         x = self.dropout(x)
61         x = self.prelu7(self.bn7(self.fc7(x)))
62         x = self.dropout(x)
63         x = self.prelu8(self.bn8(self.fc8(x)))
64         x = self.dropout(x)
65         x = torch.sigmoid(self.out(x))
66         return x
```

LISTING B.12: DNN Module for VUV feature Model